

CHAPTER 22

RADIOACTIVITY

5.2 Radioactivity

5.2.1 Detection of radioactivity

- 1 Describe the detection of alpha particles (α -particles) using a cloud chamber or spark counter and the detection of beta particles (β -particles) (β -particles will be taken to refer to β^-) and gamma radiation (γ -radiation) by using a Geiger-Müller tube and counter
- 2 Use count rate measured in counts / s or counts / minute
- 3 Know what is meant by background radiation
- 4 Know the sources that make a significant contribution to background radiation including:
 - (a) radon gas (in the air)
 - (b) rocks and buildings
 - (c) food and drink
 - (d) cosmic rays
- 5 Use measurements of background radiation to determine a corrected count rate

5.2.2 The three types of emission

- 1 Describe the emission of radiation from a nucleus as spontaneous and random in direction
- 2 Describe α -particles as two protons and two neutrons (helium nuclei), β -particles as high-speed electrons from the nucleus and γ -radiation as high-frequency electromagnetic waves
- 3 State, for α -particles, β -particles and γ -radiation:
 - (a) their relative ionising effects
 - (b) their relative penetrating powers
- 4 Describe the deflection of α -particles, β -particles and γ -radiation in electric fields and magnetic fields

5.2 Radioactivity continued

5.2.3 Radioactive decay

- 1 Know that radioactive decay is a change in an unstable nucleus that can result in the emission of α -particles or β -particles and/or γ -radiation and know that these changes are spontaneous and random
- 2 Use decay equations, using nuclide notation, to show the emission of α -particles, β -particles and γ -radiation

5.2.4 Fission and fusion

- 1 Describe the process of fusion as the formation of a larger nucleus by combining two smaller nuclei with the release of energy, and recognise fusion as the energy source for stars
- 2 Describe the process of fission when a nucleus, such as uranium-235 (U-235), absorbs a neutron and produces daughter nuclei and two or more neutrons with the release of energy
- 3 Explain how the neutrons produced in fission create a chain reaction and that this is controlled in a nuclear reactor, including the action of coolant, moderators and control rods

5.2.5 Half-life

- 1 Define the half-life of a particular isotope as the time taken for half the nuclei of that isotope in any sample to decay; recall and use this definition in calculations, which may involve information in tables or decay curves
- 2 Describe the dating of objects by the use of ^{14}C
- 3 Explain how the type of radiation emitted and the half-life of the isotope determine which isotope is used for applications including:
 - (a) household fire (smoke) alarms
 - (b) irradiating food to kill bacteria
 - (c) sterilisation of equipment using gamma rays
 - (d) measuring and controlling thicknesses of materials with the choice of radiations used linked to penetration and absorption
 - (e) diagnosis and treatment of cancer using gamma rays

5.2.6 Safety precautions

- 1 State the effects of ionising nuclear radiations on living things, including cell death, mutations and cancer
- 2 Explain how radioactive materials are moved, used and stored in a safe way, with reference to:
 - (a) reducing exposure time
 - (b) increasing distance between source and living tissue
 - (c) use of shielding to absorb radiation

1. O/N/23/11/Q36

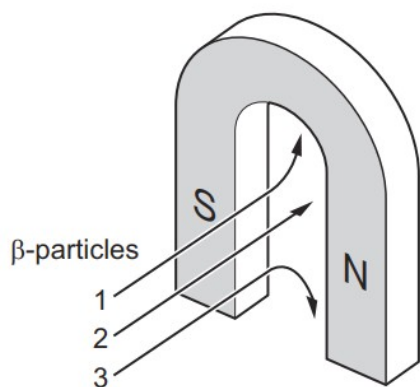
Alpha (α) particles and beta (β) particles are emitted during radioactive decay.

Which particle has the weaker ionising effect and which has the weaker penetrating power?

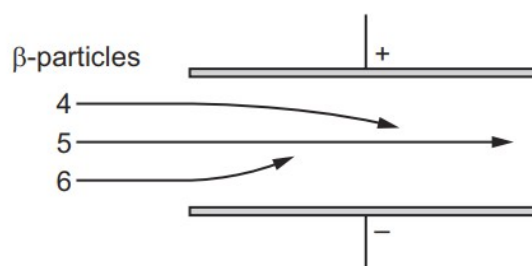
	weaker ionising effect	weaker penetrating power
A	α	α
B	α	β
C	β	α
D	β	β

2. O/N/23/11/Q37

A beam of beta (β) particles passes into a magnetic field and into an electric field.



into magnetic field



into electric field

Which numbers show the paths of the beam through the fields?

- A** 1 and 4 **B** 1 and 6 **C** 2 and 5 **D** 3 and 6

3. O/N/23/11/Q38

The isotope radon-222 has a half-life of 3.8 days.

Which statement about a sample containing 1.0 mg of radon-222 is correct?

- A** After 1.9 days there is 0.50 mg of radon-222 left.
B After 3.8 days there is 0.50 mg of radon-222 left.
C Its half-life is half that of a 2.0 mg sample.
D There is no radon-222 left after 7.6 days.

4. O/N/23/12/Q39

The half-life of a radioactive isotope is 2.5 minutes.

This isotope is the only radioactive isotope in a radioactive sample.

Between time $t = 0$ and $t = 7.5$ minutes, the mass of this isotope in the sample decreases by $70\text{ }\mu\text{g}$.

What is the mass of this isotope at $t = 0$?

- A** $75\text{ }\mu\text{g}$ **B** $80\text{ }\mu\text{g}$ **C** $210\text{ }\mu\text{g}$ **D** $560\text{ }\mu\text{g}$

5. M/J/23/12/Q36

A sample of a radioactive isotope with a very long half-life is placed next to a detector. Sheets of different materials are inserted, one at a time, between the radioactive sample and the detector.

The background count rate is subtracted from the readings on the detector and the corrected count rates are recorded in the table.

type of material	<u>corrected count rate</u> counts/s
no sheet inserted	450
thin aluminium sheet	381
thin lead sheet	285
thin paper sheet	452

Which types of radiation are emitted by the radioactive sample?

- A** alpha-particles and beta-particles only
B alpha-particles only
C beta-particles and gamma radiation only
D gamma radiation only

6. M/J/23/12/Q37

In a fission reactor, which particle causes a uranium-235 nucleus to split?

- A** alpha-particle
B gamma ray
C neutron
D proton

7. M/J/23/12/Q38

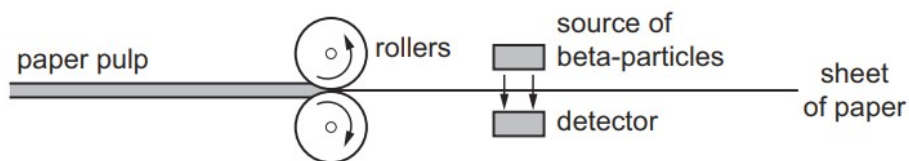
What is the purpose of moderators in nuclear reactors?

- A** to absorb all the neutrons in order to stop the chain reaction
B to produce more neutrons in order to create a chain reaction
C to remove thermal energy in order to control the chain reaction
D to slow down neutrons in order to continue the chain reaction

8. M/J/23/11/Q37

The diagram shows how the thickness of paper is measured during manufacture.

If the sheet is too thick, fewer beta-particles can reach the detector.



Why are beta-particles used for this purpose?

- A** Alpha-particles are all stopped by the paper.
- B** Alpha-particles make the paper radioactive.
- C** Gamma radiation is all stopped by the paper.
- D** Gamma radiation is only produced by sources with a short half-life.

9. M/J/23/11/Q38

A sample of tritium has a mass of 32 mg.

Tritium has a half-life of 12 years.

How much of the sample of tritium decays in 36 years?

- A** 4.0 mg **B** 8.0 mg **C** 24 mg **D** 28 mg

10. O/N/22/11/Q36

Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

What are these three types of radiation?

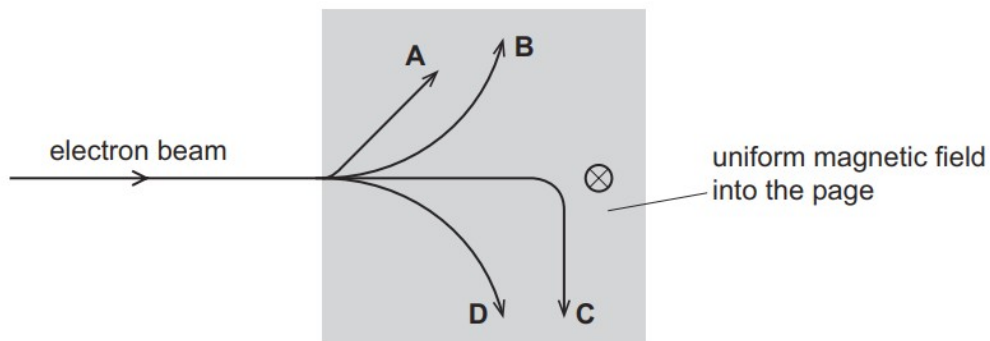
	helium nuclei	electromagnetic waves	electrons
A	alpha	gamma	beta
B	beta	gamma	alpha
C	gamma	alpha	beta
D	gamma	beta	alpha

11. O/N/22/11/Q37

A beam of electrons is fired into a uniform magnetic field as shown.

The direction of the magnetic field is into the page.

Which path do the electrons follow?



12. O/N/22/11/Q38

Which statement about all three types of radioactive emissions is correct?

- A** They are all completely absorbed by a thin aluminium sheet.
- B** They are all deflected by electric fields.
- C** They are all deflected by magnetic fields.
- D** They all cause ionisation of gases.

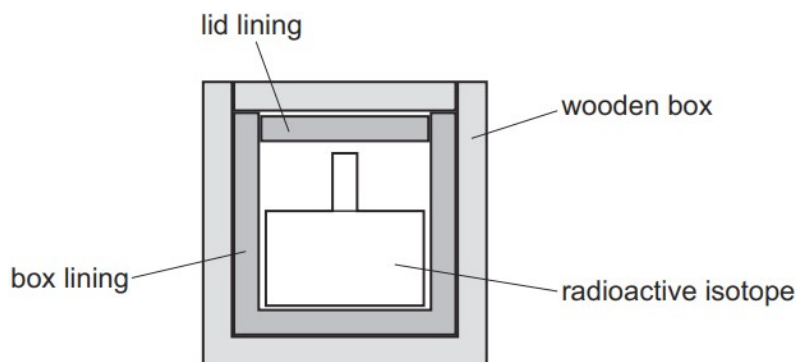
13. O/N/22/11/Q39

What is meant by the half-life of a radioactive substance?

- A** half the time for all the unstable nuclei to decay
- B** half the time it takes an unstable nucleus to emit radiation
- C** the time for half the unstable nuclei to decay
- D** the time for the nucleon number to halve

14. O/N/22/11/Q40

Radioactive isotopes that emit alpha-particles, beta-particles and gamma rays are stored in wooden boxes that are lined with a material that prevents most radiation escaping from the box.



Which material is used for the lining?

- A** copper
- B** glass
- C** lead
- D** plastic

15. O/N/22/12/Q39

Different radioactive emissions have different characteristics.

A student lists the following characteristics for an alpha-particle.

characteristic P It has a positive charge.

characteristic Q It has a high penetrating ability.

characteristic R It is a component of the electromagnetic spectrum.

Which characteristics are correct?

- A** P only
- B** R only
- C** P and Q
- D** Q and R

16. O/N/22/12/Q40

A wooden object is believed to be 6000 years old.

Which isotope, found in the object, is used to determine the age of the object?

- A** ^{12}C
- B** ^{13}C
- C** ^{14}C
- D** ^{15}C

17. O/N/21/11/Q38

Technetium-99m is a radioactive isotope used in medical scanning. It is injected into the body and its emissions are detected outside the body.

Which characteristics of technetium-99m make it suitable for use in medical scanning?

- A** It has a long half-life and emits alpha radiation.
- B** It has a long half-life and emits gamma radiation.
- C** It has a short half-life and emits alpha radiation.
- D** It has a short half-life and emits gamma radiation.

18. O/N/21/11/Q39

Which statement about the production of electricity in a nuclear power station is correct?

- A** In the reactor, the main reaction occurs when protons hit uranium nuclei.
- B** The process taking place in the reactor is called nuclear fusion.
- C** The reactor produces energy to boil water and to produce steam.
- D** Carbon dioxide is the major waste product from the reactor.

19. O/N/21/12/Q38

A radioactive sample contains an isotope that emits alpha particles.

Which quantity stays constant?

- A** the half-life of the isotope
- B** the mass of the sample
- C** the number of neutrons in the sample
- D** the rate of decay of the isotope

20. M/J/21/12/Q39

What does an alpha-particle consist of?

- A** two electrons and four neutrons only
- B** two protons and two neutrons only
- C** two protons and four neutrons only
- D** two protons, two electrons and two neutrons

21. M/J/21/12/Q40

The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A 8 minutes
- B 9 minutes
- C 18 minutes
- D 24 minutes

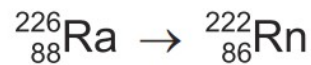
22. M/J/21/11/Q37

Which statement about nuclear fusion is correct?

- A Nuclear fusion occurs at low temperatures.
- B Nuclear fusion occurs only between heavy nuclei.
- C Nuclear fusion occurs in the formation of many stars.
- D Nuclear fusion powers most electricity-generating stations.

23. M/J/21/11/Q38

In one radioactive decay, radium-226 decays to radon-222 as shown.



Which particles are also produced?

- A both an alpha-particle and a beta-particle
- B an alpha-particle only
- C a beta-particle only
- D a neutron

24. O/N/20/11/Q40

An isotope P is radioactive and has a half-life of 7.0 years. A sample initially contains 0.016 kg of P.

After how long will the sample contain 0.0020 kg of P?

- A 7.0 years
- B 14 years
- C 21 years
- D 28 years

25. O/N/20/12/Q39

What is the definition of the half-life of a radioactive element?

- A half of the time it takes for one atom to decay
- B half of the time taken for all the atoms in a sample to decay
- C the time taken for half of the atoms in a sample to decay
- D the time taken for the nucleon number (mass number) to halve

26. M/J/20/11/Q38

Radioactivity is used in several activities that take place in hospitals.

Which activity **never** uses radioactivity?

- A cooking meals
- B diagnosing illnesses
- C sterilising equipment
- D treating illnesses

27. M/J/20/11/Q39

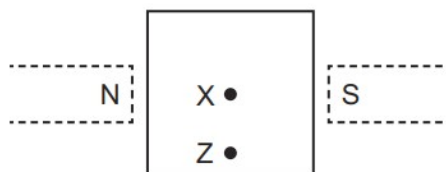
$^{14}_6\text{C}$ is a natural radioactive isotope of carbon and is present in all living things. It has a half-life of 5700 years.

How old is a bone that is found to have only $\frac{1}{4}$ of the natural proportion of $^{14}_6\text{C}$?

- A 5700 years
- B 11 400 years
- C 17 100 years
- D 22 800 years

28. O/N/19/11/Q40

A narrow beam of radiation from a radioactive source travels out of the page through the point X.



The poles of a magnet are placed on either side of the beam. Radiation is now detected both at X and at Z.

Which types of radiation are present in the beam?

- A alpha and beta radiation
- B alpha and gamma radiation
- C alpha, beta and gamma radiation
- D beta and gamma radiation

29. O/N/19/12/Q39

Radioactive substances are used as tracers in medical examinations.

A liquid tracer is injected into a patient. A detector 20 cm above the patient shows where the radioactive substance is in the body.

What is the best radioactive substance to use?

- A** an alpha-particle emitter with a half-life of a few hours
- B** an alpha-particle emitter with a half-life of a few years
- C** a gamma ray emitter with a half-life of a few hours
- D** a gamma ray emitter with a half-life of a few years

30. O/N/19/12/Q40

An isotope of lead, $^{212}_{82}\text{Pb}$, is radioactive. After three decays, two beta-particles and an alpha-particle have been emitted. Another radioactive isotope of lead is formed.

What is the mass number (nucleon number) of the new isotope of lead?

- A** 200 **B** 208 **C** 212 **D** 216

31. M/J/19/12/Q40

A proton has charge q and mass m .

What is the charge and what is the mass of an alpha-particle?

	charge	mass
A	q	$2m$
B	$2q$	$2m$
C	$2q$	$4m$
D	$4q$	$2m$

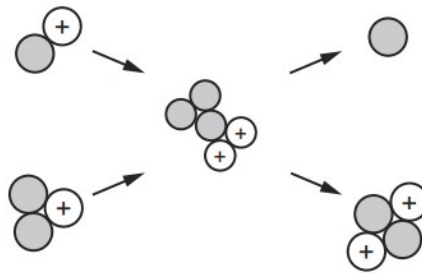
32. O/N/18/11/Q39

What does the alpha-radiation given off by radioactive nuclei consist of?

- A** fast-moving protons
- B** helium nuclei
- C** microwaves
- D** radio waves

33. O/N/18/11/Q37

The diagram shows a nuclear reaction.



The shaded particles are uncharged.

What are the products shown in this reaction?

- A** a helium atom and a neutron
- B** a helium atom and a proton
- C** a helium nucleus and a neutron
- D** a helium nucleus and a proton

34. O/N/18/12/Q38

Which statement about the reactor in a nuclear power station is correct?

- A** In the reactor, the main reaction occurs when protons hit uranium nuclei.
- B** The process taking place in the reactor is called nuclear fusion.
- C** The reactor produces energy to boil water and to produce steam.
- D** Carbon dioxide is the major waste product from the reactor.

35. M/J/18/11/Q38

A radioactive isotope of hydrogen has a half-life of 12 years.

A sample contains 40 million atoms of this isotope.

How many atoms of this isotope are left after 24 years?

- A** 20 million
- B** 10 million
- C** 5 million
- D** zero

36. M/J/18/11/Q37

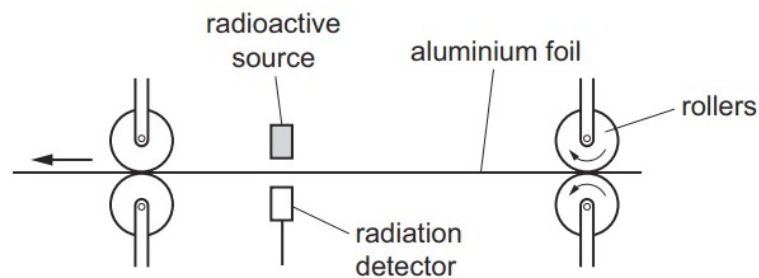
Which of alpha, beta and gamma radiations are waves?

- A** alpha only
- B** beta only
- C** gamma only
- D** alpha and beta

37. M/J/18/11/Q36

Thin aluminium foil is used in cooking.

The diagram shows a system used to control the thickness of aluminium foil being made in a factory.



What is the most suitable radioactive source for the system?

	source	half-life of source
A	beta	1 hour
B	beta	6 years
C	gamma	1 hour
D	gamma	6 years

38. O/N/17/11/Q40

Which types of radiation may be emitted by radioactive decay?

- A** alpha-particles and gamma rays
- B** microwaves and infra-red
- C** radio waves and microwaves
- D** ultra-violet and X-rays

39. O/N/17/12/Q37

A sample contains atoms of an isotope that has a half-life of 7.2 years.

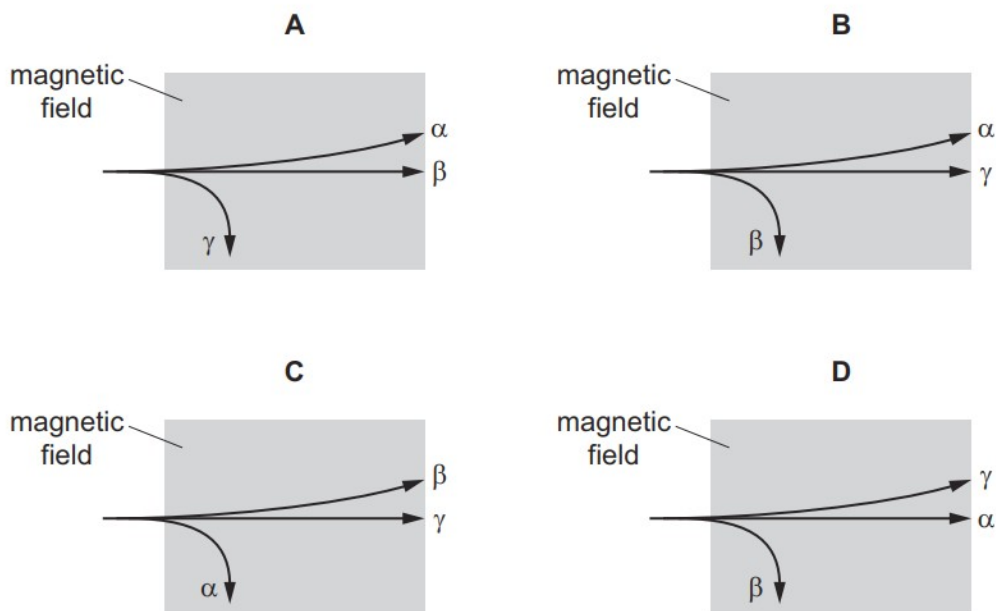
Which quantity halves every 7.2 years?

- A** the mass of the sample
- B** the nucleon (mass) number
- C** the number of atoms of the isotope
- D** the proton (atomic) number

40. O/N/17/12/Q38

A beam, consisting of alpha-particles (α), beta-particles (β), and gamma rays (γ), passes into a magnetic field.

Which diagram shows their paths in the magnetic field?



41. O/N/17/12/Q40

A nucleus of phosphorus $^{32}_{15}\text{P}$ emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

	nucleon number (mass number)	proton number (atomic number)
A	28	13
B	31	14
C	31	15
D	32	16

42. M/J/17/12/Q39

Which row is correct for nuclear fission and for nuclear fusion?

	fission	fusion
A	produces larger nuclei	is the energy source of a star
B	produces larger nuclei	releases energy in a power station
C	produces smaller nuclei	is the energy source of a star
D	produces smaller nuclei	releases energy in a power station

43. M/J/17/12/Q40

What happens to the proton number (atomic number) of a nucleus as it emits an alpha-particle?

- A** It decreases by one.
- B** It decreases by two.
- C** It decreases by four.
- D** It does not change.

44. O/N/16/11/Q38

Which types of radiation may be emitted by radioactive nuclei?

- A** beta and gamma
- B** microwaves and infra-red
- C** radio waves and microwaves
- D** ultra-violet and X-rays

45. O/N/16/11/Q39

A freshly made sample of radioactive material gives a count rate of 8000 counts per minute. After 20 days, it gives a count rate of 500 counts per minute.

What is the half-life of the material?

- A** 4.0 days
- B** 5.0 days
- C** 20 days
- D** 80 days

46. O/N/16/11/Q40

Which number is always equal to the proton number of a neutral atom of a radioactive isotope?

- A** the number of electrons in the atom
- B** the number of neutrons in the atom
- C** the number of nucleons in the atom
- D** the number of particles in the atom

47. O/N/16/12/Q40

A nucleus of uranium $^{238}_{92}\text{U}$ decays to thorium by emitting an alpha-particle.

What is the resulting thorium nucleus?

- A** $^{234}_{90}\text{Th}$ **B** $^{236}_{90}\text{Th}$ **C** $^{238}_{93}\text{Th}$ **D** $^{239}_{92}\text{Th}$

48. M/J/16/12/Q37

Which nucleus is produced when americium-241 ($^{241}_{95}\text{Am}$) emits an alpha-particle?

- A** $^{237}_{93}\text{Np}$ **B** $^{237}_{97}\text{Bk}$ **C** $^{245}_{93}\text{Np}$ **D** $^{245}_{97}\text{Bk}$

49. M/J/16/12/Q38

Which states the three types of radiation emitted by radioactive isotopes in order of their ionising effect from highest to lowest?

- A** alpha-particles, beta-particles, gamma-rays
B alpha-particles, gamma-rays, beta-particles
C beta-particles, gamma-rays, alpha-particles
D gamma-rays, beta-particles, alpha-particles

50. M/J/16/12/Q39

Which statement about the half-life of a radioactive isotope is correct?

- A** Half-life changes as the isotope decays.
B Half-life is the time it takes for the nucleon number of the isotope to halve.
C Half-life is half the time it takes for the number of nuclei of the isotope to decrease to zero.
D Half-life is the time it takes for the number of nuclei of the isotope to decrease by half.

51. M/J/16/11/Q38

Which type of radiation consists of electrons?

- A** alpha-particles
B beta-particles
C gamma-rays
D X-rays

52. M/J/16/11/Q39

A sample of a radioactive isotope produces a count rate of 10 000 counts per minute. The half-life of the isotope is one day.

What was the count rate of the sample two days ago?

- A 2500 counts per minute
- B 5000 counts per minute
- C 20 000 counts per minute
- D 40 000 counts per minute

53. O/N/15/11/Q39

Which type of electromagnetic radiation is produced during radioactive decay?

- A alpha-particles
- B beta-particles
- C gamma-rays
- D X-rays

54. O/N/15/11/Q40

Which row states where nuclear fusion occurs and what nuclear fusion is?

	nuclear fusion occurs in	nuclear fusion is
A	a power station	the joining of small nuclei
B	a power station	the splitting of large nuclei
C	a star	the joining of small nuclei
D	a star	the splitting of large nuclei

55. O/N/15/12/Q39

A radioactive isotope ^{14}C emits beta-particles.

A sample of this isotope is left for a period of time that is equal to its half-life.

Which two quantities decrease to **half** of their initial value during this time?

- A the decay rate and the number of protons
- B the mass of the sample and the number of ^{14}C nuclei
- C the number of ^{14}C nuclei and the decay rate
- D the number of protons and the mass of the sample

56. M/J/15/12/Q38

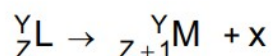
A nucleus of $^{215}_{84}\text{Po}$ decays by emitting an alpha-particle and the resulting nucleus then decays by emitting a beta-particle.

What is the nucleon number and proton number of the final nucleus?

	nucleon number	proton number
A	211	81
B	211	83
C	212	81
D	212	83

57. M/J/15/12/Q39

A radioactive material decays by this process:



What is particle x?

- A** an electron
- B** a helium nucleus
- C** a neutron
- D** a proton

58. M/J/15/12/Q40

The half-life of a radioactive isotope is 24 hours. A sample of this isotope produces an initial count rate of 720 counts per second.

How long does it take for the count rate to fall to 90 counts per second?

- A** 3 hours
- B** 72 hours
- C** 96 hours
- D** 192 hours

59. M/J/15/11/Q38

A factory continuously produces plastic sheets. A radioactive isotope and a detector are used to check the thickness of the sheets.

What is the most suitable source to use?

- A** an alpha source with a half-life of a few minutes
- B** an alpha source with a half-life of several years
- C** a beta source with a half-life of a few minutes
- D** a beta source with a half-life of several years

60. M/J/15/11/Q39

A source contains a radioactive material.

Without the radioactive source present, a detector records a background count rate of 20 counts per minute.

This source is placed in a fixed position near the detector. Initially a count rate of 520 per minute is recorded.

What count rate is recorded after a time of **two** half-lives of the radioactive source?

- A** 125 counts per minute
- B** 130 counts per minute
- C** 135 counts per minute
- D** 145 counts per minute

61. M/J/15/11/Q40

A nucleus contains 94 protons and 240 nucleons. It emits an alpha-particle.

How many protons and how many neutrons are in the nucleus produced?

	number of protons	number of neutrons
A	90	144
B	90	236
C	92	144
D	92	236

62. O/N/14/11/Q37

Three types of radiation emitted by unstable nuclei are electrons, helium nuclei and electromagnetic radiation.

What are these types of radiation?

	electrons	helium nuclei	electromagnetic radiation
A	alpha	beta	gamma
B	alpha	gamma	beta
C	beta	alpha	gamma
D	beta	gamma	alpha

63. O/N/14/11/Q38

The energy emitted by the Sun is released when the nuclei of an element fuse together.

Which nuclei, when fusing together, release most of the energy in the Sun?

- A carbon
- B helium
- C hydrogen
- D uranium

64. O/N/14/11/Q39

A sample of wood contains 9.0×10^{16} nuclei of carbon-14. The nuclei undergo radioactive decay with a half-life of 5600 years.

How many carbon-14 nuclei remain in this sample after 16 800 years?

- A 1.1×10^2 B 1.1×10^{16} C 3.0×10^{16} D 4.5×10^{16}

65. O/N/15/12/Q38

People working with radioactive materials use a photographic film badge covered in paper. The badge is used to monitor the level of their exposure to radiation.

Which radiation is detected?

- A alpha-particles only
- B beta-particles only
- C gamma-rays and beta-particles
- D gamma-rays only

66. O/N/14/12/Q39

When a radioactive atom decays by alpha-particle emission, its nucleus loses

- A 1 proton only.
- B 1 proton and 1 electron.
- C 2 protons and 2 electrons.
- D 2 protons and 2 neutrons.

67. M/J/14/12/Q39

Which row states the nature and range of beta-particles in air?

	nature	range in air
A	electromagnetic radiation	1-10 cm
B	electromagnetic radiation	10-100 cm
C	electron	1-10 cm
D	electron	10-100 cm

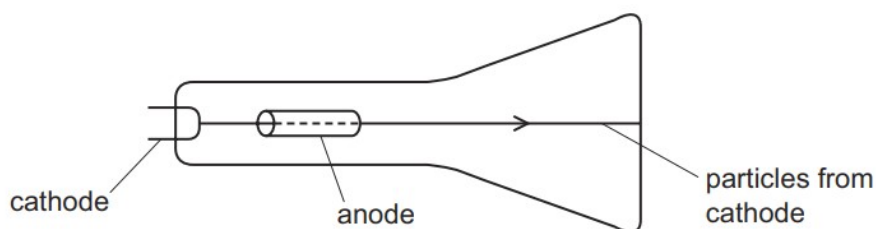
68. M/J/14/12/Q40

Which particle has the smallest mass?

- A alpha-particle
- B electron
- C neutron
- D proton

69. M/J/14/11/Q38

The diagram shows part of a cathode-ray oscilloscope.



Which particles are emitted by the cathode?

- A electrons
- B helium nuclei
- C neutrons
- D protons

70. M/J/14/11/Q39

Which row gives the range and electrical charge of an alpha-particle?

	range in air	electrical charge
A	a few centimetres	negative
B	a few centimetres	positive
C	a few metres	negative
D	a few metres	positive

71. M/J/14/11/Q40

$^{14}_6\text{C}$ represents a nuclide of the element carbon and $^{14}_7\text{N}$ a nuclide of nitrogen.

How does a neutral atom of $^{14}_7\text{N}$ differ from a neutral atom of $^{14}_6\text{C}$?

- A The nitrogen atom has one electron less than the carbon atom.
- B The nitrogen atom has one neutron more than the carbon atom.
- C The nitrogen atom has one proton less than the carbon atom.
- D The nitrogen atom has one proton more than the carbon atom.

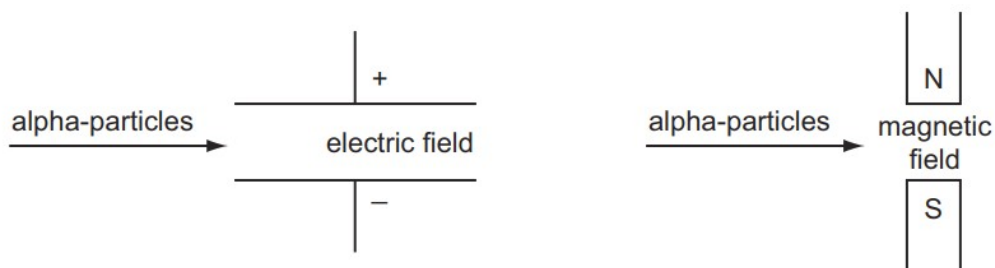
72. O/N/13/11/Q37

What are the characteristics of an alpha-particle?

	charge	ionising effect
A	negative	strong
B	negative	weak
C	positive	strong
D	positive	weak

73. O/N/13/11/Q38

Alpha-particles pass through an electric field or a magnetic field.



How is the path of the particles affected by these fields?

	electric field	magnetic field
A	deflected	deflected
B	deflected	undeflected
C	undeflected	deflected
D	undeflected	undeflected

74. O/N/13/12/Q36

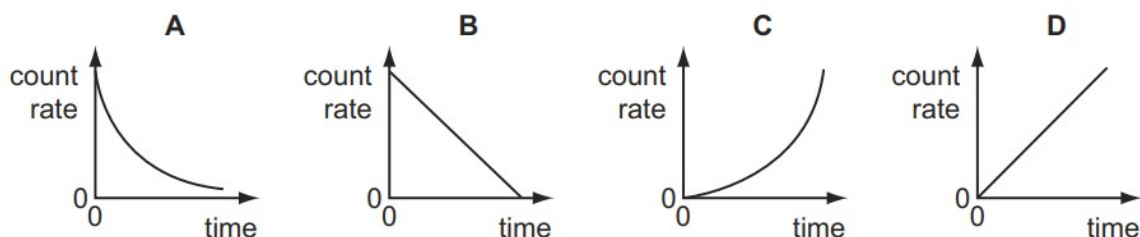
Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

What are these three types of radiation?

	helium nuclei	electromagnetic waves	electrons
A	alpha	gamma	beta
B	beta	gamma	alpha
C	gamma	alpha	beta
D	gamma	beta	alpha

75. O/N/13/12/Q38

Which graph shows how the count rate of a radioactive isotope varies with time?



76. O/N/13/12/Q39

For which purpose is the radioactive isotope carbon-14 used?

- A dating of ancient bones
- B fuel for a nuclear power station
- C killing cancerous cells
- D operating mobile telephones

77. M/J/13/12/Q39

The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A 8 minutes
- B 9 minutes
- C 18 minutes
- D 24 minutes

78. M/J/13/12/Q40

In the Geiger-Marsden experiment, a beam of alpha-particles is fired at a thin sheet of gold in a vacuum. The majority of the alpha-particles pass straight through the sheet without being deflected.

What does this show?

- A The alpha-particle is uncharged.
- B The alpha-particle is very large.
- C The nucleus is positively charged.
- D The nucleus is very small.

79. M/J/13/11/Q39

A school keeps radioactive sources for use in radioactivity experiments in a laboratory.

The background radiation is measured at the start of an experiment.

Which statement is correct?

- A** The background radiation is caused by the school's radioactive sources in the laboratory.
- B** The background radiation is present when there are no radioactive sources in the laboratory.
- C** The background radiation is radiation that is not detected in radioactivity experiments.
- D** The background radiation is the same in laboratories in different countries.

80. O/N/12/11/Q39

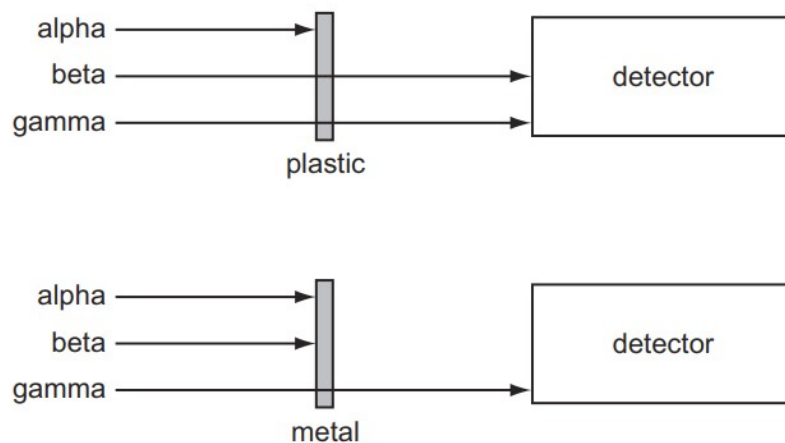
In nuclear1....., hydrogen nuclei2..... to form helium nuclei, releasing energy.

Which words correctly complete gaps 1 and 2?

	1	2
A	fission	join together
B	fission	split apart
C	fusion	join together
D	fusion	split apart

81. O/N/12/12/Q39

The diagram shows the radioactive emissions that pass through a piece of plastic and a piece of metal of the same thickness.



Which types of radioactive emission can distinguish between the plastic and the metal?

- A** alpha, beta and gamma
- B** alpha only
- C** beta only
- D** gamma only

82. O/N/12/11/Q40

Which nucleus is produced when thorium-223 ($^{223}_{90}\text{Th}$) emits an alpha-particle?

- A $^{219}_{88}\text{Ra}$ B $^{219}_{92}\text{U}$ C $^{227}_{88}\text{Ra}$ D $^{227}_{92}\text{U}$

83. O/N/12/12/Q40

The background count rate in a laboratory is 10 counts/min.

The measured count rate from a radioactive source is 410 counts/min, which includes the background.

The half-life of the source is 5 days.

What is the measured count rate after 15 days?

- A 10 counts/min
B 50 counts/min
C 60 counts/min
D 110 counts/min

84. M/J/12/12/Q39

What is the safest way to dispose of a large quantity of highly radioactive waste?

- A burning it on a fire
B burying it in dry rock deep underground
C pouring it down the drain
D pumping it into a river

85. M/J/12/12/Q40

The count rate from a radioactive material falls from 400 counts per second to 50 counts per second in 12 minutes.

What is its half-life?

- A 8 minutes
B 6 minutes
C 4 minutes
D 3 minutes

86. M/J/12/11/Q40

An isotope X is radioactive and has a half-life of 4 years. A sample initially contains 8000 atoms of X.

After how many years will the sample contain 1000 atoms of X?

- A 4 B 8 C 12 D 16

87. M/J/12/11/Q39

When dealing with radioactive materials there are possible dangers.

Which statement is correct?

- A** Beta-particles can pass through skin and damage body cells.
- B** Materials that emit only alpha-particles must be kept in thick lead containers.
- C** Radioactive materials are safe to handle after two half-lives.
- D** Sources of gamma radiation are dangerous because they have long half-lives.

88. O/N/11/11/Q39

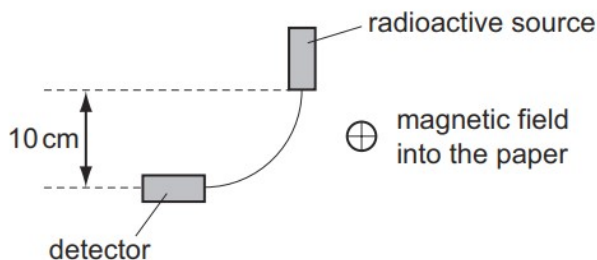
The isotope barium-140 has a half-life of 13 days. A sample of this isotope is kept for 13 days.

Which quantity halves during this time?

- A** the number of atoms of barium-140 in the sample
- B** the number of electrons in an atom of barium-140
- C** the number of nucleons in an atom of barium-140
- D** the number of protons in a nucleus of barium-140

89. O/N/11/11/Q40

In a laboratory experiment, particles from a radioactive source are deviated by a magnetic field and reach a detector.



Which particles are deviated and reach the detector?

- A** alpha-particles and gamma-rays
- B** alpha-particles only
- C** beta-particles and gamma-rays
- D** beta-particles only

90. M/J/11/12/Q38

A radioactive nuclide ${}^{238}_{92}\text{U}$ decays into thorium by emitting an alpha-particle.

The thorium then decays into protactinium by emitting a beta-particle.

What is the symbol for protactinium?

- A** ${}^{230}_{90}\text{Pa}$ **B** ${}^{234}_{89}\text{Pa}$ **C** ${}^{234}_{90}\text{Pa}$ **D** ${}^{234}_{91}\text{Pa}$

91. M/J/11/12/Q40

Tritium is a radioactive isotope of hydrogen with a half-life of 12 years.

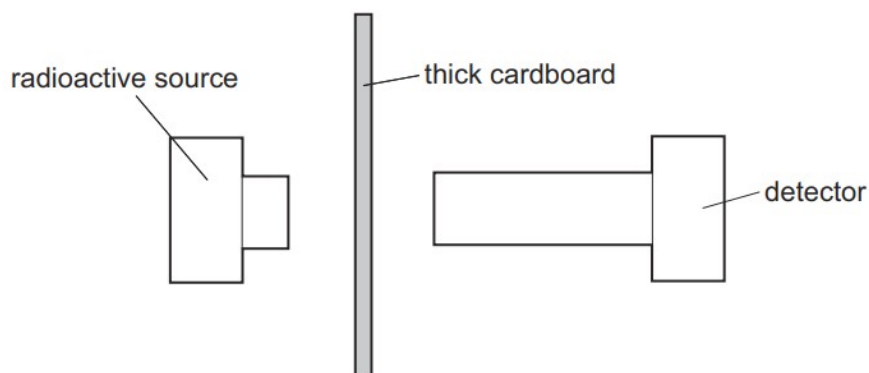
If a sample starts with 40 million atoms of tritium, how many atoms of tritium will be left after 12 years?

- A** 40 million
B 20 million
C 10 million
D 5 million

92. O/N/10/11/Q38

A student investigates a radioactive source that emits only alpha-particles. Without any source nearby, the detector shows a low reading.

The source and thick cardboard are placed near the detector, as shown.



What is the reading on the detector now, and why?

	detector reading	reason
A	low	background radiation is detected
B	low	some alpha-particles pass through cardboard
C	zero	alpha-particles are all absorbed by the cardboard
D	zero	background radiation is all absorbed by the cardboard

93. O/N/10/11/Q40

The table shows details of two samples of radioactive nuclides X and Y.

nuclide	number of radioactive atoms at time = 0	half-life
X	16 000	1 day
Y	2 000	2 days

After how many days will the number of atoms of nuclide X be equal to the number of atoms of nuclide Y?

- A** 2 days **B** 4 days **C** 6 days **D** 8 days

94. M/J/10/12/Q40

When a sample of a radioactive nuclide decays, the count rate falls from 1200 to 150 in three minutes.

What is the half-life of the radioactive nuclide?

- A** 0.75 minutes
B 1.0 minutes
C 3.0 minutes
D 9.0 minutes

95. O/N/09/Q39

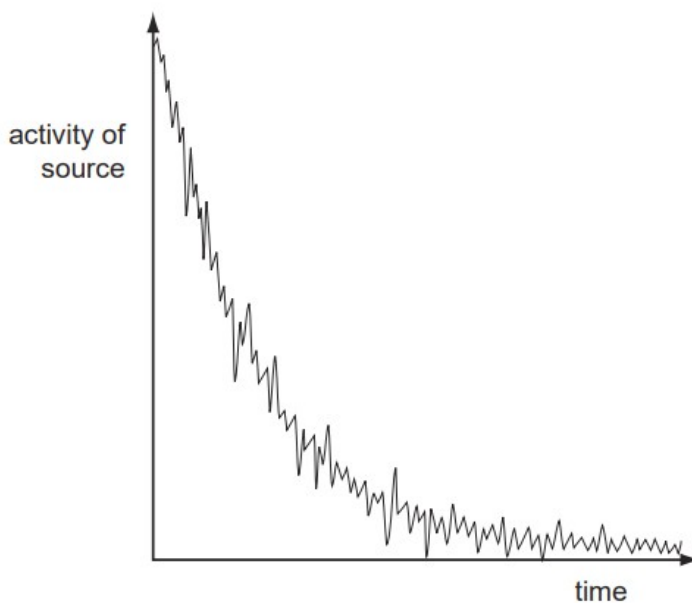
A radioactive isotope has a half-life of 6000 years.

How much time passes before the rate of emission from a sample of this isotope falls to $\frac{1}{16}$ of its original value?

- A** 6000 years
B 18 000 years
C 24 000 years
D 96 000 years

96. O/N/09/Q38

The activity of a radioactive source is measured over a period of time. The graph shows the decay curve.



Why is the curve **not** smooth?

- A** Background radiation has not been subtracted.
- B** Radioactive decay is a random process.
- C** The half-life is not constant.
- D** The temperature is changing.

97. M/J/09/Q39

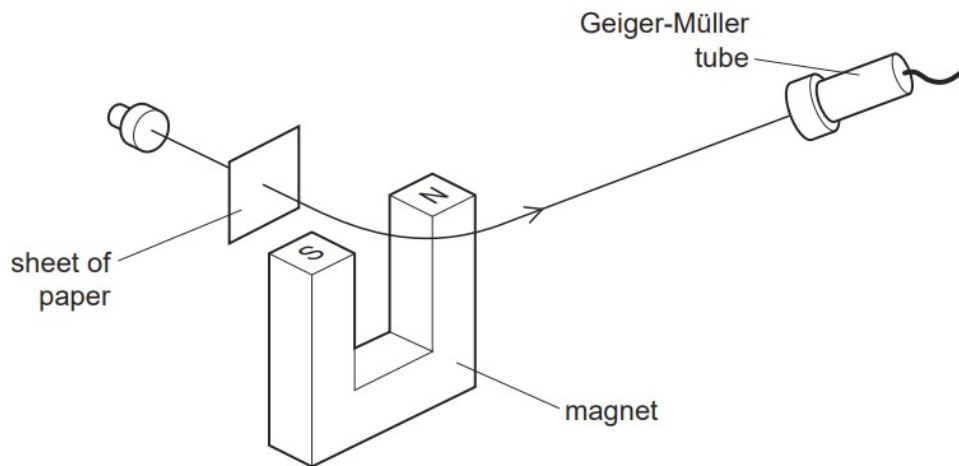
The count-rate from a radioactive source falls from 400 to 50 in 3.0 minutes.

What is the half-life?

- A** 0.75 minutes
- B** 1.0 minutes
- C** 2.7 minutes
- D** 8.0 minutes

98. M/J/09/Q38

A radioactive source emits alpha-particles, beta-particles and gamma-rays. A Geiger-Müller tube and counter detect the emissions, which pass through a thin sheet of paper and a strong magnetic field.



What is detected by the Geiger-Müller tube?

- A** alpha-particles and beta-particles
- B** alpha-particles only
- C** beta-particles and gamma-rays
- D** beta-particles only

99. O/N/08/Q38

What occurs in the decay of a radioactive nucleus?

- A** The nucleus absorbs another nucleus.
- B** The nucleus absorbs at least one form of radiation.
- C** The nucleus always splits into two equal fragments.
- D** The nucleus emits at least one form of radiation.

100. M/J/08/Q38

Which travels in a straight line across a magnetic field?

- A** alpha-particle
- B** electron
- C** gamma-ray
- D** proton

101. M/J/08/Q39

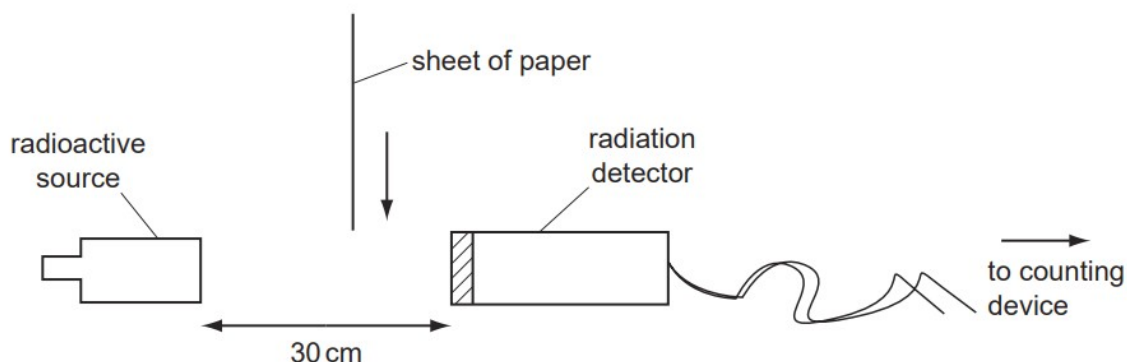
In the treatment of brain cancer, a patient's head is enclosed in a helmet containing a number of radioactive sources. The radiation from each source is directed towards the cancer.

Which nuclide is the most suitable for these sources?

	nuclide	radiation	half-life
A	caesium-137	gamma	30 years
B	sodium-24	beta	15 hours
C	strontium-90	beta	29 years
D	californium-246	alpha	36 hours

102. O/N/07/Q38

An experimenter sets up the following apparatus, in air, to discover whether a radioactive source is emitting alpha-particles.



The experimenter moves the paper between the source and the detector. There is no difference in the count-rate.

Why is this?

- A** Paper does not absorb alpha-particles.
- B** The count-rate due to the background radiation is too high.
- C** The radioactive source is too far from the detector.
- D** The source emits beta-particles and alpha-particles.

103. O/N/07/Q39

The half-life of a radioisotope is 2400 years. The activity of a sample is 720 counts/s.

How long will it take for the activity to fall to 90 counts/s?

- A** 300 years **B** 2400 years **C** 7200 years **D** 19 200 years

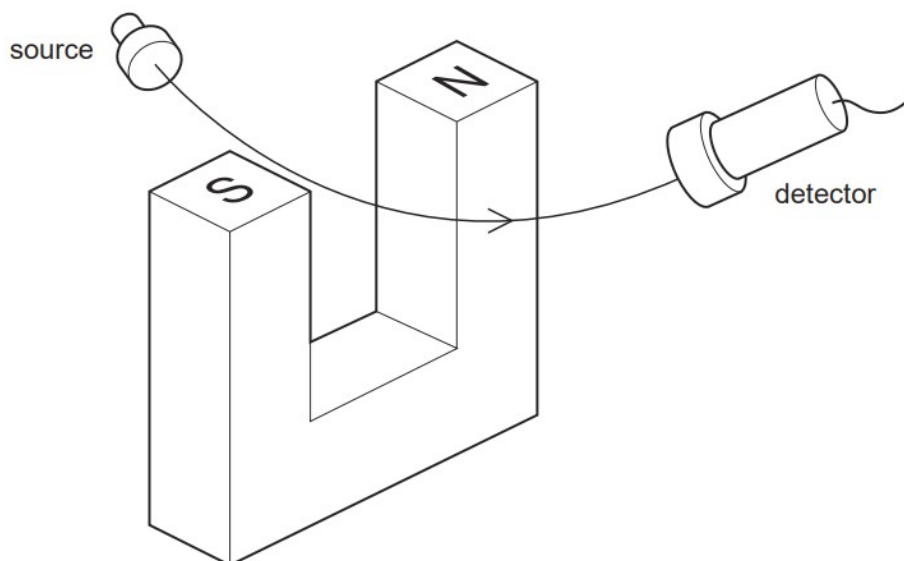
104. M/J/07/Q38

Which statement is true for all three types of radioactive emission (alpha-particles, beta-particles and gamma-rays)?

- A They are completely absorbed by a thin aluminium sheet.
- B They are deflected by electric fields.
- C They emit light.
- D They ionise gases.

105. M/J/07/Q39

The diagram shows one type of radiation passing between the poles of a strong magnet and being detected.



Which type of radiation is being detected?

- A alpha-particles only
- B beta-particles only
- C gamma-rays only
- D alpha-particles and beta-particles

106. O/N/06/Q38

The half-life of a radioactive material is 24 years.

The activity of a sample falls to a fraction of its initial value after 72 years.

What is the fraction?

- A $\frac{1}{3}$
- B $\frac{1}{4}$
- C $\frac{1}{6}$
- D $\frac{1}{8}$

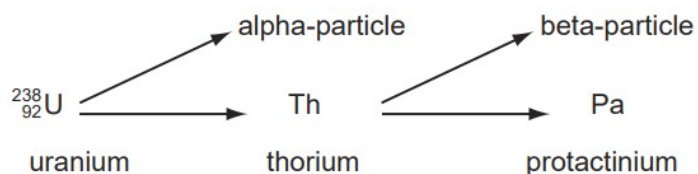
107. O/N/06/Q39

Which is the correct comparison of the penetrating power and ionising power of alpha-particles and gamma radiation?

	greater penetrating power	greater ionising power
A	alpha	alpha
B	alpha	gamma
C	gamma	alpha
D	gamma	gamma

108. O/N/06/Q40

The uranium nucleus ${}_{92}^{238}\text{U}$ emits an alpha-particle to become thorium, which then emits a beta-particle to become protactinium.



What is the proton number (atomic number) of protactinium?

- A** 95 **B** 91 **C** 90 **D** 89

109. M/J/06/Q39

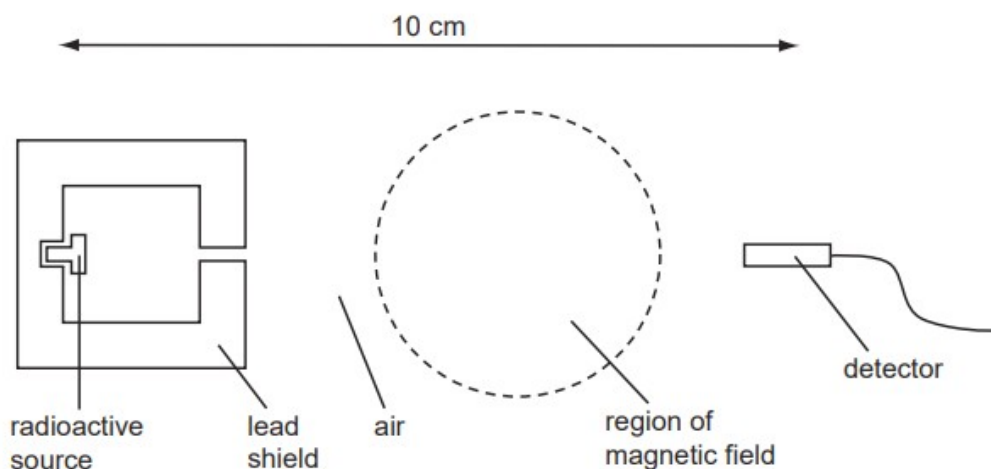
A detector is used to measure the count-rate near a radioactive source. The reading is 4000 counts per minute. After 30 minutes the count-rate has fallen to 500 counts per minute.

What is the half-life of the radioactive source? You may ignore the effects of background radiation.

- A** 3 minutes
B 5 minutes
C 6 minutes
D 10 minutes

110. M/J/06/Q38

A student investigates the emission from an unknown radioactive source. The source is 10 cm in front of a detector. A strong magnetic field between the source and the detector is then switched on.



The results are shown.

	average count per minute
without magnetic field	4500
with magnetic field	2000
background radiation	50

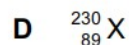
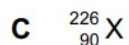
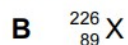
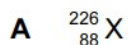
Which radioactive source produced these results?

source	emissions from source
A	alpha-particles and gamma-rays only
B	beta-particles only
C	beta-particles and gamma-rays only
D	gamma-rays only

111. O/N/05/Q38

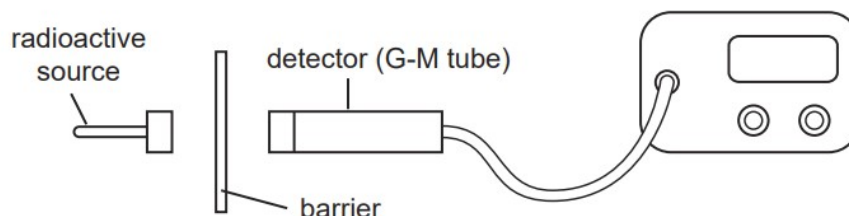
A nucleus is represented by ${}_{91}^{230}\text{Z}$. It emits one alpha-particle and then one beta-particle.

What is the resulting nucleus X?



112. O/N/05/Q37

The diagram shows the apparatus used in an experiment in which barriers of various materials are placed in turn between different radioactive sources and a detector.



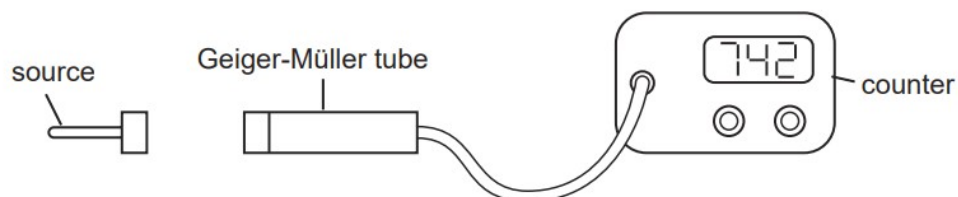
The table shows the count rates recorded by the detector for four sources.

Which source emits alpha-particles only?

source	count rate / counts per minute			
	no barrier	paper	thin aluminium	thick lead
A	200	200	200	30
B	200	30	30	30
C	1200	600	200	30
D	1200	1200	30	30

113. M/J/05/Q38

A radioactive source is placed 3 cm from a Geiger-Müller tube in air. The average count rate is 742 counts / minute.



Then, in three experiments, measurements are taken with sheets of different materials placed between the source and the tube. The results are recorded in the table.

material of sheet between source and tube	thin card	aluminium foil	thick lead
average count rate / counts per minute	273	275	68

Which types of radiation does the source emit?

- A** alpha and beta only
- B** alpha and gamma only
- C** beta and gamma only
- D** alpha, beta and gamma

114. M/J/05/Q39

Which equation shows a nuclear fission reaction?

- A** ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He}$
- B** ${}^{14}_7\text{N} + \text{neutron} \rightarrow {}^{15}_7\text{N}$
- C** ${}^{46}_{21}\text{Sc} \rightarrow {}^{46}_{21}\text{Sc} + \text{gamma ray}$
- D** ${}^{239}_{92}\text{U} \rightarrow {}^{95}_{38}\text{Sr} + {}^{141}_{54}\text{Xe} + 3 \text{ neutrons}$

115. O/N/04/Q39

A nucleus consists of 90 protons and 144 neutrons.

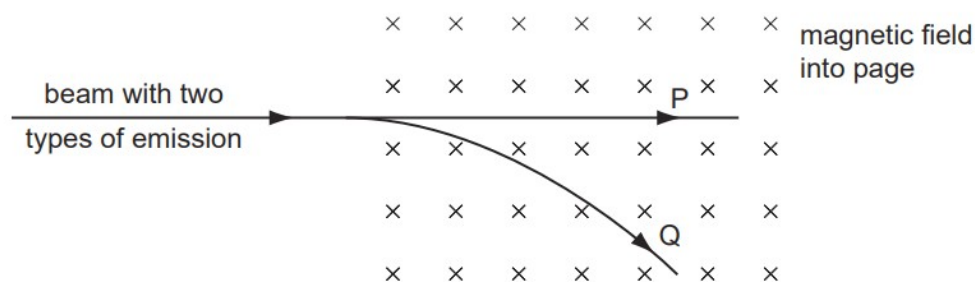
After emitting two beta-particles followed by an alpha-particle, this nucleus has

- A** 86 protons and 140 neutrons.
- B** 86 protons and 142 neutrons.
- C** 90 protons and 140 neutrons.
- D** 90 protons and 142 neutrons.

116. M/J/04/Q38

Two types of emission from a radioactive source are separated by passing them through a magnetic field.

The deflections are shown in the diagram.



What are the emissions P and Q?

	emission P	emission Q
A	alpha-particles	gamma-rays
B	beta-particles	gamma-rays
C	gamma-rays	alpha-particles
D	gamma-rays	beta-particles

117. M/J/04/Q40

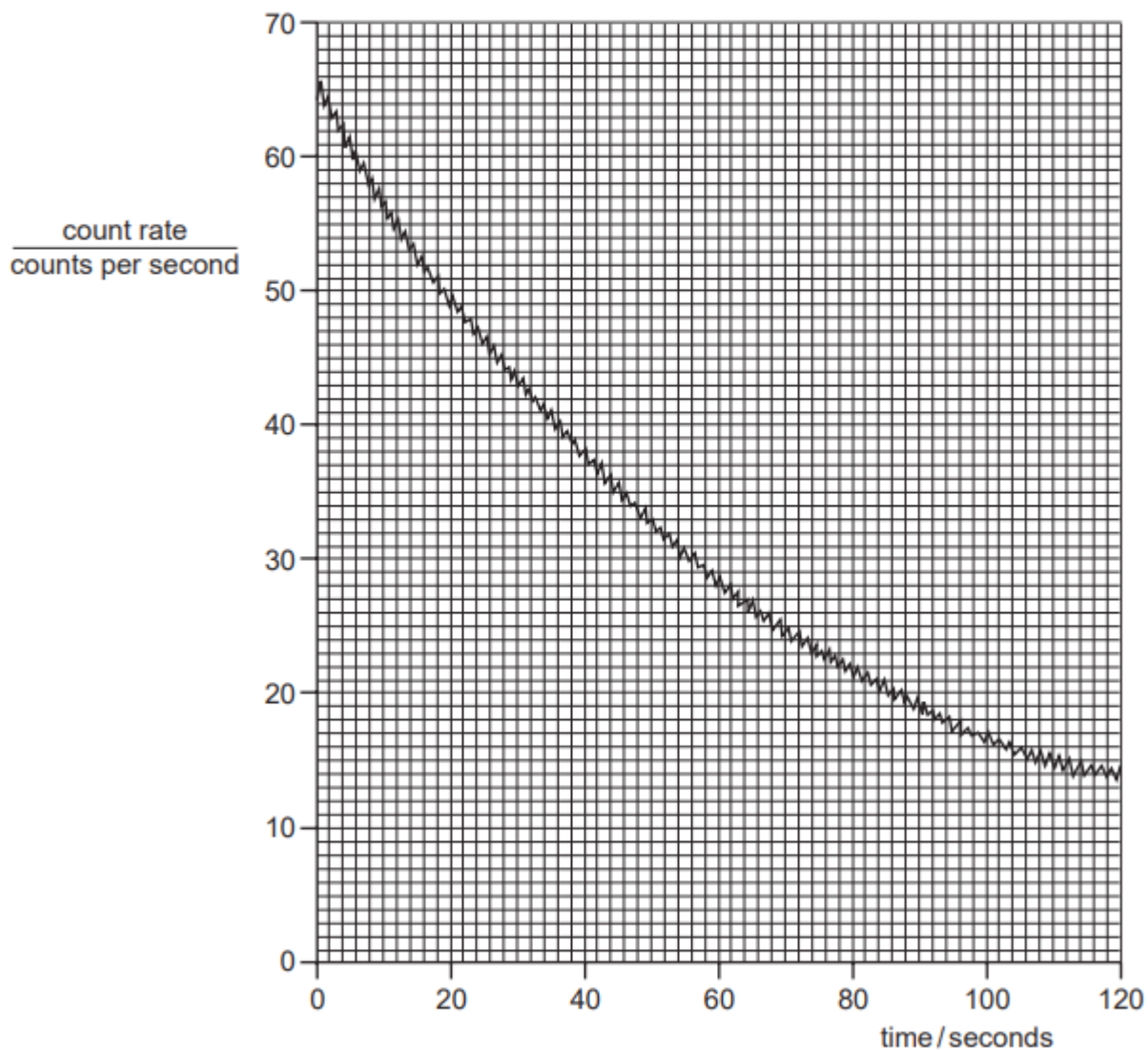
Ra decays with a half-life of 1600 s.

Rn decays with a half-life of 52 s.

Po decays with a half-life of 9.1 s.

Pb decays with a half-life of 10.6 h.

The changing count rate for one of these radioactive nuclides is shown in the graph.



From the half-life shown by the graph, which was the decaying radioactive nuclide?

A Ra

B Rn

C Po

D Pb

ANSWERS

1. C	48. A	97. B
2. D	49. A	98. D
3. B	50. D	99. D
4. B	51. B	100. A
5. C	52. D	101. A
6. C	53. C	102. C
7. D	54. C	103. C
8. A	55. C	104. D
9. D	56. B	105. B
10. A	57. A	106. D
11. D	58. B	107. C
12. D	59. D	108. B
13. C	60. D	109. D
14. C	61. C	110. C
15. A	62. C	111. C
16. C	63. C	112. B
17. D	64. B	113. B
18. C	65. C	114. D
19. A	66. D	115. C
20. B	67. D	116. D
21. D	68. B	117. B
22. C	69. A	
23. B	70. B	
24. C	71. D	
25. C	72. C	
26. A	73. A	
27. B	74. A	
28. D	75. A	
29. C	76. A	
30. B	77. D	
31. C	78. D	
32. B	79. B	
33. C	80. C	
34. C	81. C	
35. B	82. A	
36. C	83. C	
37. B	84. B	
38. A	85. C	
39. C	86. C	
40. B	87. A	
41. D	88. A	
42. C	89. D	
43. B	90. D	
44. A	91. B	
45. B	92. A	
46. A	93. C	
47. A	94. B	
	95. C	
	96. B	

1. O/N/23/22/Q9

In space, an interstellar cloud of dust and gas collapses to form a protostar. The cloud contains hydrogen.

- (a) Describe the energy transfers that take place as the cloud collapses and forms the protostar.

.....
.....
..... [2]

- (b) The collapse results in a nuclear reaction in the protostar that involves the isotope hydrogen-3 (${}^3_1\text{H}$).

The isotope hydrogen-3 (${}^3_1\text{H}$) is radioactive. It decays by beta particle emission.

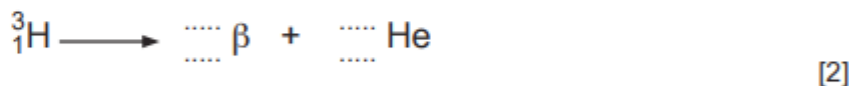
- (i) State how the nuclei of isotopes of the same element are different.

.....
..... [1]

- (ii) Explain why hydrogen-3 cannot decay by alpha particle emission.

.....
..... [1]

- (iii) Complete the equation for the decay of hydrogen-3 to an isotope of helium (He).



- (iv) Explain how the nuclear reaction in the protostar stops further collapse as the protostar becomes a stable star.

.....
.....
.....
.....
..... [3]

- (c) The cloud of dust and gas that collapses also contains atoms of the heaviest elements.

State which part of the life cycle of a star is responsible for the production of the heaviest elements.

.....
..... [1]

2. O/N/23/22/Q10

The isotope iodine-131 emits two types of nuclear radiation. Gamma radiation is one of the types emitted.

A sample of iodine-131 produces a narrow beam of radiation which passes into a magnetic field.

Fig. 10.1 shows some of the paths taken in the magnetic field by the radiation emitted from the sample.

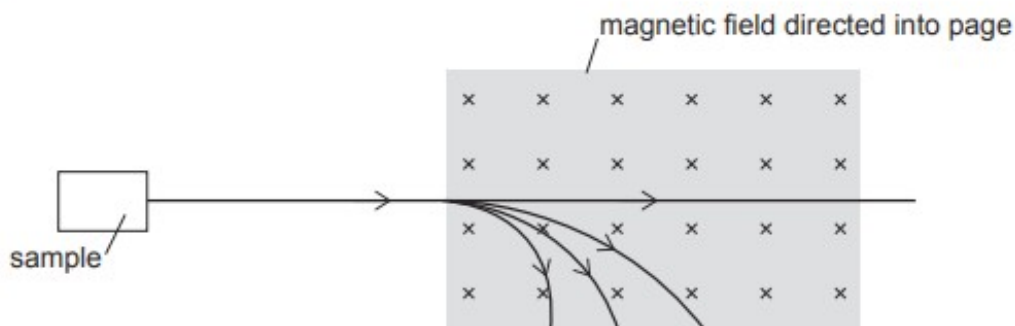


Fig. 10.1

(a) The direction of the magnetic field is into the page.

(i) On Fig. 10.1, mark with a gamma symbol (γ) any of the paths taken by the gamma radiation. [1]

(ii) Determine the other type of radiation emitted by iodine-131.

Explain how you obtain your answer.

type of radiation

explanation

.....
.....
.....

[4]

(iii) In another experiment, the beam of radiation from the sample is incident on a 5 mm thick sheet of lead.

State what happens to the two types of radiation in the beam.

.....
.....
..... [2]

(b) The gamma radiation produced has a frequency of 8.8×10^{19} Hz.

The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = m [2]

[Total: 9]

3. O/N/23/21/Q8

Uranium-235 ($^{235}_{92}\text{U}$) is the isotope of uranium that is used as the fuel in a nuclear power station.

- (a) The isotope of uranium that is most abundant in nature is uranium-238 ($^{238}_{92}\text{U}$).

State how a neutral atom of uranium-235 differs from a neutral atom of uranium-238.

.....
..... [1]

- (b) In the reactor of a nuclear power station, a nucleus of uranium-235 absorbs a slow-moving neutron and undergoes fission.

- (i) State what happens to the nucleus of an atom during fission.

.....
..... [1]

- (ii) Explain how fission can lead to a chain reaction in the reactor.

.....
.....
.....
.....
.....
.....
.....
..... [4]

- (iii) State the purpose of the moderator in the reactor.

.....
..... [1]

- (iv) Explain the action of the control rods in the reactor.

.....
.....
..... [2]

[Total: 9]

4. M/J/23/22/Q9

A radioactive source emits α -particles, β -particles and γ -radiation.

- (a) (i) State which type of radiation produces the strongest ionising effect.

..... [1]

- (ii) State which type of radiation is deflected most by a magnetic field.

..... [1]

- (b) Fig. 9.1 shows a Geiger-Müller (G.M.) tube and counter. A radioactive source is placed 10 cm from the G.M. tube.

In Fig. 9.2 a piece of metal 5 mm thick is placed between the source and the G.M. tube. The readings on the counter have been corrected for background radiation and show the count rate due to the source.

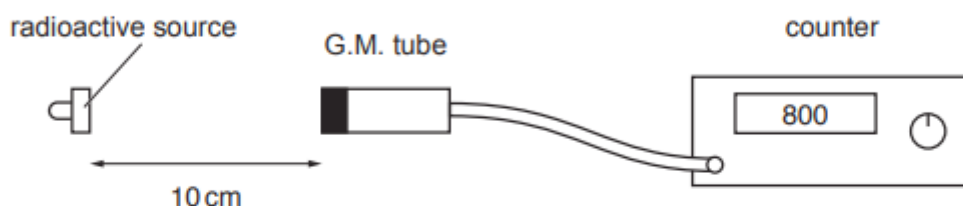


Fig. 9.1

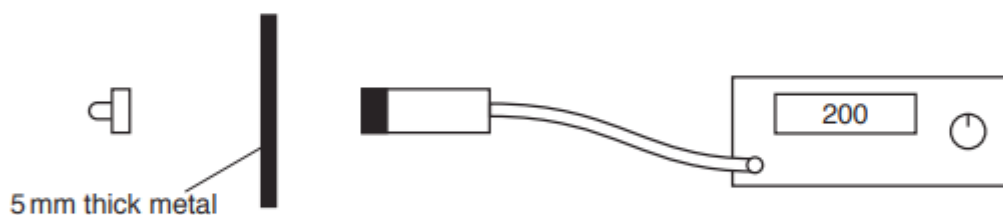


Fig. 9.2

- (i) Explain how the readings show that the source emits β -particles and γ -radiation.

.....
.....
.....
..... [3]

- (ii) State why the readings **cannot** be used to show that the source emits α -particles.

.....
..... [1]

- (c) Describe **one** way that a radioactive source is moved safely in a school laboratory.

.....
..... [1]

5. M/J/23/21/Q10

Carbon-14 is an isotope of carbon. It undergoes radioactive decay with a half-life of 6000 years. The age of a sample of wood is found using the carbon-14 that it contains.

(a) Describe what happens during radioactive decay.

.....
.....
..... [2]

(b) The count rate of the carbon-14 in the sample of wood is initially 1600 counts/minute.

On Fig. 10.1 draw a graph to show how the count rate will vary over the next 24 000 years.

The initial count rate is already marked with an x.

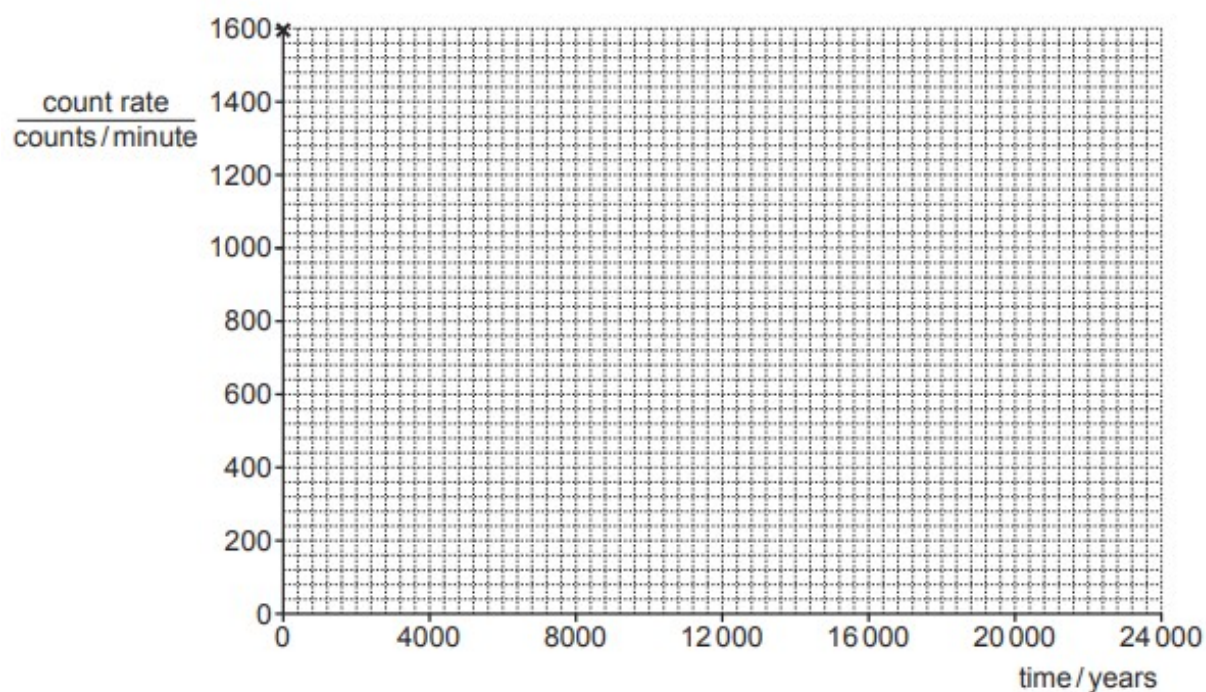


Fig. 10.1

[2]

- (c) Name the apparatus used to measure the radiation from the sample of wood and describe how it is used to measure the count rate.

apparatus used

how the apparatus is used

.....

..... [2]

- (d) Describe how the count rate from a sample of wood is used to find its age.

.....

.....

.....

.....

.....

..... [2]

[Total: 8]

6. O/N/22/22/Q9

Protactinium-234 (${}_{91}^{234}\text{Pa}$) is a radioactive isotope of protactinium that decays to uranium-234 (${}_{92}^{234}\text{U}$).

- (a) Compare the nuclide notation ${}_{91}^{234}\text{Pa}$ with the nuclide notation ${}_{92}^{234}\text{U}$ and deduce what this shows about what is emitted from a nucleus of protactinium-234 as it decays to uranium-234.

Place a tick (✓) in the appropriate boxes of Table 9.1 to show what is deduced from comparing the nuclide notations.

Table 9.1

	yes	no	it is not possible to tell
an alpha-particle is emitted			
a beta-particle is emitted			
a gamma ray is emitted			

[2]

- (b) The most abundant isotope of protactinium is protactinium-231.

- (i) Explain, by referring to their nuclear compositions, why protactinium-231 and protactinium-234 are both isotopes of the same element.

.....
 [1]

- (ii) Explain, by referring to their nuclear compositions, why protactinium-231 and protactinium-234 are different isotopes of that element.

.....
 [1]

- (c) A teacher places a radiation detector on a bench in a school laboratory and switches it on.

- (i) The teacher measures and records the background radiation count rate.

Describe what is meant by 'background radiation' and state **two** significant sources of the count rate recorded by the teacher.

background radiation

.....

source 1

source 2

[3]

- (ii) The teacher moves a sample of protactinium-234 so that it is next to the detector.

Suggest **one** precaution that ensures that the sample is moved in a safe way.

.....
..... [1]

- (iii) The count rate is measured every 20 s with the sample present, and then corrected for background radiation.

Fig. 9.1 shows a graph of the corrected count rate against time for the protactinium-234 sample.

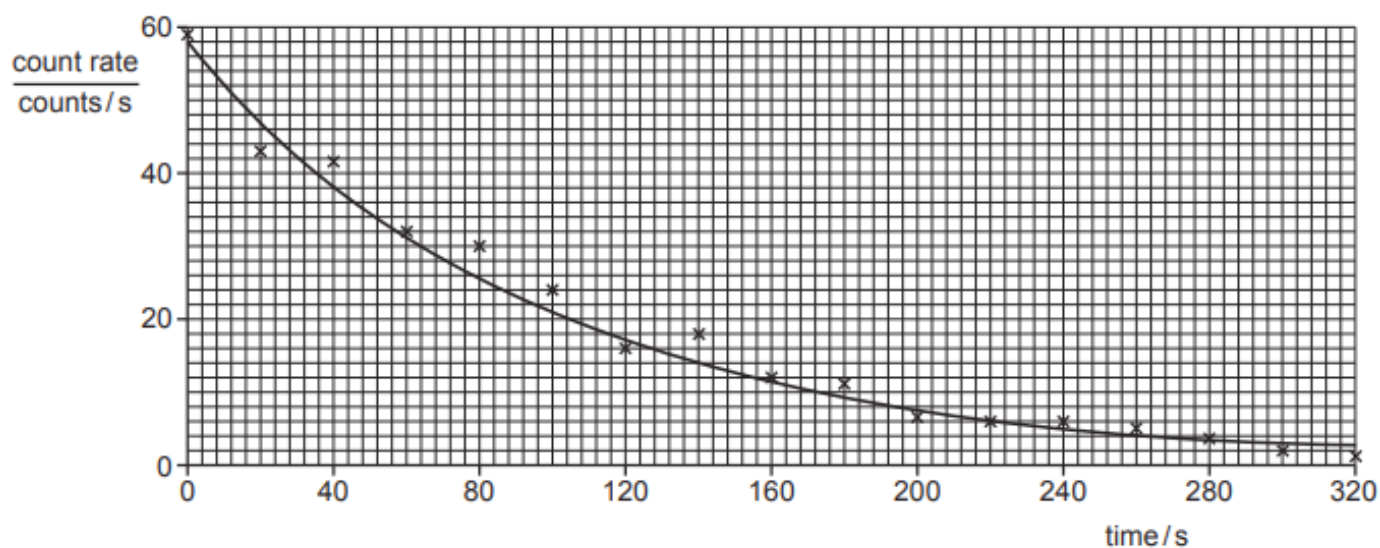


Fig. 9.1

The curve is the best-fit line.

Explain why many of the crosses do not lie on the curve.

.....
.....
..... [2]

- (iv) Using Fig. 9.1, determine the half-life of protactinium-234.

Show your working.

half-life = [3]

- (v) The uranium-234 formed from the protactinium-234 is also radioactive. Its half-life is many thousands of years.

Explain why the radiation from uranium-234 does **not** affect the count rates measured in this experiment.

.....
.....
..... [2]

[Total: 15]

7. M/J/22/22/Q9

Thorium-229 is a radioactive isotope used in several medical applications that involve alpha-particles and beta-particles.

- (a) During ionisation, a helium atom becomes a helium ion.

Fig. 9.1 shows a diagram of a helium (He^+) ion.

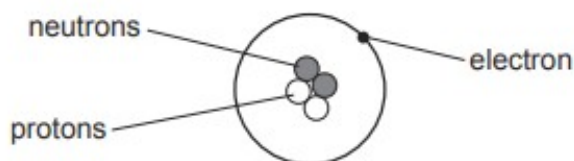


Fig. 9.1

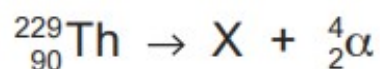
- (i) State how the structure of a helium atom differs from the structure of the helium ion.

.....
 [1]

- (ii) State how the structure of an alpha-particle differs from the structure of the helium ion.

.....
 [1]

- (iii) A nucleus of thorium-229 ($^{229}_{90}\text{Th}$) decays by alpha (α) emission to a nucleus of element X.



The nucleus of X then decays to a nucleus of Y by beta (β) emission.



Complete Table 9.1 to show the number of protons and neutrons in a nucleus of X and in a nucleus of Y.

Table 9.1

nucleus	number of protons	number of neutrons
X		
Y		

[4]

- (b) Experiments can show that a sample of a material is radioactive.

Describe the apparatus and the procedure used to show that a sample emits **both** alpha-particles and beta-particles.

You may draw a diagram of the apparatus, if you wish.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) (i) State what is meant by the half-life of thorium-229.

.....

..... [2]

- (ii) A sample of pure thorium-229 contains 4.0×10^{14} atoms. After 22 000 years, the number of atoms of thorium-229 in the sample is 5.0×10^{13} .

Determine the half-life of thorium-229.

Show your working.

half-life = [3]

8. M/J/22/21/Q9

Table 9.1 shows details of seven different nuclides.

Table 9.1

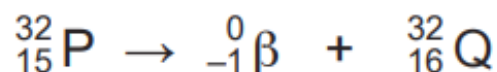
nuclide	radiation emitted	half-life
hydrogen-2	none	–
hydrogen-3	beta	12 year
francium-223	beta	22 min
iridium-192	gamma	74 day
phosphorus-32	beta	14 day
radon-222	alpha	4 day
technetium-99	gamma	6 hour

- (a) (i) Hydrogen-2 and hydrogen-3 are isotopes of the element hydrogen.

Define the term isotope.

.....
 [2]

- (ii) The equation for the decay of phosphorus-32 (P-32) as it emits a beta particle is:



Explain whether the equation shows that Q is another isotope of phosphorus. Q is not the chemical symbol for the atom.

.....
 [1]

- (b) One of the sources in Table 9.1 is used in a medical procedure to detect unusual bone structures. It is injected into a patient and the radiation emitted is detected outside the body.

- (i) State which source in Table 9.1 is most suitable for this type of medical use.

..... [1]

- (ii) Explain **two** reasons for your choice

.....

 [2]

(c) Radon gas is one natural source of background radiation.

Some causes of background radiation are man-made, for example, X-rays.

(i) State one other **natural** source of background radiation.

..... [1]

(ii) State one other **man-made** source of background radiation.

..... [1]

(iii) State one **harmful** effect of background radiation.

..... [1]

(d) One fusion reaction that occurs is:



(i) Complete the equation to show the missing proton and nucleon numbers. [1]

(ii) Deduce the name of particle X.

..... [1]

(iii) Suggest where this fusion reaction takes place.

..... [1]

(e) Compare the properties of alpha-particles and beta-particles in terms of their:

- ability to penetrate through materials
- ionising effects
- deflection in a magnetic field.

.....
.....
.....
.....
.....
.....
..... [3]

[Total: 15]

9. O/N/21/22/Q6

Isotope X is radioactive. It decays by alpha-particle emission to a stable isotope.

- (a)** State how a nucleus of X changes when it emits an alpha-particle.

.....

.....

..... [2]

- (b)** There is a radiation detector in a laboratory where there are no radioactive samples.

The detector is switched on and shows an average count rate of 22 counts/minute.

- (i)** State why the radiation detector shows a count rate.

.....

..... [1]

- (ii)** A sample of isotope X is placed 2cm from the detector and the reading displayed is 8000 counts/minute.

The sample is moved a distance of 10cm from the detector. The reading returns to an average value of 22 counts/minute.

Explain why the reading returns to the original value.

.....

.....

..... [2]

- (c)** An alpha-particle passes into a region where there is a magnetic field. In the magnetic field, a force acts on the alpha-particle so that it follows a circular path. Fig. 6.1 shows that the particle passes through point J.

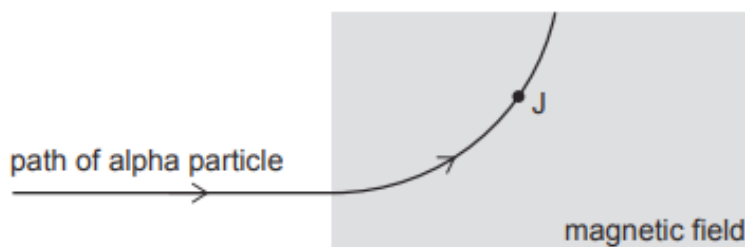


Fig. 6.1

- (i)** On Fig. 6.1, draw an arrow through point J to show the direction of the force on the alpha-particle at J. [1]

- (ii) Determine the direction of the magnetic field and mark a tick in the box (✓) that indicates this direction.

☐	to the left
☐	to the right
☐	towards the top of the page
☐	towards the bottom of the page
☐	into the page
☐	out of the page

[1]

- (iii) Explain whether this force does work on the alpha-particle as the particle moves along the circular path.

.....

.....

..... [2]

[Total: 9]

10. O/N/21/21/Q10

- The isotope yttrium-90 ($^{90}_{39}\text{Y}$) is radioactive. It is a beta-particle emitter that decays to product Q. Product Q is stable.

(a) State one feature that is common to all isotopes of yttrium.

..... [1]

(b) Describe how a **neutral atom** of Q differs from a **neutral atom** of yttrium-90.

.....

 [3]

(c) A sample of yttrium-90 is placed close to a radiation detector in a laboratory. There are no other radioactive samples in the laboratory. A counter records the count rate.

Fig. 10.1 is a graph of the count rate plotted against time.

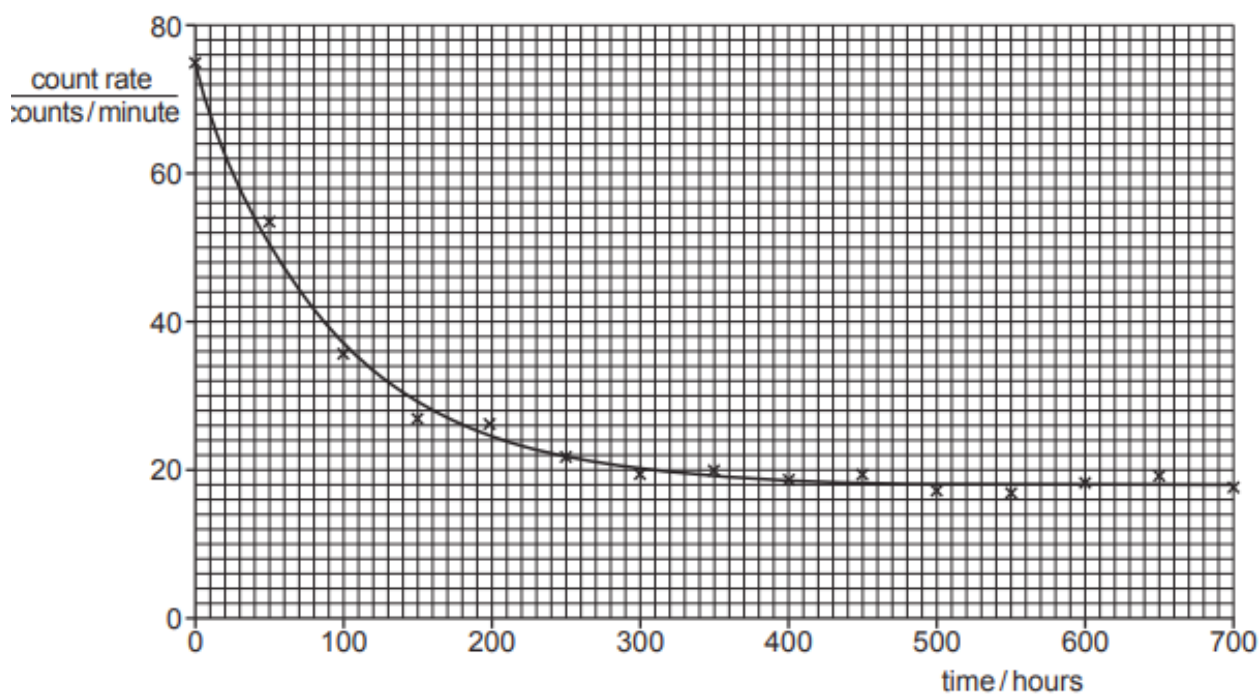


Fig. 10.1

(i) Using Fig. 10.1, determine the average background count rate.

average background count rate = [1]

(ii) Suggest **two** different origins for the background count.

1.

2.

[2]

(iii) Using Fig. 10.1, determine the half-life of yttrium-90. Show how the answer is obtained.

half-life = [4]

(iv) Many of the points plotted in Fig. 10.1 do not lie on the best-fit line.

Explain why.

.....

.....

..... [2]

(d) A beam of beta-particles, travelling in a vacuum, enters the region between two parallel, metal plates. One plate is negatively charged and the other is positively charged.

Fig. 10.2 shows the arrangement.

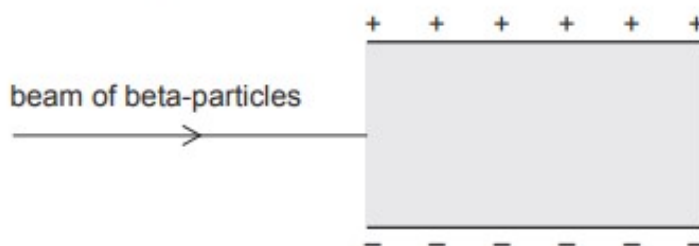


Fig. 10.2

On Fig. 10.2, draw the path taken by the beta-particles as they travel between the two plates.

[2]

11. M/J/21/22/Q11

A highly radioactive source that emits beta-particles is placed a few centimetres away from a detector, as shown in Fig. 11.1.

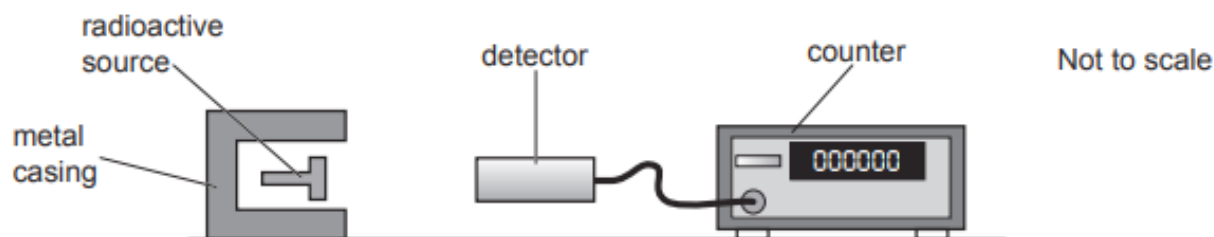


Fig. 11.1

- (a) State the name of the particle which has the same mass and charge as a beta-particle.

..... [1]

- (b) State and explain why the metal casing in Fig. 11.1 is used.

..... [2]

- (c) State and explain what happens to the number of particles detected in a minute as the radioactive source is moved:

- (i) a few centimetres **further** away from the source

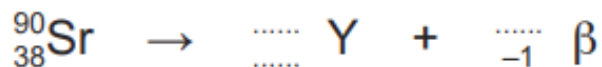
..... [2]

- (ii) more than a metre away from the source.

..... [2]

- (d) A nucleus of strontium-90 (Sr-90) decays by beta emission to a nucleus of yttrium (Y).

Complete the decay equation for this decay.



[3]

- (e) Nuclear fusion and nuclear fission both release large amounts of energy.

- (i) Describe how the process of nuclear fusion differs from the process of nuclear fission.

.....

.....

.....

.....

..... [3]

- (ii) Describe the conditions needed for nuclear fusion to take place.

.....

.....

.....

..... [2]

[Total: 15]

12. M/J/21/21/Q10

Table 10.1 contains details of the nature and some properties of alpha, beta and gamma emissions.

Table 10.1

	alpha	beta	gamma
nature	2 protons and 2 neutrons		
charge		negative	
penetrating power		stopped by 5 mm of aluminium	

- (a) Complete Table 10.1 by filling in the missing details. [6]
- (b) Surgical instruments in sealed plastic bags are placed in thin plastic boxes. A conveyor belt takes the boxes close to a cobalt-60 source which sterilises the instruments.

This is shown in Fig. 10.1.

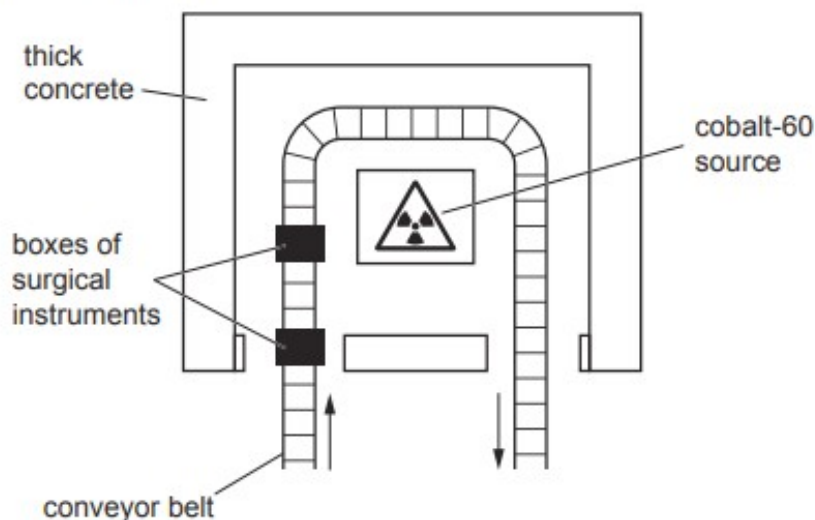


Fig. 10.1 (viewed from above)

The cobalt-60 source is a radioactive isotope of cobalt that emits gamma-radiation.

- (i) Describe what is meant by the term *isotope*.

.....
 [2]

- (ii) Suggest a property of gamma-radiation that enables it to sterilise the instruments in the bags in the boxes.

..... [1]

(iii) State why a source emitting only alpha-radiation cannot be used in this way.

.....
..... [1]

(iv) The half-life of cobalt-60 is 5.3 years.

Explain why a source with a half-life of 5.3 minutes is unsuitable for use in this application.

.....
..... [1]

(c) Geiger and Marsden performed an experiment in which alpha-particles were fired at a thin film of gold.

(i) Fig. 10.2 shows an alpha-particle passing close to the nucleus of a gold atom.

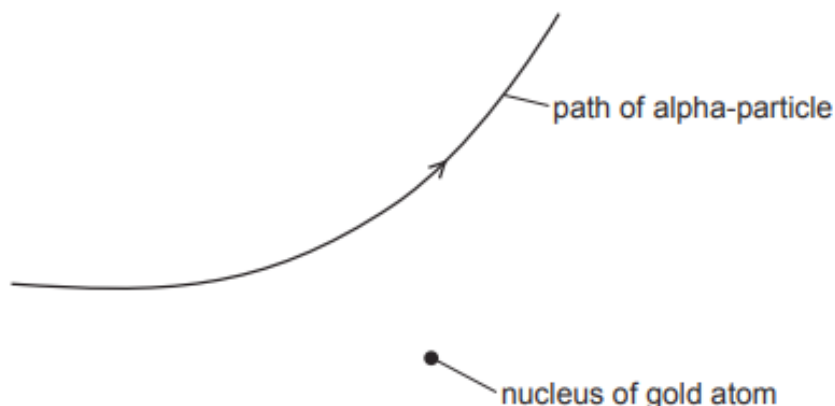


Fig. 10.2

Explain why the alpha-particle is deflected.

.....
..... [2]

(ii) In the experiment, most of the alpha-particles pass straight through the foil without deflection.

Explain, using ideas about the structure of the atom, why this happens.

.....
..... [2]

[Total: 15]

13. O/N/20/22/Q6

Phosphorus-32 ($^{32}_{15}\text{P}$) is an isotope of phosphorus that undergoes radioactive decay.

(a) The most common isotope of phosphorus is phosphorus-31.

(i) Describe the structure and composition of a neutral atom of phosphorus-31.

.....

 [2]

(ii) State how an atom of phosphorus-32 differs from an atom of phosphorus-31.

.....
 [1]

(b) Phosphorus-32 decays by beta-particle emission to a stable isotope of sulfur. The half-life for this decay is 2.0 weeks.

(i) State how a nucleus of this isotope of sulfur is different to a nucleus of phosphorus-32.

.....
 [1]

(ii) At time $t=0$, a radioactive sample contains 3.2×10^{11} atoms of phosphorus-32.

At the same moment, the sample contains no atoms of sulfur. This is shown by the cross on Fig. 6.1.

On Fig. 6.1, plot a graph to show how the total number of sulfur atoms in the sample changes with t and draw a suitable curve. [3]

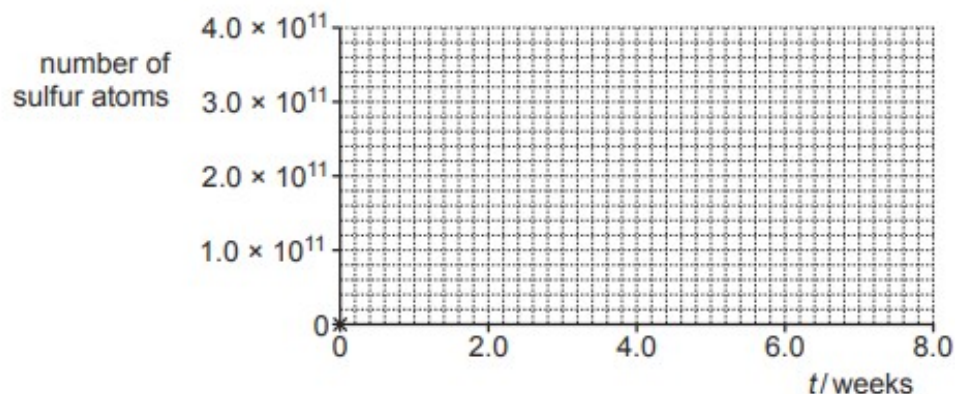


Fig. 6.1

(c) State **two** precautions taken when storing or moving radioactive materials.

1.

.....

2.

.....

[2]

[Total: 9]

14. M/J/20/22/Q7

Radon-222 is an isotope of radon which undergoes a series of radioactive decays.

Fig. 7.1 is a diagram showing the proton number (atomic number) and nucleon number (mass number) of nuclei involved in the series of decays.

The point P represents a nucleus of radon-222.

Starting at P, a nucleus of radon-222 decays to Q; then from Q to R; then from R to S; then from S to T; and finally from T to U.

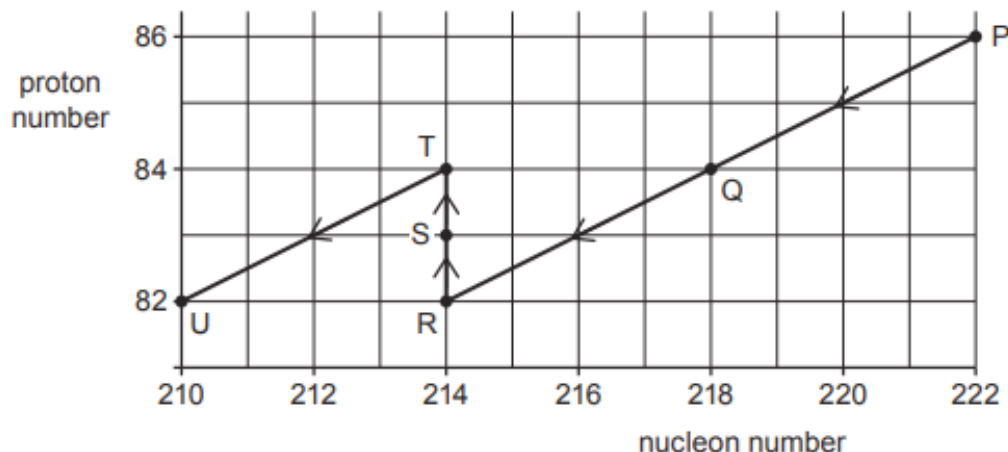


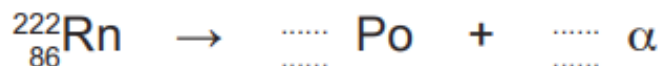
Fig. 7.1

- (a) (i) State two points on Fig. 7.1 which represent isotopes of the same element.
..... [1]
- (ii) Different isotopes of the same element have different atomic compositions.
State how the composition of their atoms is different.
.....
..... [1]
- (b) A nucleus of radon-222 emits an alpha-particle as it decays.
- (i) The radioactive decay of a single nucleus is random.
Explain what is meant by the *random* radioactive decay of a nucleus.
.....
.....
..... [1]

- (ii) In nuclide notation, radon-222 is written as $^{222}_{86}\text{Rn}$.

When a nucleus of radon-222 emits an alpha-particle (α), it decays to an isotope of polonium (Po).

Complete the decay equation below for this decay.



[2]

- (c) (i) State the name of the particle emitted as a nucleus of R decays to a nucleus of S.

..... [1]

- (ii) Describe the change in the composition of a nucleus of R as it decays to a nucleus of S.

.....

.....

..... [2]

[Total: 8]

15. M/J/20/21/Q7

Two isotopes of carbon are carbon-12 and carbon-14.

One of these isotopes, carbon-14, undergoes radioactive decay.

- (a) Describe what is meant by *radioactive decay*.

.....
 [2]

- (b) Carbon-12 has a proton number (atomic number) of 6 and a nucleon number (mass number) of 12.

Complete Table 7.1 for a neutral atom of each of these two isotopes.

Table 7.1

	carbon-12	carbon-14
number of protons	6	
number of neutrons		
number of electrons		

[2]

- (c) A sample of carbon-14 is contained in a thin aluminium container of thickness 0.2 mm.

Radiation from the sample is detected outside the container.

When the thickness of the aluminium is increased to 6 mm, no radiation from the sample is detected outside the container.

- (i) State the type of radiation which is stopped by increasing the thickness of the aluminium.

..... [1]

- (ii) Explain how you know that the sample does **not** emit one other type of radiation.

.....

 [2]

[Total: 7]

16. O/N/19/22/Q10

All the isotopes of the gas radon are radioactive.

- (a) State **one** similarity and **one** difference between the nuclei of two different isotopes of radon.

similarity

difference

[2]

- (b) The isotope radon-222 decays by alpha-particle emission to an isotope of polonium (Po). The proton number (atomic number) of polonium is 84.

- (i) Determine the number of neutrons in an atom of the polonium isotope.

number of neutrons = [2]

- (ii) Determine the number of protons in an atom of radon-222.

number of protons = [2]

- (iii) Describe how a neutral atom of helium (${}^4_2\text{He}$) differs from an alpha-particle.

.....

..... [1]

- (c) In an experiment to collect a small quantity of helium, a sample of radon-222 is enclosed in an inner glass tube which has a very thin wall. Fig. 10.1 shows that this tube is placed inside a container that is initially evacuated.

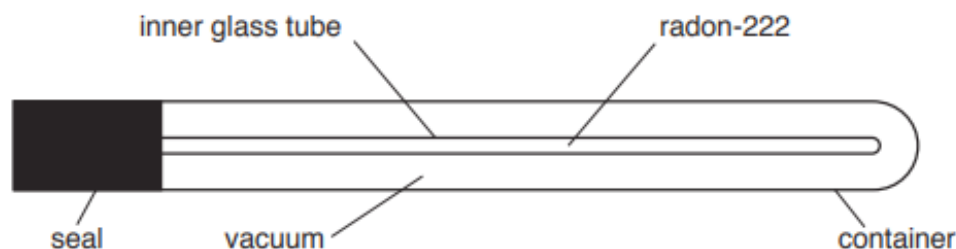


Fig. 10.1

Both the container and the inner glass tube are sealed.

As the radon-222 decays, alpha-particles pass through the thin wall of the inner glass tube.

Fig. 10.2 shows how the total number of alpha-particles produced by the radioactive decay of the radon-222 changes as time passes.

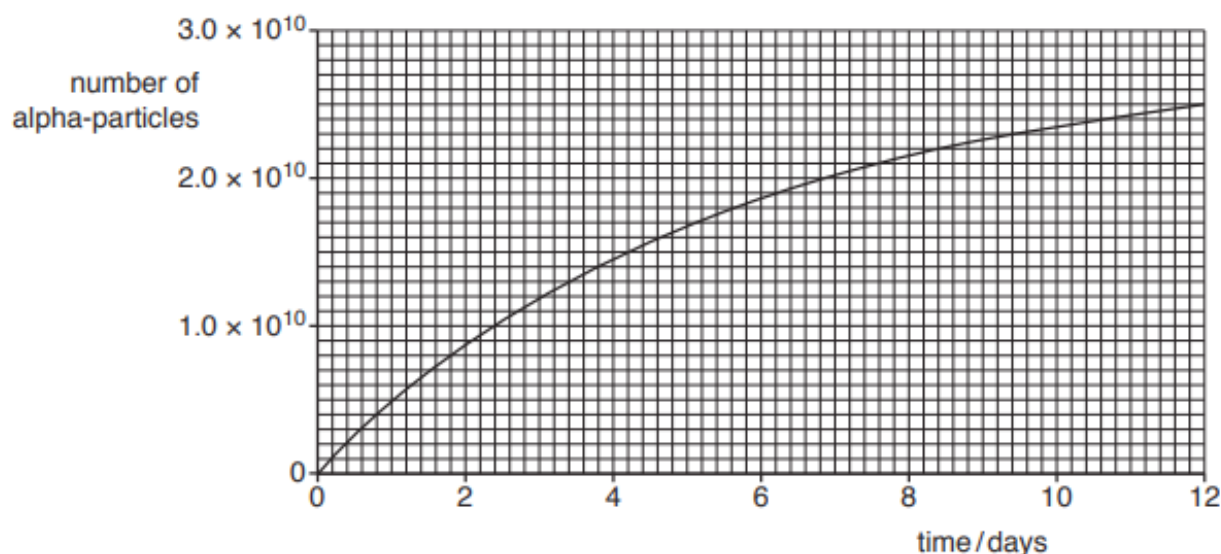


Fig. 10.2

- (i) Use Fig. 10.2 to determine the number of alpha-particles produced in 7.6 days.

..... [1]

- (ii) Initially, there are 2.8×10^{10} atoms of radon-222 in the inner glass tube. As each radon-222 atom decays, it produces an alpha-particle.

Calculate the number of radon-222 atoms that remain after 7.6 days.

number of atoms remaining = [1]

- (iii) Using the number of radon-222 atoms present initially and the number present after 7.6 days, calculate the half-life of radon-222.

half-life = [3]

- (d) The alpha-particles become helium atoms which are collected in the vacuum shown in Fig. 10.1.

Explain, in terms of the properties of alpha-particles, why the wall of the inner glass tube must be extremely thin.

.....

..... [1]

- (e) In some parts of the world, radon-222 accumulates in the air in buildings and is breathed in by people.

Explain why the presence of an alpha-emitter in the lungs is particularly hazardous.

.....

.....

..... [2]

[Total: 15]

17. O/N/19/21/Q7

Table 7.1 shows the composition of four neutral atoms and one atom that has been ionised.

These five particles are labelled P, Q, R, S and T.

Table 7.1

	P	Q	R	S	T
number of electrons	17	18	18	19	20
number of neutrons	20	20	24	23	21
number of protons	17	19	18	19	20

(a) The charge on an electron is $-1.6 \times 10^{-19} \text{C}$.

(i) State which of P, Q, R, S and T is the ionised atom.

..... [1]

(ii) State the size and the sign of the charge on the ionised atom.

charge = [1]

(b) State which pair from P, Q, R, S and T are isotopes of the same element.

..... [1]

(c) One of the atoms is a radioactive isotope that decays to produce one of the other atoms.

(i) Deduce which atom decays in this way.

..... [1]

(ii) Deduce which atom is the product of this decay.

..... [1]

(iii) Deduce which type of radioactive decay occurs.

..... [1]

(iv) The half-life for this radioactive decay is 33 years.

A radioactive sample contains a large number of the atoms that decay in this way.

Determine the fraction of this number of atoms that will decay in 99 years.

fraction that decays = [3]

18. M/J/19/22/Q11

- (a) Stars are formed inside a cloud of gas and dust in space.

When a star forms, the cloud first becomes smaller as it collapses.

- (i) State what causes the gas cloud to collapse.

..... [1]

- (ii) Eventually nuclear fusion occurs in the star.

State **two** conditions required for nuclear fusion to occur in the star.

1.

2.

[2]

- (b) When a large star explodes, nuclei of heavy elements are formed.

One such nucleus is uranium-235 ($^{235}_{92}\text{U}$).

- (i) State the name and the number of each type of particle found in one nucleus of $^{235}_{92}\text{U}$.

name of particle number found in nucleus

name of particle number found in nucleus

[2]

- (ii) State one feature that is the same for the nuclei of all isotopes of uranium.

.....

..... [1]

- (iii) A nucleus of $^{235}_{92}\text{U}$ in a nuclear reactor can undergo nuclear fission.

Describe the process of nuclear fission in a nuclear reactor.

.....

.....

..... [2]

- (iv) State one difference between nuclear fusion and nuclear fission.

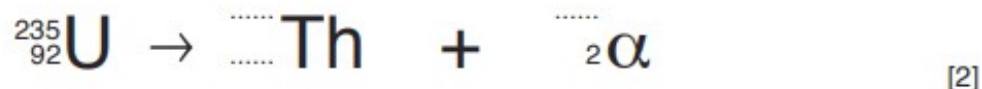
.....

..... [1]

- (v) The nucleus of $^{235}_{92}\text{U}$ is not stable.

It decays by the emission of an alpha-particle (α) into a nucleus of thorium (Th).

Complete the nuclear equation that represents this decay.



- (c) Geiger and Marsden carried out an experiment to investigate the scattering of alpha-particles. Fig. 11.1 shows a diagram of the apparatus that they used.

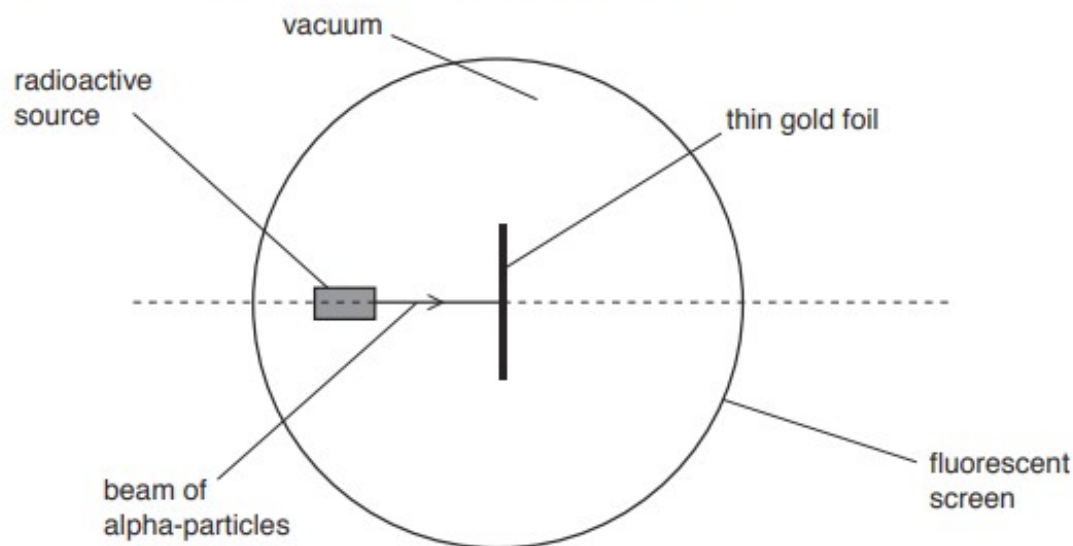


Fig. 11.1

When an alpha-particle strikes the fluorescent screen a flash of light is emitted.

- (i) On Fig. 11.1, mark with a letter P where most flashes occur on the screen. [1]

- (ii) Describe what is seen on the other parts of the screen.

.....
..... [1]

- (iii) State two conclusions about atoms of gold that resulted from this experiment.

.....
.....
.....
..... [2]

[Total: 15]

19. M/J/19/21/Q11

A teacher carries out an experiment on radioactivity using the apparatus shown in Fig. 11.1.

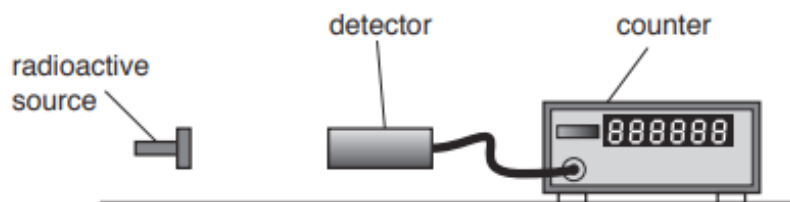


Fig. 11.1 (not to scale)

The count rate is a measurement of the count during one minute.

(a) (i) State the name of one type of radiation detector.

..... [1]

(ii) State one safety precaution the teacher must take when using a radioactive source.

.....
..... [1]

(b) The teacher states that the emissions from the radioactive source are random in time.

(i) State what is meant by *random in time*.

.....
..... [1]

(ii) Describe how the apparatus is used to show that the emissions are random in time.

.....
..... [2]

(c) (i) State what is meant by *background radiation*.

.....
..... [1]

(ii) The teacher measures the count rate due to the source.

Describe how the count rate is corrected for the background radiation.

.....
.....
..... [2]

- (d) The corrected count rate is recorded for various distances d between the source and the detector. Fig. 11.2 shows a graph of the results obtained.

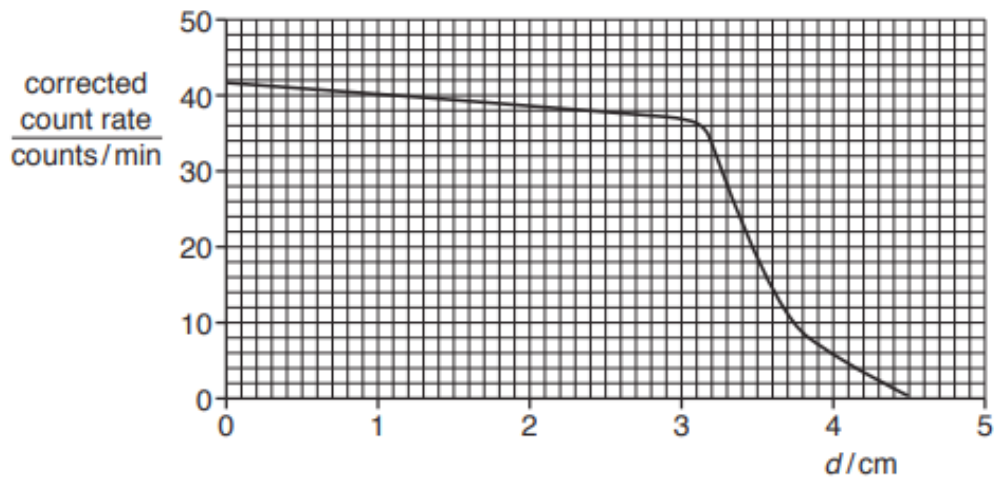


Fig. 11.2

- (i) Determine the maximum distance that the radiation travels in air.

..... [1]

- (ii) State the name of the radiation emitted by the radioactive source and give a reason for your answer.

radiation is

reason

..... [2]

- (iii) As the radiation from the source travels in the air it causes ionisation.

Describe what is meant by *ionisation* of the air.

.....

.....

..... [1]

- (e) The half-life of another radioactive source is 40 minutes.

Initially, the corrected count rate from this source is 300 counts/minute.

On Fig. 11.3, plot suitable points and draw a graph of the corrected count rate as it changes with time.

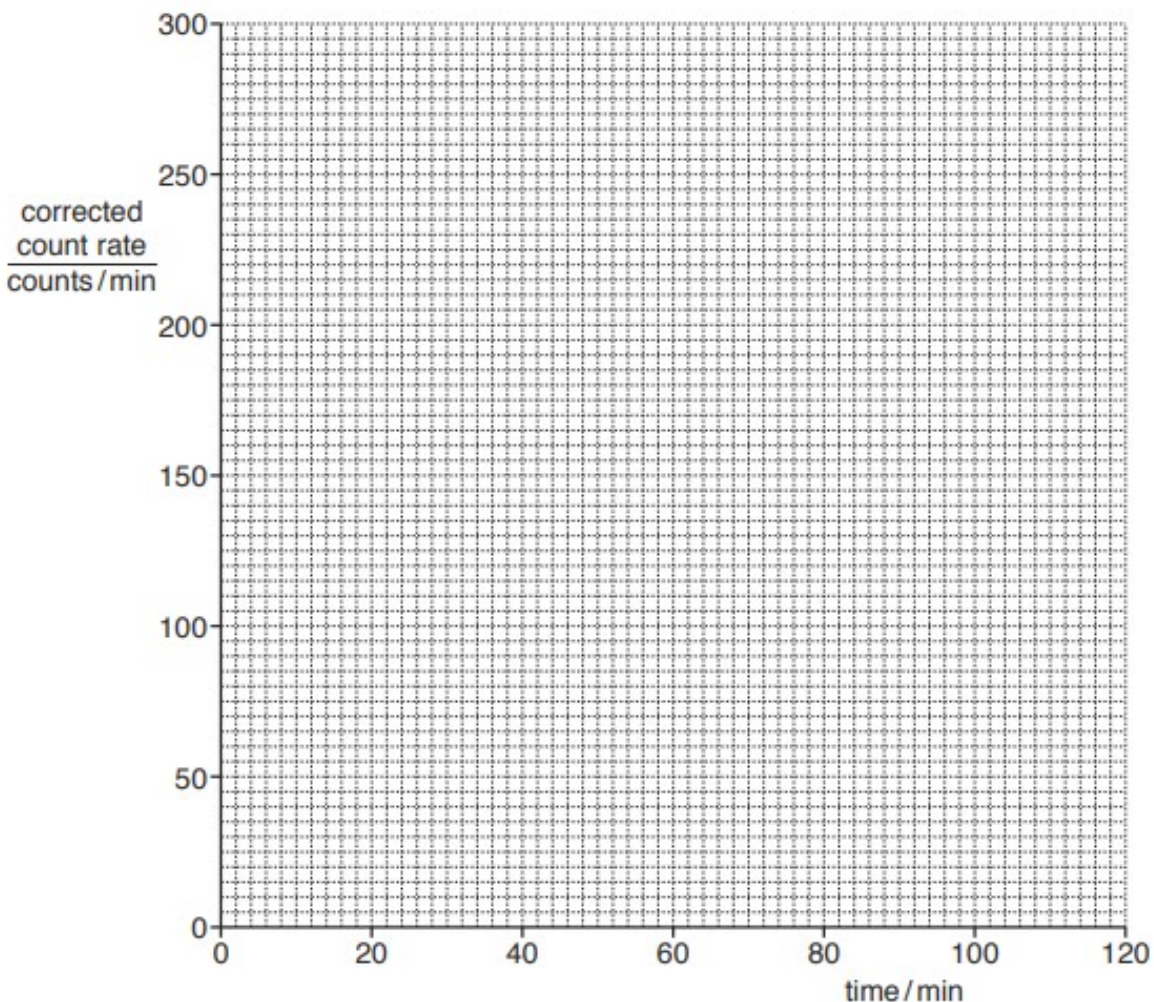


Fig. 11.3

[3]

[Total: 15]

20. O/N/18/22/Q6

Fig. 6.1 shows a radiation detector placed on a laboratory bench.

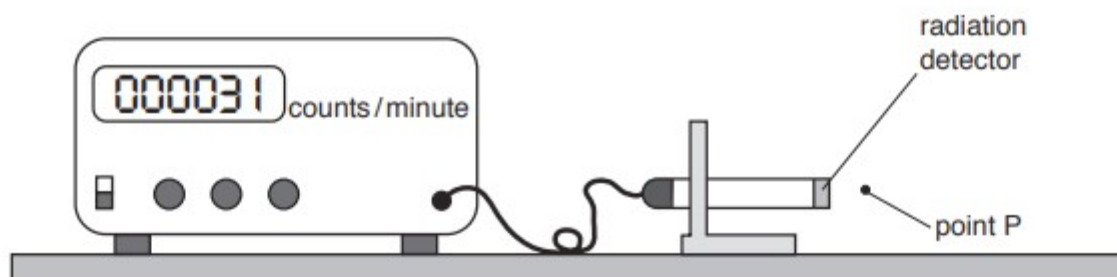


Fig. 6.1

The detector is switched on and six readings of the count rate are recorded.

The table in Fig. 6.2 shows the readings obtained.

reading number	1	2	3	4	5	6
$\frac{\text{count rate}}{\text{counts/minute}}$	31	36	29	32	31	33

Fig. 6.2

- (a) Using all the readings obtained, determine an average value for the background count rate.

background count rate = [1]

- (b) Fig. 6.1 shows a point P which is a very short distance from the end of the radiation detector. A sample of the radioactive isotope cobalt-60 is placed at P. The average value of the count rate obtained is now 975 counts/minute.

The average count rate is determined with different objects between the radiation detector and the sample. The table in Fig. 6.3 shows the results obtained.

object	$\frac{\text{average count rate}}{\text{counts / minute}}$
no object	975
four sheets of paper	976
0.50mm thickness sheet of aluminium	117
2.0cm thickness sheet of lead	52

Fig. 6.3

(i) Indicate, by placing ticks (✓) in the appropriate boxes, the radiation emitted by cobalt-60.

☐

alpha-particles

☐

beta-particles

☐

gamma rays

[1]

(ii) This radiation is produced when a nucleus of cobalt-60 ($^{60}_{27}\text{Co}$) decays into a nucleus of the daughter product X. Product X is not radioactive.

Determine

1. the number of protons in a nucleus of X,

number of protons = [1]

2. the number of neutrons in a nucleus of X.

number of neutrons = [1]

(c) The half-life of cobalt-60 is 5.3 years.

(i) State what is meant by *half-life*.

.....

 [2]

(ii) When there is a lead sheet between the detector and the sample, the average count rate is obtained from six readings taken at one-minute intervals. The six readings are given in the table in Fig. 6.4.

reading number	1	2	3	4	5	6
$\frac{\text{count rate}}{\text{counts/minute}}$	61	46	43	56	49	57

Fig. 6.4

There are reasons for suggesting that the variation in these readings is random and not because the number of cobalt-60 atoms in the sample is decreasing.

State two of these reasons.

1.

 2.
 [2]

21. O/N/18/21/Q10

- (a) Two neutral atoms are atoms of different isotopes of the same element.

State

- (i) one similarity in the structures of these two atoms,

..... [1]

- (ii) one difference between the structures of these two atoms.

..... [1]

- (b) The isotope plutonium-239 decays by the emission of an alpha-particle (symbol α) to an isotope of uranium (symbol U). The half-life for this decay is 2.4×10^4 years.

- (i) Complete the equation for this decay.



- (ii) The count rate measured due to a pure sample of plutonium-239 is 6400 counts / second.

Calculate the count rate after a period of 1.2×10^5 years has elapsed.

count rate = [3]

- (iii) Fig. 10.1 shows an alpha-particle entering a uniform magnetic field.



Fig. 10.1

1. On Fig. 10.1, draw the path of the alpha-particle in the magnetic field. [2]

2. Explain why the alpha-particle follows the path you drew in (b)(iii)1.

.....
 [1]

(c) In a nuclear reactor at a power station, plutonium-239 is made to undergo nuclear fission.

(i) State what is meant by *nuclear fission*.

.....
..... [1]

(ii) State what happens to a plutonium-239 nucleus in the nuclear reactor to cause fission.

.....
..... [1]

(iii) State one advantage of generating electricity in a nuclear power station rather than using

1. an oil-fired power station,

.....
..... [1]

2. an array of wind turbines.

.....
..... [1]

22. M/J/18/22/Q11

A smoke detector contains a small radioactive source of americium-241. The source emits alpha-particles.

- (a) (i) In the space below, draw a diagram of an apparatus that can be used to show that the source emits alpha-particles.

[2]

- (ii) Describe how this apparatus is used.

.....

.....

.....

.....[2]

- (iii) Explain how the results of the test show that the source emits alpha-particles.

.....

.....

.....

.....[2]

- (b) The smoke detector works because alpha-particles from the source ionise the air.

Compare the relative ionising effects and penetrating powers of alpha-particles, beta-particles and gamma rays.

ionising effects

.....

penetration effects

.....[2]

(c) Americium-241 has a half-life of 430 years.

(i) Radium-224 is another element that emits alpha-particles. It has a half-life of 3.6 days.

Explain why americium-241 is more suitable as the radioactive source in a smoke detector than radium-224.

.....
.....
.....[1]

(ii) A smoke detector contains 8.0×10^{11} atoms of americium-241.

Calculate the time taken for the number of atoms of americium-241 to fall to 1.0×10^{11} .

time =[3]

(d) When used correctly, the radioactive source in the smoke detector is less harmful than background radiation.

(i) State one harmful effect of background radiation.

.....[1]

(ii) A radioactive source is picked up using a long-handled tool. Explain why this is safer than using a short-handled tool.

.....
.....
.....
.....[2]

23. O/N/17/21/Q8

A sample of the radioactive isotope radon-222 decays by the emission of alpha-particles.

(a) As alpha-particles travel through air, the air is ionised.

(i) Describe the composition of an alpha-particle.

.....
.....
..... [1]

(ii) Explain how an alpha-particle ionises air.

.....
.....
..... [2]

(iii) State how the relative ionising effect of alpha-particles compares with that of

1. beta-particles,

.....

2. gamma rays.

..... [1]

(b) The half-life of radon-222 is 3.3×10^5 s.

The number of radon-222 atoms in the sample is 4.8×10^4 .

(i) Determine the time that it takes for the number of radon-222 atoms in this sample to decrease to 1.5×10^3 .

time = [3]

(ii) Suggest one reason why, in practice, the time for the number of radon-222 atoms to decrease to 1.5×10^3 may differ slightly from the value obtained in **(b)(i)**.

.....
..... [1]

24. M/J/17/22/Q11

One source of background radiation is cosmic rays.

The cosmic rays that enter the Earth's atmosphere are known as primary cosmic rays. Most of these rays are fast-moving protons or fast-moving alpha-particles.

Both the Earth's magnetic field and the Earth's atmosphere reduce the number of primary cosmic rays that reach the Earth's surface.

- (a) (i)** Name a source of background radiation apart from cosmic rays.

.....
.....[1]

- (ii)** State one effect of background radiation.

.....
.....[1]

- (iii)** State the number of protons and the number of neutrons in an alpha-particle.

number of protons

number of neutrons

[2]

- (iv)** Explain how the Earth's atmosphere reduces the number of alpha-particles that reach the surface.

.....
.....[1]

- (v)** Fig. 11.1 shows an area where there is a uniform magnetic field into the page.

On Fig. 11.1, draw a diagram to show the deflection of an alpha-particle by this magnetic field.

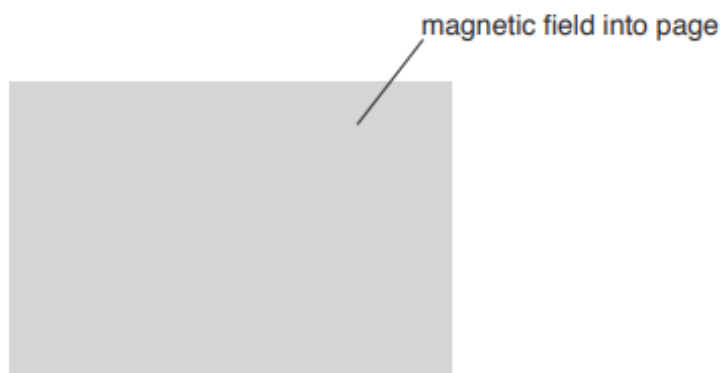


Fig. 11.1

[2]

- (b) Primary cosmic rays produce carbon-14 in the atmosphere. The half-life of carbon-14 is 5700 years.

(i) Define the term *half-life*.

.....

.....[2]

- (ii) A specimen of wood is found to contain 25 000 atoms of carbon-14. An otherwise identical specimen taken from a living tree is found to contain 200 000 atoms of carbon-14.

Determine the age of the specimen that contains 25 000 atoms of carbon-14.

age =[2]

- (iii) State why it is not possible to use the amount of carbon-14 to date a specimen that is 570 000 years old.

.....

.....[1]

- (c) Carbon-14 is one of the isotopes of carbon. Another isotope is carbon-12.

(i) Describe how a nucleus of carbon-14 differs from a nucleus of carbon-12.

.....

.....

.....[2]

(ii) Describe one similarity in the structure of the nuclei of these isotopes.

.....

.....

.....[1]

25. M/J/17/21/Q8

A doctor has the choice of two isotopes, $^{131}_{53}\text{I}$ and $^{125}_{53}\text{I}$, to inject into a patient.

These isotopes emit radiation from deep inside the body. The radiation is detected outside the body and provides information to help the doctor find out what is wrong with the patient.

The isotope $^{131}_{53}\text{I}$ decays into a stable isotope of xenon by emitting a beta-particle (β) and a gamma ray (γ).

(a) State what is meant by a *gamma ray*.

.....
..... [1]

(b) Complete the equation to show the radioactive decay of $^{131}_{53}\text{I}$.



[2]

(c) When a nucleus of $^{125}_{53}\text{I}$ decays, only a gamma ray is emitted.

Suggest why it is better for the doctor to inject the isotope $^{125}_{53}\text{I}$ into the patient.

.....
.....
.....
..... [2]

26. O/N/16/22/Q7

Plutonium-238 is a radioactive isotope that decays by alpha-particle emission. It also emits gamma-rays.

- (a) The nuclide notation for plutonium-238 is ${}_{94}^{238}\text{Pu}$.

Describe the structure and composition of a neutral atom of plutonium-238.

.....

.....

.....

.....[3]

- (b) Alpha-particles, beta-particles and gamma-rays all cause ionisation. The ionisation of substances in the human body can be hazardous.

(i) State how the ionising effect of beta-particles compares with

1. the ionising effect of alpha-particles,

.....[1]

2. the ionising effect of gamma-rays.

.....[1]

(ii) State two precautions that are taken when radioactive substances are moved in a safe way.

1.

.....

2.

.....

[2]

27. O/N/16/21/Q8

Sodium-24 and sodium-23 are two of the isotopes of sodium. Sodium-24 is a radioactive isotope that emits beta-particles and gamma-rays as it decays.

(a) The nuclide notation for the isotope sodium-24 is ${}^{24}_{11}\text{Na}$.

(i) Describe the composition and structure of a neutral atom of sodium-24.

.....

.....

.....

.....

.....[3]

(ii) State how the composition of a nucleus of sodium-24 differs from the composition of a nucleus of sodium-23.

.....

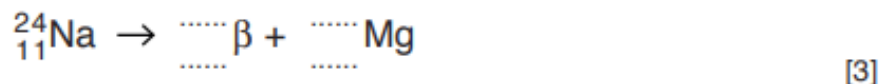
.....[1]

(b) A nucleus of sodium-24 decays. It emits a beta-particle and produces a nucleus of an isotope of magnesium (Mg).

(i) State the name of the particle that is identical to a beta-particle.

.....[1]

(ii) Complete the nuclide equation for the emission of a beta-particle (β) by sodium-24.



(c) When it decays, sodium-24 also emits gamma-rays.

State what gamma-rays are.

.....

.....

.....[2]

- (d) Fig. 8.1 shows a beam of beta-particles and a beam of gamma-rays entering the electric field between two oppositely charged plates.

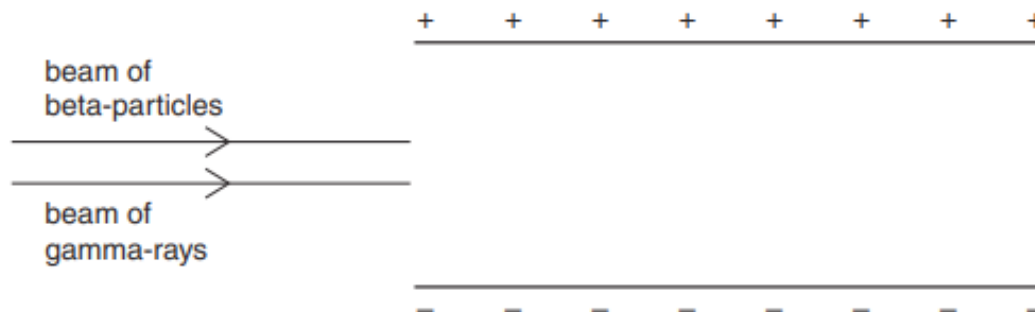


Fig. 8.1 (not to scale)

- (i) On Fig. 8.1, draw the path of the beta-particles in the electric field between the charged plates. [1]
- (ii) On Fig. 8.1, draw the path of the gamma-rays in the electric field between the charged plates. [1]
- (iii) Explain why the path taken by the beta-particles differs from that taken by the gamma-rays.
-[1]
- (e) Sodium-24 is sometimes used in diagnosing medical conditions. The half-life of sodium-24 is 15 hours.

Explain one reason why a half-life of 15 hours makes the isotope suitable for use in the human body.

.....[2]

28. M/J/16/22/Q11

Fig. 11.1 represents a nuclear fusion reaction.

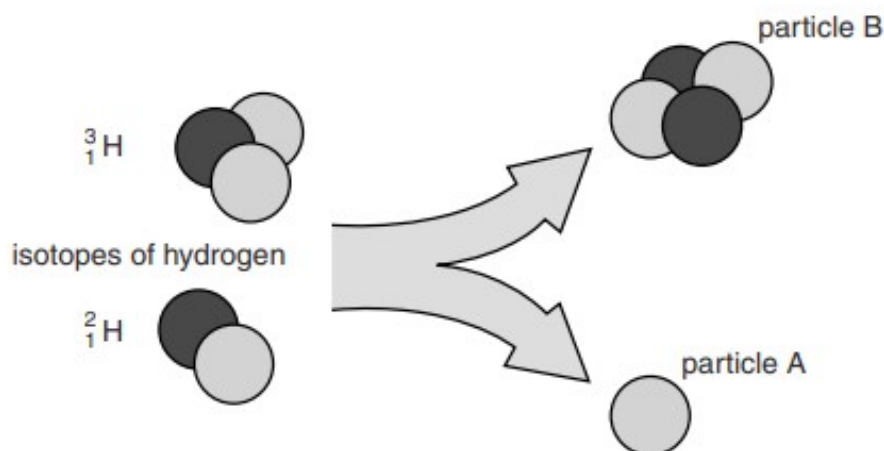


Fig. 11.1

At very high temperatures, nuclei of the two isotopes ${}^2_1\text{H}$ and ${}^3_1\text{H}$ fuse together. Energy is produced and two new particles are formed, particle A and particle B.

(a) Explain what is meant by *isotopes* of hydrogen.

.....

[2]

(b) Using Fig. 11.1, state

- (i) the number of neutrons in a nucleus of the isotope ${}^3_1\text{H}$,
- (ii) the name of particle A,
- (iii) the proton number of particle B,
- (iv) the nucleon number (mass number) of particle B.

[4]

(c) A very high temperature is needed to force nuclei together.

Explain why.

.....

[2]

- (d) A star forms from a large cloud of gas and dust in space.

Describe what happens as the star forms.

.....

.....

.....

.....

.....[3]

- (e) The isotope of hydrogen ${}^3_1\text{H}$ has a half-life of 12 years.

- (i) State what is meant by *half-life*.

.....

.....[2]

- (ii) A sample contains 16 000 atoms of ${}^3_1\text{H}$.

Calculate the number of atoms of ${}^3_1\text{H}$ present in the sample after 48 years.

number =[2]

29. O/N/15/22/Q11

The proton number (atomic number) of the element lead is 82. The isotope lead-209 ($^{209}_{82}\text{Pb}$) is radioactive and decays by the emission of beta-particles.

(a) Describe the composition and structure of a neutral atom of lead-209.

.....

.....

.....

..... [3]

(b) A nucleus of lead-209 emits a beta-particle.

(i) State how the composition of the nucleus produced differs from the original nucleus.

.....

..... [2]

(ii) Complete the table in Fig. 11.1 to show the relative ionising effects of the three types of ionising radiation produced by radioactive decay.

least strongly ionising $\longrightarrow$ most strongly ionising		

Fig. 11.1

[2]

(iii) A sample containing lead-209 nuclei is used to produce a beam of beta-particles. The beta-particles enter a magnetic field. The magnetic field is perpendicular to the direction of travel of the beta-particles.

The beta-particles travel from left to right. Fig. 11.2 shows that the direction of the magnetic field is out of the page.



Fig. 11.2

On Fig. 11.2, sketch the path of the beta-particles in the magnetic field.

[2]

(c) An industrial technician uses a detector to measure the background count rate in a laboratory.

(i) State what is meant by *background radiation*.

.....
..... [1]

(ii) Suggest two major sources of background radiation.

1.
 2.
- [2]

(iii) The average reading for the background count measured by the technician is 16 counts/minute.

He then brings a sample that contains a radioactive isotope of lead close to the detector and he finds that the new count rate is 92 counts/minute. The half-life of this isotope is 3.3 hours.

Determine the count rate measured using the detector after 6.6 hours.

count rate = [3]

30. O/N/15/21/Q11

Fig. 11.1 shows the proton number and the nucleon number of the nuclei of five neutral atoms.

	neutral atom 1	neutral atom 2	neutral atom 3	neutral atom 4	neutral atom 5
proton number	16	17	18	18	19
nucleon number	35	35	38	40	39

Fig. 11.1

(a) State the two atoms in Fig. 11.1 that

(i) are different isotopes of the same element,

..... [1]

(ii) contain the same number of neutrons,

..... [1]

(iii) contain the same number of electrons.

..... [1]

(b) The nucleus of atom 1 is radioactive and when it decays, it emits a beta-particle.

State, for the nucleus produced by this decay

(i) the proton number,

proton number = [1]

(ii) the nucleon number.

nucleon number = [1]

- (c) A sample contains a large number of radioactive nuclei that emit beta-particles. The sample is placed near to a radiation detector in a laboratory and the count rate is recorded over a long period of time.

Fig. 11.2 shows the count rate recorded by the detector during this period.

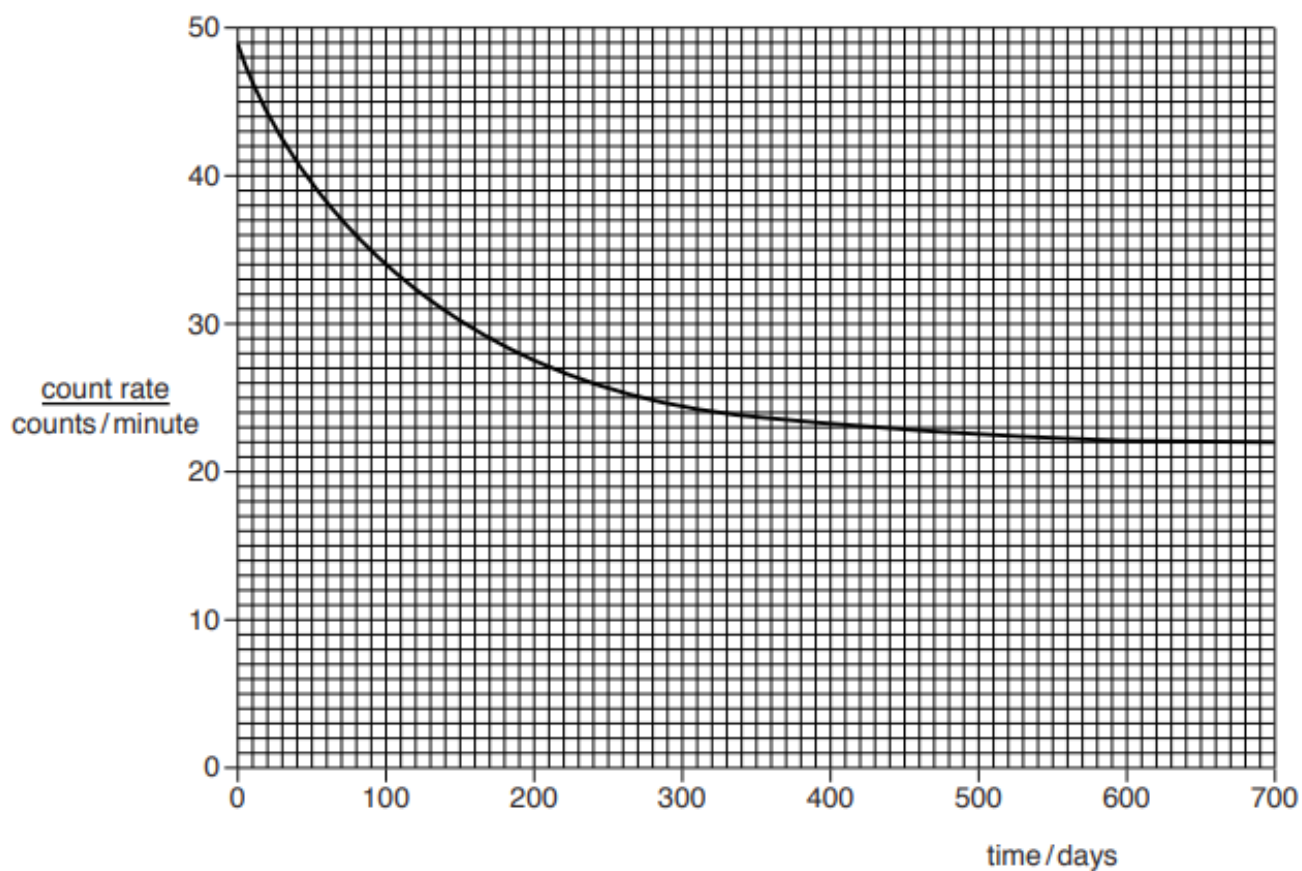


Fig. 11.2

- (i) Suggest two major sources of background radiation.

1.

2.

[2]

(ii) Using Fig. 11.2,

1. determine a value for the background count rate in the laboratory,

background count rate = [1]

2. determine the count rate, at time = 0, due to the sample on its own,

count rate due to sample = [1]

3. estimate the half-life of the radioactive atoms in the sample.

half-life = [2]

(d) A student suggests a change to the experiment in (c). He suggests completely encasing the radiation detector in lead before the sample that emits beta-particles is placed near to it. The effect on the count rate of background radiation may then be ignored.

- (i) Describe and explain the effect on the background count rate recorded by the detector of completely encasing the radiation detector in lead.

.....
.....
..... [2]

- (ii) State and explain whether the student's suggestion is a sensible one.

.....
.....
..... [2]

31. M/J/15/21/Q11

- (a) A student makes a model of an atom. The model contains 24 electrons, 25 protons and 26 neutrons. Some of these particles are inside a nucleus at the centre of the model.

- (i) Determine the nucleon number (mass number) of the atom.

.....[1]

- (ii) Explain why the model represents a charged atom.

.....

[2]

- (iii) The student makes a new model of a different isotope of the same element.

Describe the nucleus of this new model.

.....

[2]

- (b) Americium-241 is radioactive. Its nuclide notation is $^{241}_{94}\text{Am}$.

- (i) Determine the number of neutrons in a nucleus of americium-241.

number =[1]

- (ii) A nucleus of americium-241 emits an α -particle and decays to uranium-237.

Complete the nuclear equation for the decay of americium-241.



- (c) Geiger and Marsden studied the structure of gold atoms. Fig. 11.1 shows a version of their apparatus. Alpha-particles strike a thin gold foil.

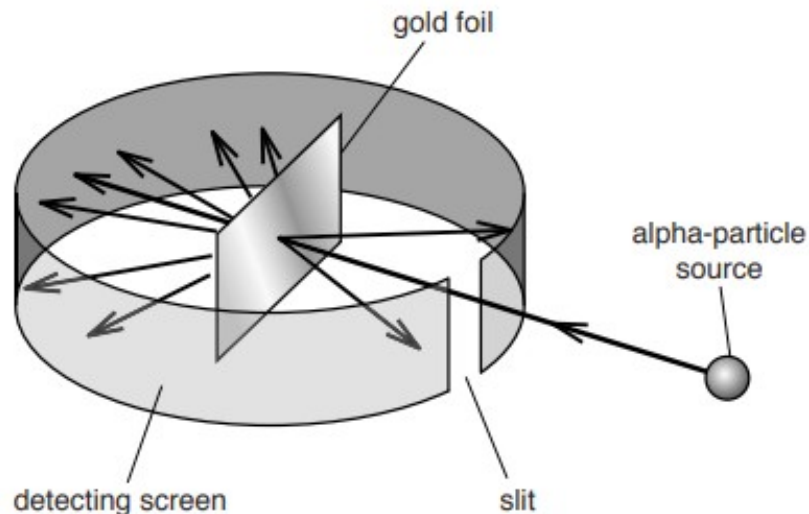


Fig. 11.1

The apparatus shown is in a container from which all the air is removed.

- (i) Suggest why it is necessary to remove all the air from the container.

.....
.....
.....[1]

- (ii) The alpha-particles are emitted from the source at random.

Explain why most of the alpha particles from the source do not reach the gold foil.

.....
.....[1]

- (iii) Fig. 11.2 shows a model of an atom of the gold foil, with its nucleus at the centre.

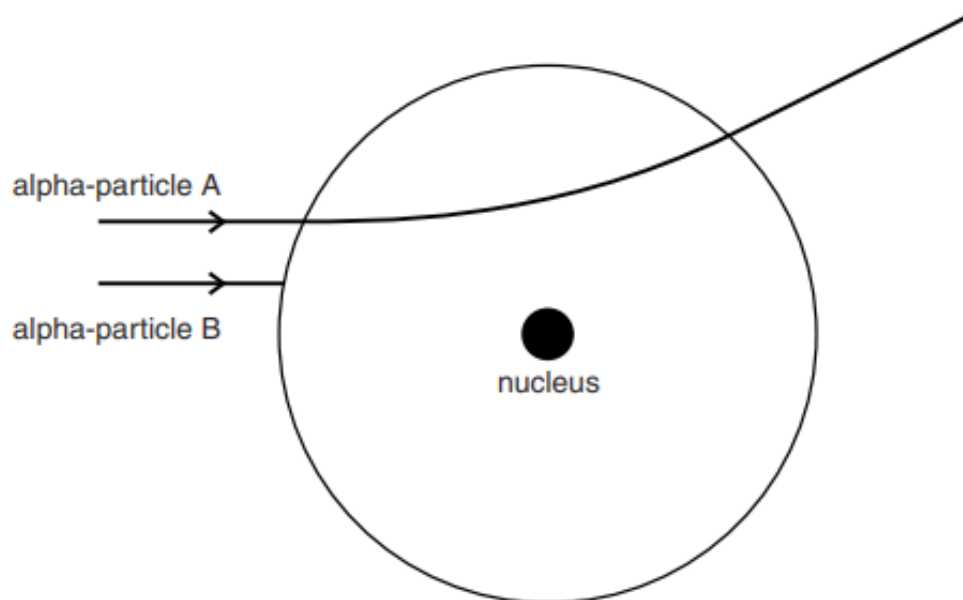


Fig. 11.2 (not to scale)

The alpha-particle labelled A is deflected by the nucleus, as shown.

On Fig. 11.2, complete the path of the alpha-particle labelled B.

[1]

- (iv) Explain how the alpha-particle scattering experiment provides evidence for the existence of a small nucleus inside the atom.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

32. O/N/14/22/Q11

Strontium-90 ($^{90}_{38}\text{Sr}$) is a radioactive isotope. A nucleus of strontium-90 decays by the emission of a beta-particle (β).

(a) Define the term *isotope*.

.....
[2]

(b) For a neutral atom of strontium-90, state

(i) the number of electrons,

number of electrons =[1]

(ii) the number of neutrons.

number of neutrons =[1]

(c) When a nucleus of strontium-90 decays, it becomes a nucleus of yttrium (symbol Y).

Complete the equation for this decay.



(d) A sample initially contains 6.0×10^8 atoms of strontium-90.

(i) The half-life of strontium-90 is 29 years.

Calculate the number of strontium-90 atoms that remain in the sample after 87 years.

number of atoms =[2]

(ii) Describe an experiment that shows that the sample emits only beta-particles.

.....
.....
.....
.....
.....
.....[4]

(iii) A teacher says that the emission of beta-particles from strontium-90 is random.

1. Explain what is meant by *random* in this context.

.....
.....[1]

2. State **two** ways in which the emission of beta-particles from strontium-90 is random.

.....
.....[2]

33. O/N/14/21/Q8

The nuclide notation for the radioactive isotope boron-12 is ${}^{12}_{5}\text{B}$.

- (a) In the space below, draw a labelled diagram to illustrate the structure of a neutral atom of this isotope. Show all the particles in the atom.

[4]

- (b) As boron-12 decays, it emits a beta-particle. A new atom is produced.

Determine

- (i) the proton number (atomic number) of the new atom,

proton number =[1]

- (ii) the nucleon number (mass number) of the new atom.

nucleon number =[1]

34. M/J/14/22/Q8

- (a) An atom consists of electrons surrounding a nucleus made up of protons and neutrons.

State which of these particles

- (i) have an equal and opposite charge,

..... [1]

- (ii) have almost equal mass.

..... [1]

- (b) The nuclide notation for carbon-14 is $^{14}_6\text{C}$. Carbon-14 decays by beta emission to a stable isotope of nitrogen (N).

- (i) Write numbers in the empty boxes below to show the nuclide notation for this isotope of nitrogen.



[2]

- (ii) The half-life of carbon-14 is 5700 years.

A sample of wood from a living tree contains 2.4×10^{12} atoms of carbon-14.

A similar sample of the same size is taken from an old piece of wood. It contains 6.0×10^{11} atoms of carbon-14.

Calculate the age of the old piece of wood.

age = [2]

35. M/J/14/21/Q8

A hospital laboratory uses a small sample of a radioactive isotope of iodine, $^{131}_{53}\text{I}$.

- (a) (i) Describe the structure of the nucleus of an atom of this isotope.

.....
..... [2]

- (ii) The sample is radioactive. Describe what happens in radioactive decay.

.....
.....
..... [2]

- (b) The count in one minute from the source is measured several times. Fig. 8.1 shows the readings obtained.

2686	2759	2847	2799
------	------	------	------

Fig. 8.1

- (i) Suggest why the readings are different.

.....
..... [1]

- (ii) The half-life of $^{131}_{53}\text{I}$ is 8.0 days. Estimate the count in one minute obtained from the sample after 24 days.

count = [2]

36. O/N/13/22/Q8

A smoke detector containing an alpha-particle source is fixed to the ceiling in a room. The alpha-particles ionise the air between two metal plates so that a battery produces a current between the plates.

If smoke enters the detector, it reduces the current between the plates and an alarm sounds.

Fig. 8.1 shows the arrangement.

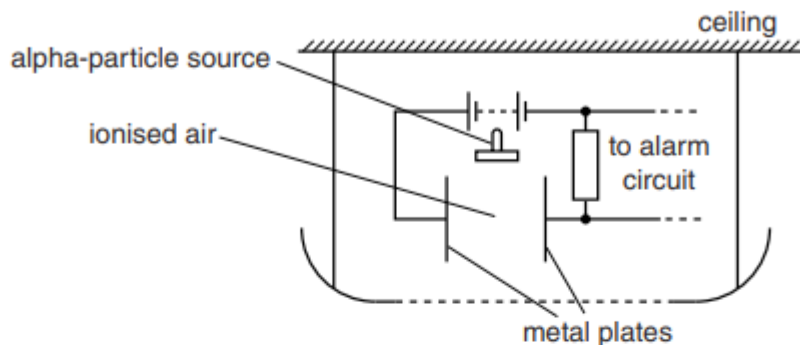


Fig. 8.1

- (a) A technician removes the source from the smoke detector and measures the count rate of the source.

- (i) Suggest two precautions that the technician should take when working with the source.

1.
2.

[2]

- (ii) The technician measures the count rate, in counts/s, five times. On each occasion, the arrangement of the equipment is the same.

The values obtained are: 81 77 80 83 and 79.

Suggest why the values are different.

.....
.....[1]

- (b) Radioactive sources that emit beta-particles or gamma-rays are not used in smoke detectors. State and explain two reasons for this.

1.
.....
2.
.....

[3]

37. O/N/13/21/Q8

A radioactive sample emits only beta-particles and gamma-rays.

- (a) Describe, with the aid of a labelled diagram, an experiment that shows that the sample does **not** emit alpha-particles.

.....

.....

.....

.....

..... [3]

- (b) State two safety precautions that must be taken when working with the radioactive sample.

1.

2. [2]

38. M/J/13/21/Q11

(a) The nucleus of an atom of carbon-14 is represented as $^{14}_6\text{C}$.

(i) Describe the structure of an atom of carbon-14.

.....
.....
..... [3]

(ii) Carbon has several isotopes.

Describe, in terms of the particles within the nucleus, how isotopes are

1. different from one another,

..... [1]

2. the same as one another.

..... [1]

(b) Carbon-14 decays by beta emission to a stable isotope of nitrogen.

Carbon-14 has a half-life of 5700 years and is used to find the age of objects by carbon dating.

(i) A pure sample of carbon-14 initially contains 8.0×10^{20} atoms.

Calculate the number of atoms of carbon and of nitrogen present in the sample after 11 400 years.

number of atoms of carbon =

number of atoms of nitrogen = [3]

(ii) Explain why carbon-14 is not used to find the age of a piece of coal that has been in the ground for millions of years.

.....
..... [1]

(c) A teacher sets up the apparatus shown in Fig. 11.1.

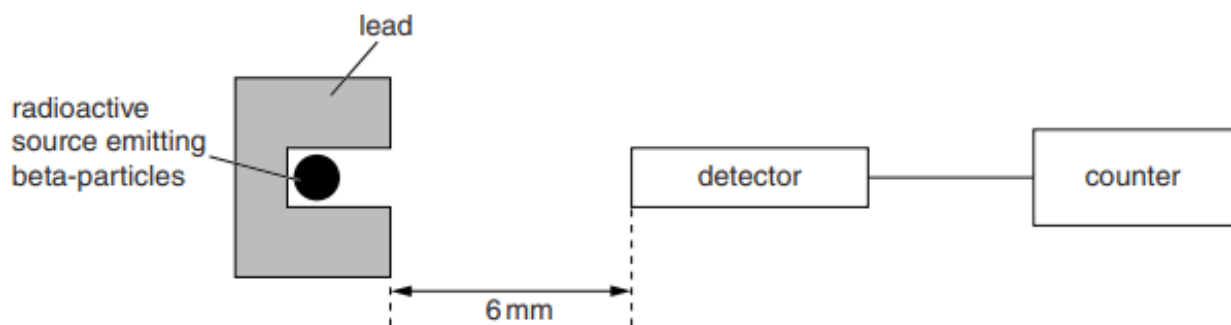


Fig. 11.1

The teacher also has available a number of sheets of aluminium, each of which is 1 mm thick.

- (i) Describe how the apparatus and the sheets are used to demonstrate that beta-particles are stopped by 5 mm of aluminium.

In your account, make clear how the background count is measured and used.

.....

.....

.....

.....

.....

..... [4]

- (ii) Suggest why the lead in Fig. 11.1 is placed around the radioactive source during the demonstration.

.....

..... [1]

- (iii) The radioactive source in Fig. 11.1 is changed for one that emits gamma-rays. Describe why the apparatus cannot be used to find the thickness of aluminium needed to stop gamma-rays.

.....

..... [1]

39. O/N/12/22/Q7

A small piece of metal taken from a nuclear reactor is radioactive.

(a) Describe a method to determine whether the piece of metal emits gamma-rays.

.....

.....

.....

.....

.....

..... [3]

(b) State two safety precautions that must be taken when handling the piece of metal.

1.

.....

2.

.....

[2]

40. M/J/12/22/Q10

To find out whether a radioactive source emits alpha-particles, beta-particles or gamma-rays, the source is placed in front of a radiation detector, as shown in Fig. 10.1.

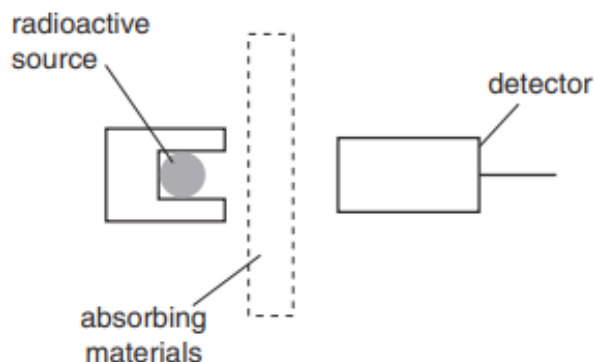


Fig. 10.1

Different absorbing materials are placed between the source and the detector. The detector measures the number of counts per minute.

(a) State what is meant by

(i) an alpha-particle,

.....
.....[2]

(ii) a gamma-ray.

.....
.....[2]

(b) (i) Suggest why, in this experiment,

1. the distance between the source and the detector is only a few centimetres,

.....
.....[2]

2. the half-life of the radioactive source is longer than a few minutes.

.....
.....[1]

(ii) State one precaution taken when using a radioactive source.

.....
.....[1]

(c) Fig. 10.2 shows the results obtained.

source present	material between source and detector	counts per minute
no	none	10
yes	none	1200
yes	thin paper	820
yes	5 mm aluminium	820

Fig. 10.2

Using information from Fig. 10.2, state and explain whether the source emits

(i) alpha-particles,

.....
.....[1]

(ii) beta-particles,

.....
.....[2]

(iii) gamma-rays.

.....
.....[1]

(d) There is a count recorded even when no source is present. This is caused by background radiation.

State two sources of background radiation.

1.
2.
[2]

(e) Describe one effect on the human body of a very high level of radiation.

.....
.....[1]

41. M/J/12/21/Q11

Fig. 11.1 shows part of a smoke detector.

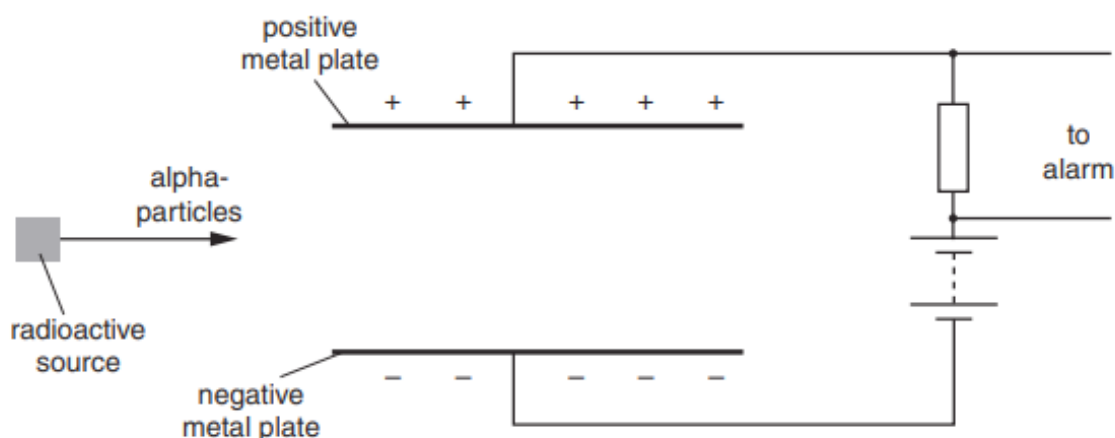


Fig. 11.1 (not to scale)

Alpha-particles from the radioactive source ionise the air between the metal plates. There is then a small current in the resistor. When smoke enters the detector, the current changes and the alarm sounds.

- (a) Explain why a source emitting only gamma-rays is unsuitable for use in the smoke detector.

.....
[1]

- (b) There is an electric field between the metal plates.

- (i) On Fig. 11.1, draw the electric field between the plates. [3]

- (ii) The alpha-particles in Fig. 11.1 are deflected slightly as they pass through the electric field.

Explain why the alpha-particles are deflected and state the direction of the deflection.

.....

[2]

- (c) State and explain how the path of gamma-rays differs from the path of alpha-particles as they pass through an electric field.

.....

[2]

- (d) The source used is the radioactive isotope americium-241, which has a half-life of 500 years.
- (i) A sample of americium-241 is left for 1500 years. Determine the fraction of the original number of atoms of americium-241 left in the sample after this time.

fraction =[2]

- (ii) Another isotope of the same element, americium-242, has a half-life of 16 hours. Explain why the half-life of americium-242 makes this isotope unsuitable for use in a smoke detector.

.....
.....
.....[1]

- (iii) Atoms of americium-241 and americium-242 contain protons, neutrons and electrons.

1. State **two** similarities in the structure of an atom of americium-241 and an atom of americium-242.

.....
.....
.....[2]

2. Describe the difference between an atom of americium-241 and an atom of americium-242.

.....
.....
.....[2]

42. O/N/11/22/Q11

Fig. 11.1 illustrates a neutron hitting a uranium-235 nucleus.

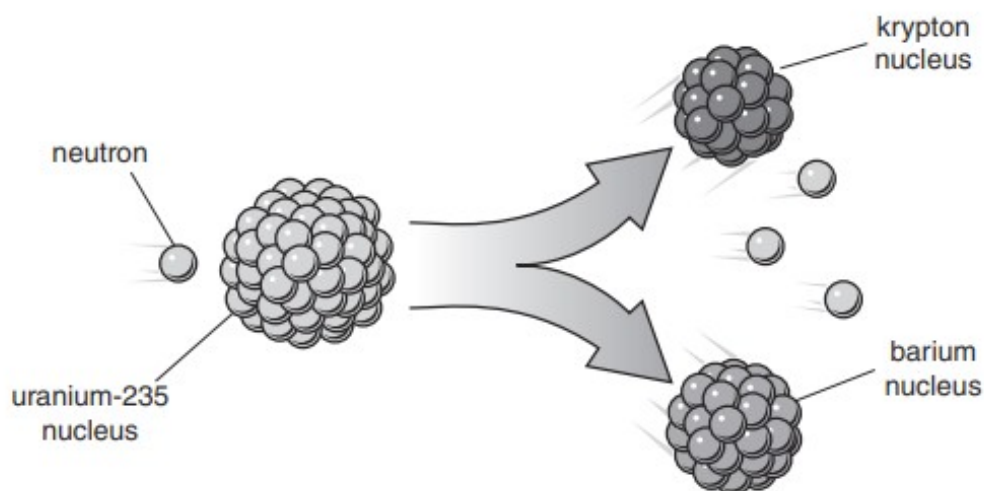


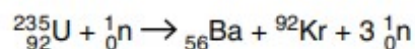
Fig. 11.1

The uranium-235 nucleus splits into a nucleus of barium (Ba) and a nucleus of krypton (Kr) and three neutrons are released.

(a) State the name of this process.

.....[1]

(b) This process may be represented by a nuclear equation. An incomplete version of this equation is shown below.



(i) Calculate

1. the number of neutrons in a nucleus of uranium-235,

number of neutrons =[1]

2. the proton number (atomic number) of krypton,

proton number =[1]

3. the nucleon number (mass number) of the barium nucleus.

nucleon number =[1]

- (ii) During this process, there is a decrease in mass. Energy is released. The decrease in mass is 3.1×10^{-28} kg and the speed of light is 3.0×10^8 m/s.

Calculate the energy released in this process.

energy =[3]

- (c) A nuclear power station generates electrical energy. In the power station, steam is used to drive a turbine. Describe how the splitting of uranium-235 is used to produce the steam. You may include a block diagram.

.....
.....
.....
.....
.....[5]

- (d) Some of the waste products from a nuclear power station are radioactive and have very long half-lives.

- (i) State the meaning of *half-life*.

.....
.....[2]

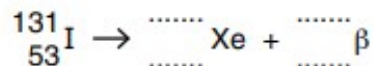
- (ii) Describe one safety precaution that is taken when radioactive waste products are handled.

.....
.....[1]

43. O/N/11/21/Q8

- (a) One radioactive isotope of iodine is $^{131}_{53}\text{I}$. As a nucleus of this isotope decays, it emits a beta-particle (symbol: β) and it becomes a nucleus of an isotope of xenon (symbol: Xe).

Complete the equation in nuclide form, for this decay.



[2]

- (b) A sample of a radioactive isotope emits both beta-particles and gamma-rays.

Fig. 8.1 shows these two types of radiation entering a magnetic field.

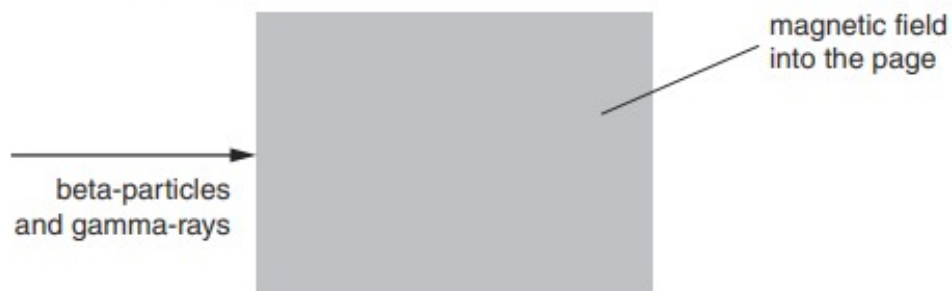


Fig. 8.1

The direction of the magnetic field is into the page.

On Fig. 8.1, mark **and** label the path taken by

- (i) the beta-particles, [1]

- (ii) the gamma-rays. [1]

- (c) Emission from a radioactive source is a random process.

State two ways in which the process is random.

1.

2.

[2]

44. M/J/11/22/Q8

Fig. 8.1 illustrates the process that occurs in the core of a nuclear reactor.

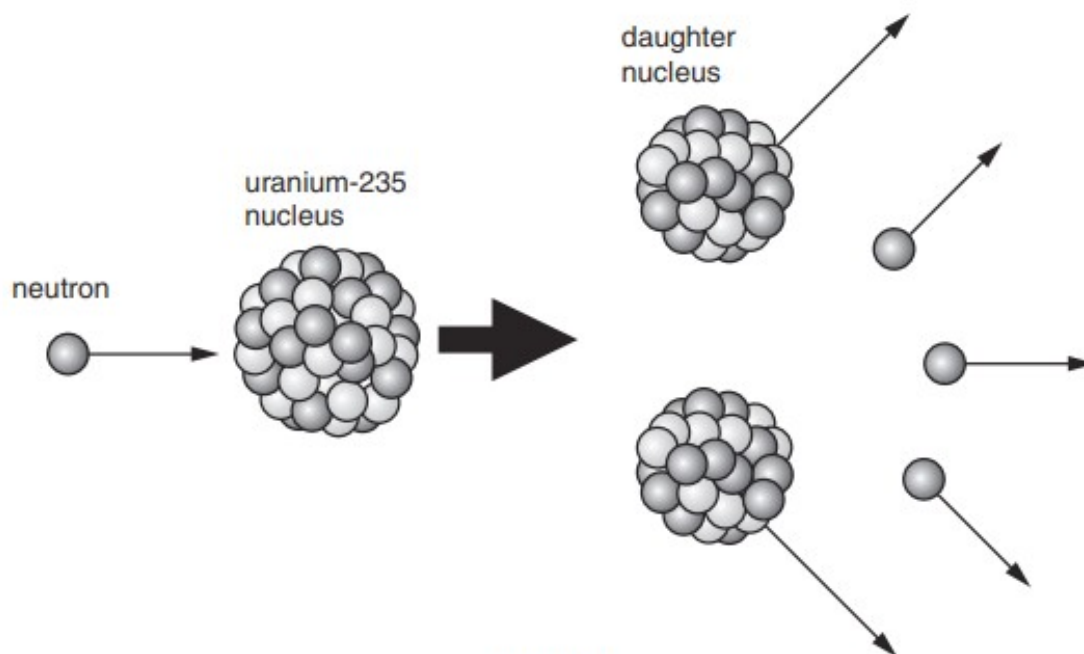


Fig. 8.1

(a) State the name of the process illustrated in Fig. 8.1.

.....[1]

(b) Describe what happens during this process.

.....
.....
.....[2]

(c) Some of the waste from a nuclear reactor is radioactive with a long half-life.

Explain what is meant by

(i) *radioactive*,

.....
.....[1]

(ii) *a long half-life*.

.....
.....[2]

45. M/J/11/21/Q11

An atom of one isotope of sodium contains 11 protons, 13 neutrons and 11 electrons.

- (a) (i) State the nucleon number (mass number) of this isotope.

..... [1]

- (ii) Explain why the number of protons is equal to the number of electrons in a neutral atom.

.....

..... [1]

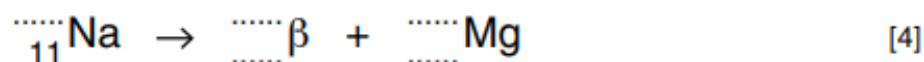
- (b) A nucleus of this isotope undergoes radioactive decay by the emission of a beta-particle.

- (i) State the energy change that occurs during radioactive decay.

.....

..... [2]

- (ii) Complete the nuclear equation that represents this decay by filling in the appropriate numbers in the blank spaces. Na is the chemical symbol for sodium.



- (c) At the start of an experiment, the count rate due to a sample of this isotope of sodium is found to be 1200 counts per minute. This is shown on Fig. 11.1 by a point plotted at time $t = 0$.

The half-life of this isotope is 15 hours.

- (i) On Fig. 11.1, draw a smooth line to show how the count rate is expected to change between $t = 0$ and $t = 30$ hours. [3]

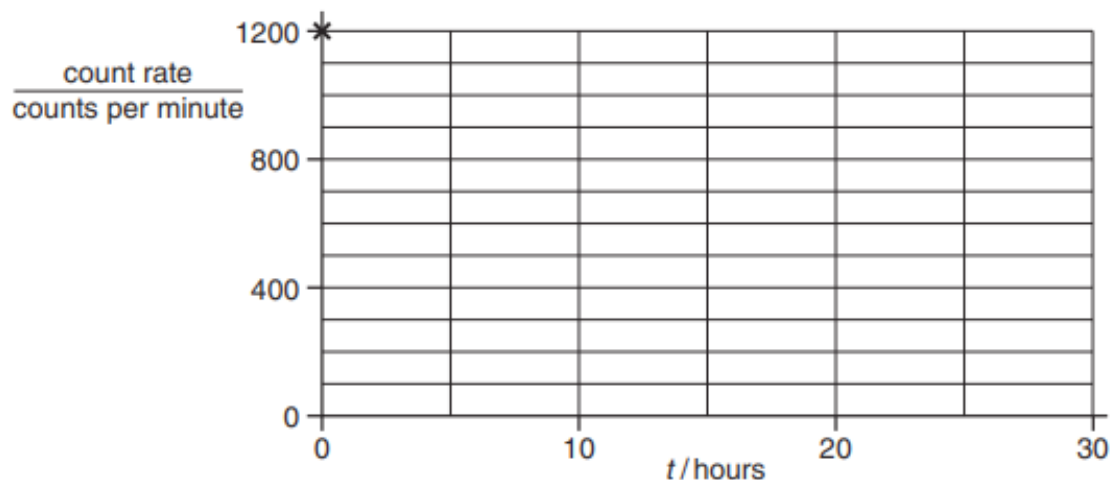


Fig. 11.1

- (ii) Explain why readings taken in an actual experiment may not follow a smooth line.

.....
..... [1]

- (iii) Determine the count rate due to this isotope 30 hours before the start of the experiment.

count rate = [1]

- (iv) In the space below, draw a labelled diagram of the apparatus used to measure the count rate of the radioactive isotope.

[2]

46. O/N/10/21/Q10

An atom of phosphorus (chemical symbol P) is composed of 15 electrons, 17 neutrons and 15 protons.

(a) For this atom, state

(i) the proton number (atomic number), [1]

(ii) the nucleon number (mass number). [1]

(b) This phosphorus atom is radioactive. As it decays to an atom of sulfur (chemical symbol S), it emits a beta-particle (symbol β).

(i) Write a nuclear equation for this decay.

[3]

(ii) Explain what is meant by a *beta-particle*.

.....

..... [2]

(iii) A sample of radioactive material contains many of these phosphorus atoms. Describe an experiment to investigate whether the sample emits only beta-particles. A diagram may be included.

.....

.....

.....

.....

.....

.....

..... [4]

(c) This isotope of phosphorus has a radioactive half-life of 14.3 days.

(i) Explain what is meant by *radioactive half-life*.

.....
.....
..... [2]

(ii) A solution containing such phosphorus atoms is used in a medical procedure. When it is prepared, the solution has an activity of 1400 counts/second. The solution cannot be used when its activity falls below 350 counts/second.

Calculate the maximum time between preparing and using the solution.

time = [2]

47. M/J/10/22/Q8

A radioactive rock emits gamma-rays. A teacher plans an experiment to show a class that the emission of gamma-rays from the rock is random in time.

(a) (i) State the apparatus that the teacher needs to detect the gamma-rays.

.....
.....[1]

(ii) State one safety precaution that the teacher must take.

.....
.....[1]

(iii) Describe how the experiment is performed.

.....
.....
.....
.....
.....
.....[2]

(b) Describe what is meant by a *gamma-ray*.

.....
.....[2]

48. M/J/10/21/Q11

(a) Complete the table of Fig. 11.1.

type of radiation	nature of radiation	charge
alpha-particle	two protons and two neutrons	positive
beta-particle		
gamma-ray		

Fig. 11.1

[4]

(b) Technetium-99 is a radioactive isotope that emits gamma-rays and it has a half-life of 6.0 hours.

In a hospital, a solution containing this isotope is injected into a patient. The solution travels around the body and the isotope attaches to cancer cells. A few hours after the injection, an instrument is used to detect the gamma-rays emitted by the isotope. A high reading on the instrument indicates the position of cancer cells.

(i) Explain what is meant by *half-life*.

.....

[2]

(ii) Explain why an isotope emitting gamma-rays is used rather than an isotope emitting alpha-particles.

.....

[2]

49. O/N/09/Q8

(a) State the number of

(i) electrons in a neutral atom of the isotope ${}^7_3\text{Li}$,

number of electrons = [1]

(ii) nucleons in an atom of the isotope ${}^{208}_{82}\text{Pb}$,

number of nucleons = [1]

(iii) protons in an atom of the isotope ${}^{23}_{11}\text{Na}$.

number of protons = [1]

(b) An atom of the isotope chlorine-35 contains 17 protons.

For an atom of the isotope chlorine-37, state the number of

(i) protons,

number of protons = [1]

(ii) neutrons.

number of neutrons = [1]

50. M/J/09/Q8

The symbol ${}_{92}^{238}\text{U}$ represents a nucleus of uranium with nucleon number (mass number) 238 and proton number (atomic number) 92.

(a) State the meaning of *nucleon number*.

.....
.....
..... [2]

(b) A nucleus of uranium-238 decays to form a nucleus of thorium by the emission of an alpha-particle.

State

(i) the proton number of an alpha-particle,

..... [1]

(ii) the nucleon number of an alpha-particle,

..... [1]

(iii) the proton number of thorium,

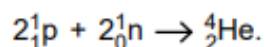
.....

(iv) the nucleon number of the thorium isotope formed.

..... [1]

51. O/N/08/Q8

The reaction that takes place at the centre of the Sun can be represented as



- (a) State the name of this type of reaction.

..... [1]

- (b) The speed of light is $3.0 \times 10^8 \text{ m/s}$.

- (i) The mass of the helium nucleus (${}^4_2\text{He}$) produced is different from the total mass of the two protons and the two neutrons. The difference is $6.6 \times 10^{-29} \text{ kg}$.

Explain why energy is released during the reaction.

.....
.....
..... [2]

- (ii) Calculate the energy released in this reaction. State the formula used.

energy = [3]

52. M/J/07/Q8

(a) Explain how it is possible for an element to have different isotopes.

.....

.....

.....

..... [1]

(b) State what is meant by the *half-life* of a radioactive isotope.

.....

.....

.....

..... [2]

(c) Fig. 8.1 shows how the number of atoms of a radioactive isotope changes with time.

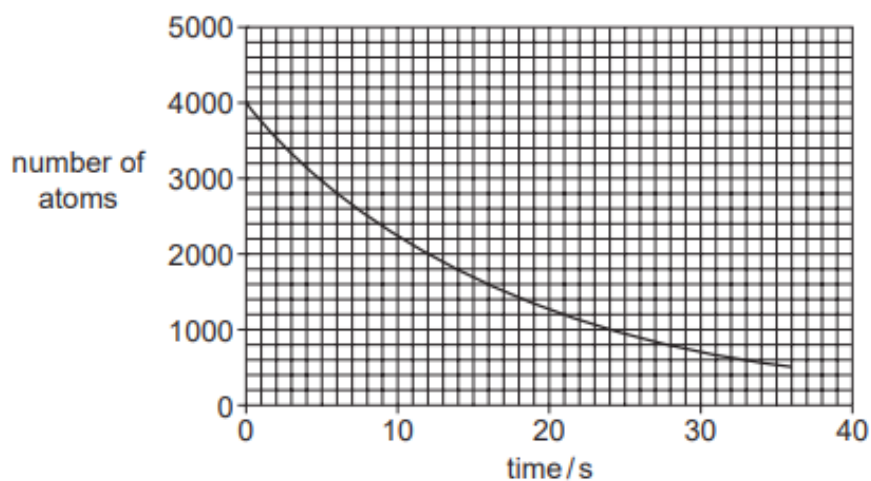


Fig. 8.1

Determine the half-life of the radioactive isotope. On Fig. 8.1, show how you obtained your result.

half-life = [2]

53. M/J/06/Q8

A radioactive source undergoes radioactive decay.

- (a) Explain what is meant by *radioactive decay*.

.....

.....

..... [2]

- (b) Gamma-rays from a radioactive source are used to treat some foods.

The gamma-rays kill bacteria in the food so that it stays fresh.

Some people are worried that food treated in this way becomes radioactive. A scientist tests three food samples. She measures the count-rate of the food before and after treatment with gamma-rays. The results are shown in the table below. The radioactive source is not present during any test.

food sample	count-rate before treatment/ counts per minute	count-rate after treatment/ counts per minute
A	29	32
B	32	28
C	30	31

- (i) Explain why there is a measured count-rate before the food is treated.

.....

.....

.....

..... [1]

- (ii) Determine and explain whether the treated food becomes radioactive.

.....

.....

.....

.....

.....

..... [2]

ANSWERS

1.

Question	Answer	Marks
9(a)	any two from: from <u>gravitational</u> potential energy or <u>gravitational</u> potential energy <u>to</u> or <u>gravitational</u> potential energy decreases kinetic energy mentioned <u>to</u> internal energy or internal energy increases	B2
9(b)(i)	different number of neutrons (in the nucleus)	B1
9(b)(ii)	(its nucleus only contains) only has one proton (alpha-particles contain two protons) or not enough protons	B1
9(b)(iii)	${}^0_{-1}(\beta)$	B1
	${}^3_2(\text{He})$	B1
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	B1
	outward force (due to high temperature / nuclear reaction) or force due to high temperature / nuclear reaction or radiation pressure	B1
	balances (gravitational) force (inwards)	B1
9(c)	(red supergiant explodes as) supernova (explosion)	B1

2.

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ and no other path indicated as γ	B1
10(a)(ii)	beta (β -radiation)	B1
	left-hand rule (mentioned)	B1
	first finger into page and thumb downwards	B1
	current opposite to motion or particles negative (and so are beta)	B1
10(a)(iii)	beta (β -radiation) completely absorbed	B1
	<u>some</u> gamma rays absorbed or (some) gamma passes through	B1
10(b)	$(\lambda =) v/f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	$3.4 \times 10^{-12} \text{ (m)}$	A1

3.

Question	Answer	Marks
8(a)	it / uranium-235 has (3) fewer neutrons (in the nucleus)	B1
8(b)(i)	the nucleus splits (and releases more neutrons)	B1
8(b)(ii)	neutrons released (when nucleus splits)	B1
	neutrons absorbed by other (uranium-235) nuclei	B1
	these nuclei split and release neutrons	B1
	this process continues (using released neutrons)	B1
8(b)(iii)	to slow down the neutrons (emitted by fission)	B1
8(b)(iv)	they / control rods absorb neutrons (without further fission)	B1
	to slow down / stop / control the chain reaction / limit the number of fissions	B1

4.

Question	Answer	Marks
9(a)(i)	alpha	B1
9(a)(ii)	beta	B1
9(b)(i)	beta particles because count rate goes down (by 600) when metal placed in position	B1
9(b)(i)	beta absorbed by metal / cannot pass through metal	B1
	gamma (because) radiation passes through metal / still gives reading with thick metal	B1
9(b)(ii)	alpha stopped by a few cm in air / alpha do not reach G.M. tube or alpha not detected by G.M. tube (end cap stops them)	B1
9(c)	use tongs / handle at distance / place absorber between source and observer / limit exposure time / place in lead box or thick metal box	B1

5.

Question	Answer	Marks
10(a)	emission from the <u>nucleus</u>	B1
	of an alpha particle or beta particle or gamma (radiation)	B1
10(b)	at least three points plotted at (6000, 800), (12 000, 400), (18 000, 200) and (24 000, 100)	C1
	all points correct and smooth curve drawn	A1
10(c)	GM (tube and) counter (and timer)	B1
	measure number of counts in a specified time (e.g. 1 min) or subtract background count or tube connected to ratemeter	B1
10(d)	measure count rate of similar (mass) sample of living / fresh wood	B1
	use the graph to find time or description of use of half-life to find age	B1

6.

Question	Answer				Marks
9(a)		yes	no	it is not possible to tell	B1
	it emits an α -particle		✓		
	it emits a β -particle	✓			
		yes	no	it is not possible to tell	B1
	it emits a γ -ray			✓	
9(b)(i)	(both have) 91 protons / equal numbers of protons (in nucleus)				B1
9(b)(ii)	(they have) different numbers of neutrons (in nucleus) / nucleons				B1

9(c)(i)	(count that is) always present or due to the environment / surroundings or not deliberately introduced or not being tested	B1
	any two from: rock sources (e.g. benches, building materials, human body, Earth crust) air sources (e.g. radon) cosmic sources (e.g. Sun, space)	B2
9(c)(ii)	use tongs / tweezers / forceps or wear lead(-lined) clothing / lead(-lined) gloves or wear goggles or reduce time of exposure or keep distance of separation large	B1
9(c)(iii)	(radioactive) emission is a random process / occurs at random intervals	B1
	measurements fluctuate (about an average value)	B1
9(c)(iv)	(coordinates of) two points from curve, e.g. (0, 58) and (70, 29)	C1
	(coordinates of) two points from curve and at least two half-lives apart, e.g. (0, 58) and (138, 14.5) or $64.0 \leq \tau \leq 74.0$	C1
	$66.0 \text{ s} \leq \tau \leq 73.5 \text{ s}$	A1
9(c)(v)	(large half-life because of) slow rate of decay	B1
	radiation / count rate due to $^{234}_{92}\text{U}$ imperceptible / very much less than that of $^{234}_{91}\text{Pa}$ or (nearly) all of measured radiation / count rate due to $^{234}_{91}\text{Pa}$	B1

7.

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) - two electrons (and ion has 1 electron) - one more electron (than the ion) - no charge (and ion positive) - same number of electrons as protons;	B1
9(a)(ii)	alpha particle has (any 1 from) : - no electrons - charge +2 (e) - only protons and neutrons or He ion has an electron	B1
9(a)(iii)	<i>X number of protons</i> 88	B1
	<i>Y number of protons</i> 89 or number of protons for X +1	B1
	<i>X number of neutrons</i> 137 or 225 – number of protons in X	B1
	<i>Y number of neutrons</i> 136 or number of neutrons for X –1	B1
9(b)	GM tube	cloud chamber
	use paper absorber and / or (thin) metal / >5 cm air or deflect with magnetic / electric field / charged plates	look at trails in cloud chamber <u>when</u> vapour condenses / cools
	with paper, reading / count drops if alpha present or alpha particles stopped by paper / >5 cm air or correct deflection and sensible detection of alpha particles in field	alpha tracks straight / thick / few cm long
	with metal, reading / count drops (further) if beta present or beta particles stopped by metal or correct deflection and sensible deflection of beta particles in field	beta tracks spidery / thin / long
9(c)(i)	time taken to halve	C1
	time for number of (radioactive) atoms / nuclei / count / activity to halve	A1
9(c)(ii)	any halving of 4.0×10^{14} seen	C1
	only 3 halvings or 3 half lives seen	C1
	7300 years	A1

8.

Question	Answer	Marks
9(b)(i)	technetium(–99)	B1
9(b)(ii)	only <u>gamma</u> detected outside body	B1
	EITHER half-life / 6 hours is short and less damage (to others / long term) / loses radioactivity quickly OR half-life / 6 hours is long enough for isotope to reach site and be detected / long enough to examine patient / does not need to be replaced	B1
9(c)(i)	cosmic rays; rocks; buildings; food and drink etc.	B1
9(c)(ii)	weapons tests; nuclear weapons; nuclear power stations; nuclear accidents, e.g. Chernobyl;	B1
9(c)(iii)	cancer / kill cells / harm DNA / RNA / cause mutations	B1
9(d)(i)	superscript 1 and subscript 0	B1
9(d)(ii)	neutron	B1
9(d)(iii)	in the Sun / star or at high temperature	B1
9(e)	alpha-particles more easily stopped (than beta)	B1
	alpha-particles cause more ionisation (than beta)	B1
	deflected in opposite directions (by magnetic field) or alpha-particles deflected less (for same speed)	B1

9.

Question	Answer	Marks
6(a)	any two from: number of protons decreases by two number of neutrons decreases by two or number of nucleons decreases by four	B2
6(b)(i)	background radiation (detected) or source of background radiation mentioned (e.g. cosmic rays; radiation in the air)	B1
6(b)(ii)	alpha-particles are absorbed by the <u>air</u> (between the source and detector)	B1
	all the alpha-particles / radiation are absorbed (by the air between the source and detector) or alpha-particles travel less than 10 cm (in air)	B1
6(c)(i)	arrow through J towards centre of circular path	B1
6(c)(ii)	into the page (i.e. fifth box ticked)	B1
6(c)(iii)	force perpendicular to direction of motion or no component of velocity along line of action of force or work done = force × distance moved in direction of motion	M1
	no work done	A1

10.

Question	Answer	Marks
10(a)	same number of protons (in nucleus) or same proton number or same number of electrons in a neutral atom or same atomic number	B1

10(b)	one more proton	B1
	one fewer neutron	B1
	one more electron	B1
10(c)(i)	18 counts / minute	B1
10(c)(ii)	any two of: rocks / building (materials) / coal / radon space / Sun / cosmic rays weapons testing / nuclear waste	B2
10(c)(iii)	one count rate – 18 or one count rate – 18 calculated e.g. 74 – 18 or 56	B1
	one count rate – 18 halved or one count rate – 18 reduced by factor 2^N e.g. 56 / 2 or 28 or 56 / 4 or 14	B1
	one count rate – 18 reduced by factor $2^N + 18$ e.g. 28 + 18 or 46 or 14 + 18 or 32	B1
	65 hours	B1
10(c)(iv)	(radioactive decay is) random	B1
	in time or impossible to predict when a beta-particle is emitted	B1
10(d)	correct upward curved path between plates	M1
	no more than 1 cm straight at beginning	A1

11.

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case or prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out or <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected or <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1
11(d)	proton number of Y = 39	B1
	'nucleon' number of $\beta = 0$	B1
	nucleon numbers of Y and β add to 90	B1
11(e)(i)	<i>in (nuclear) fusion</i> <u>nuclei</u> come together	B1
	<i>in (nuclear) fission</i> a <u>nucleus</u> splits	B1
	either in (nuclear) fusion <u>small</u> nuclei come together or in (nuclear) fission <u>large</u> nuclei split or fission caused by neutron (hitting nucleus)	B1
11(e)(ii)	high temperature or particles moving at high speed	B1
	high pressure or high density (of particles)	B1

12.

Question	Answer	Marks
10(a)	<i>alpha</i> positive (+2)	B1
	<i>alpha</i> stopped by paper or thin metal or 3–10 cm air	B1
	<i>beta</i> electron	C1
	<i>gamma</i> electromagnetic (wave / particle)	A1
	<i>gamma</i> no charge	C1
	<i>gamma</i> not stopped (easily) or stopped by a few cm of lead or stopped a few metres of concrete	A1
10(b)(i)	(nuclei / atoms with) same number of protons (in nucleus)	B1
	different number of neutrons	B1
10(b)(ii)	ionisation or can penetrate (through air / boxes) or has high energy	B1
10(b)(iii)	stopped by box / (air-tight) bag, instruments or stopped by air	B1
10(b)(iv)	decays too quickly or has to be replaced often	B1
10(c)(i)	(alpha) repelled by nucleus	B1
	alpha and nucleus are positive	B1
10(c)(ii)	most of an atom is empty space / nucleus is only a small part of atom	B1
	alpha particle rarely passes close to nucleus	B1

13.

Question	Answer	Marks
6(a)(i)	16 protons (in nucleus) and 16 electrons in shells or 16 protons (in nucleus) 16 electrons around / outside / surrounding / orbiting nucleus	B1
	15 neutrons (in nucleus)	B1
6(a)(ii)	one more neutron (and same number of protons in nucleus) or 15 neutrons in P-31 and 16 in P-32	B1
6(b)(i)	one more proton and one neutron fewer c.a.o.	B1
6(b)(ii)	point at / line through (2.0, 1.6×10^{11})	B1
	point at (8.0, 3.0×10^{11}) or any other obvious correct point	B1
	curve of positive and decreasing gradient and through their plotted points to at least 6.0 weeks	B1
6(c)	any two from: lead shielding (e.g. lead apron / gloves / container / (lead) glass screen / (lead glass) goggles / (lead glass) glasses large distance (e.g. tongs / forceps / tweezers / manipulator / carry at centre of large box) minimum time of exposure or wear film badge	B2

14.

Question	Answer	Marks
7(a)(i)	T and Q or U and R	B1
7(a)(ii)	different number of neutrons (in nucleus)	B1
7(b)(i)	time when decay occurs / particle emitted is not known / not constant or direction in which particle is emitted is not known / can be at any angle / direction	B1

7(b)(ii)	$^{218}_{84}\text{Po}$	B1
	$^4_2\alpha$ He or	B1
7(c)(i)	electron or beta particle	B1
7(c)(ii)	(R gains or S has) one more proton	B1
	(R loses or S has) one less neutron	B1

15.

Question	Answer	Marks
7(a)	emission of (at least one of) alpha, beta and gamma (particles / rays)	B1
	from the nucleus or at random	B1
7(b)	carbon-12 6, 6	B1
	carbon-14 6, 8, 6	B1
7(c)(i)	<u>beta</u> (particles)	B1
7(c)(ii)	(does not emit) gamma	B1
	gamma passes through thick container / 6 mm (aluminium)	B1

16.

Question	Answer	Marks
10(a)	similarity: (same) number of protons	B1
	difference: (different) number of neutrons	B1
10(b)(i)	(nucleon number =) $222 - 4$ or 218 or $218 - 84$	C1
	134	A1
10(b)(ii)	(proton number =) $84 + 2$	C1
	86	A1
10(b)(iii)	it / helium atom has two electrons (in orbit around the nucleus)	B1
10(c)(i)	2.1×10^{10} (alpha-particles)	B1
10(c)(ii)	calculated value of: 2.8×10^{10} – candidate's 10(c)(i) (atoms remain) (expected value: 7.0×10^9)	B1
10(c)(iii)	$7.0 \times 10^9 / 2.8 \times 10^{10}$ or 1 / 4	C1
	2 (half-lives) or 7.6 / 2	C1
	3.8 days	A1
10(d)	alpha-particles are (very) weakly penetrating	B1
10(d)	strongly ionising	B1
	(ionisation causes cell) mutations / cancers	B1

17.

Question	Answer	Marks
7(a)(i)	(atomic particle) Q	B1
7(a)(ii)	$+1.6 \times 10^{-19} \text{ C}$	B1
7(b)	(atomic particle) Q and (atomic particle) S	B1
7(c)(i)	(atomic particle) R	B1
7(c)(ii)	(atomic particle) S	B1
7(c)(iii)	beta-(particle) emission	B1
7(c)(iv)	99 / 33 or 3 half-lives	C1
	1 / 2 ³ or 1 / 8 (remain)	C1
	7 / 8	A1

18.

Question	Answer	Marks
11(a)(i)	gravity or gravitational attraction	B1
11(a)(ii)	Any 2 from: - high pressure (very) high temperature - high particle density	B1 B1
11(b)(i)	proton and 92	B1
	neutron and 143	B1
11(b)(ii)	same number of protons	B1
11(b)(iii)	neutron hits / absorbed by uranium / nucleus	B1
	<u>nucleus</u> splits or smaller <u>nuclei</u> formed and emits more neutrons	B1
11(b)(iv)	fusion – coming together and fission splitting up or fission involves large particles / nuclei and fusion small nuclei or in fusion nuclear waste / products are less radioactive / dangerous or no / few neutrons produced in fusion	B1
11(b)(v)	90 proton number of Th	B1
	231 and 4 nucleon numbers	B1
11(c)(i)	P marked where beam of alpha-particles hits screen	B1
11(c)(ii)	fewer flashes (as one moves from centre) or very few / only some flashes on same side as source / on left	B1
11(c)(iii)	Any 2 bullet points from <u>nucleus</u> is small or atom (mostly empty) space - most mass of atom in nucleus or mass is concentrated in small point / region or nucleus is dense or nucleus is massive - charge concentrated in small region or (positive) charge found in nucleus or nucleus has all the (positive) charge of the atom	B1 B1

19.

Question	Answer		Marks
11(a)(i)	Geiger Muller (tube / counter / detector)		B1
11(a)(ii)	keep distance (away from source) or use absorber or use for short time		B1
11(b)(i)	time between emissions varies / unpredictable or different readings (when repeated for same time)		B1
11(b)(ii)	repeated count for a specified time	measure time when count / event occurs	B1
	result not the same	not always the same	B1
11(c)(i)	radiation when there is no source or naturally occurring (radiation) or always present		B1
11(c)(ii)	<u>measure</u> count (rate) with <u>no source</u>		B1
	subtract from (measured) count (rate for same time)		B1
11(d)(i)	4.5 cm		B1
11(d)(ii)	alpha		M1
	range of alpha less than 5–7 cm (in air)		A1
11(d)(iii)	air atoms / molecules lose / gain electrons (to become ions)		B1
11(e)	points plotted at (0, 300) and (40, 150)		B1
	points plotted at (80, 75) and (120, 37.5)		B1
	smooth curve joining at least 3 correct points		B1

20.

Question	Answer	Marks
6(a)	32 counts / minute	B1
6(b)(i)	second (beta-particles) and third (gamma-rays) boxes ticked	B1
6(b)(ii)	1 28	B1
	2 32	B1
6(c)(i)	(average) time (taken for)	M1
	count rate / number of nuclei / number of atoms to halve	A1
6(c)(ii)	some readings are bigger than those before / readings fluctuate	B1
	half-life / 5.3 years too long or 5 / 6 minutes too short	B1

21.

Question	Answer	Marks
10(a)(i)	same number of protons or same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^4_2\alpha$	B1
	${}^{238}_{92}\text{U}$	B1
	${}^{235}_{92}\text{U}$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ or 5 (half-lives)	C1
	relevant halving or $1/2^5$ or $1/32$ or $6400/32$	C1
	200 counts / second	A1
10(b)(iii)	1 curving (only) downwards with correct curvature	M1
	no straight section (by eye) anywhere in magnetic field	A1
	2 (Fleming's) left-hand rule or motor rule or rule described or moving charge is a current	B1
10(c)(i)	splitting of a <u>nucleus</u>	B1
10(c)(ii)	neutron fired at a plutonium nucleus or plutonium nucleus absorbs neutron	B1
10(c)(iii)	1 no CO_2 / SO_2 / nitrogen oxides / greenhouse gases emitted or do not contribute to global warming / acid rain	B1
	2 reliable / not intermittent or smaller land area	B1

22.

Question	Answer	Marks
11(a)(i)	detector (need not be named) and absorber or ruler (to measure distance in air) OR cloud chamber (need not be named) and source inside	B1
	suitable detector named or labelled, e.g. GM detector, Geiger counter, cloud chamber, photographic film/paper, spark counter or counter / ratemeter connected to labelled detector	B1
11(a)(ii)	count with source (alone) or listen to clicks / sound	B1
	count with source or listen <u>to clicks</u> and suitable, named absorber or apply magnetic / electric field at right angles to beam	B1
11(a)(iii)	count decreases when absorber used or no / background radiation detected or particles found to be deflected in correct direction for alpha	B1
	(only) alpha-particles absorbed / stopped (by air / paper) or deflection is correct for alpha-particles	B1
11(b)	ionisation (in air) alpha (>) beta (>) gamma	B1
	penetration gamma (>) beta (>) alpha	B1
11(c)(i)	(detector) works for longer or does not need replacing (as often)	B1
11(c)(ii)	any halving seen	C1
	3 half-lives or three halvings seen	C1
	1290 or 1300 years	A1
11(d)(i)	kills / damages cells or causes cancer or causes mutations	B1
11(d)(ii)	ANY 2 from (source) further away (from the body) - less radiation / fewer particles pass through / reach body / hand - alpha particles stopped / reduced by air - particles spread out from source	B2

23.

Question	Answer	Marks
8(a)(i)	2 protons and 2 neutrons (joined together)	B1
8(a)(ii)	it is (positively) charged	B1
	it pulls electrons from molecules of air or knocks electrons from the molecules of air or gains electrons from air molecules	B1
8(a)(iii)	1 stronger and 2 stronger	B1
8(b)(i)	idea of halving	C1
	$(N =)4.8 \times 10^7 / 1.5 \times 10^6$ or 1 / 32 or 5 (half-lives)	C1
	$1.6 / 1.65 / 1.7 \times 10^6$ s	A1
8(b)(ii)	radioactive emission is a random process	B1

24.

Question	Answer	Marks
11(a)(i)	rocks / soil / Earth's surface / building materials / radon (gas) / waste from a nuclear power station / weapons testing	B1
11(a)(ii)	cancer / mutation / cell damage / gene damage or adds to / affects experimental readings / count rate or causes ionisation	B1
11(a)(iii)	protons 2	B1
	neutrons 2	B1
11(a)(iv)	alpha-particles absorbed / stopped by / cannot penetrate air / atmosphere or scattered (by air) or cause ionisation (and slow down)	B1
11(a)(v)	continuous curve deflected either clockwise or anticlockwise within shaded area	B1
	arrow or other indication to show anticlockwise deviation within shaded area	B1
11(b)(i)	time taken to halve	C1
	time taken for the activity / count (rate) / number of atoms / number of nuclei to halve	A1
11(b)(ii)	any halving seen, e.g. 200–100 or 3 half lives	B1
	17 100 years	B1
11(b)(iii)	too little carbon-14 left or all decayed or shows large reduction	B1
11(c)(i)	different number of neutrons or different mass or different nucleon number	C1
	carbon-14 has two more neutrons	A1
11(c)(ii)	same number of protons	B1

25.

Question	Answer	Marks
8(a)	(high frequency) electromagnetic wave / ray	B1
8(b)	top line of beta 0	B1
	Xe proton and neutron numbers both balance	B1
8(c)	gamma rays pass out of / are not stopped by body	B1
	less chance of cell or gene damage / cancer / radiation sickness / mutation / ionisation (of body tissues)	B1

26.

- (a) 94 electrons **and** 94 protons B1
 144 neutrons B1
 (only) electrons in orbit / surrounding nucleus **or** (only) protons and neutrons in nucleus B1
- (b) (i) (beta-particles) weak(er) B1
 (beta-particles) strong(er) B1
- (ii) any **two** lines from
 glasses / goggles **or** lead container / shield / clothing / gloves
 tweezers / manipulator / carry in large cardboard box
 minimise time of exposure / film badge B2

27.

- | | | |
|---------|---|----|
| (a) (i) | 11 protons and 11 electrons | B1 |
| | 13 neutrons | B1 |
| | electrons in orbit/surrounding nucleus or neutrons and protons in nucleus | B1 |
| (ii) | one more neutron (in sodium-24) or one fewer neutron in sodium-23 | B1 |
| | | |
| (b) (i) | electron | B1 |
| (ii) | ${}^0_{-1}(\beta)$ cao | B1 |
| | ${}^{24}_{12}(\text{Mg})$ | B1 |
| | ${}^{24}_{12}(\text{Mg})$ | B1 |
| | | |
| (c) | electromagnetic (radiation / rays / waves) | M1 |
| | (very) high frequency / energy or (very) short wavelength | A1 |
| | | |
| (d) (i) | path curving upwards | B1 |
| (ii) | path continues in straight line | B1 |
| (iii) | beta-particle charged or gamma-ray uncharged | B1 |
| | | |
| (e) | long enough or short enough | B1 |
| | to take measurements so the body is not irradiated for long | B1 |

28.

(a) (nucleus / nuclide / atom) with same number of protons

(b) (i) 2

(ii) neutron

(iii) 2

(iv) 4

(c) nuclei repel

or like / positive charges repel

(needs) high kinetic energy / speed (to overcome repulsion)

(d) ANY 3 lines from

(dust / gas) collapses / comes together / clusters / condenses

gravitational attraction or gravity mentioned

B1

temperature rises or KE (dust / gas) increases

B1

(nuclear) fusion occurs

B1

equilibrium established as radiation pressure / outward

force balances inward force

B1

(e) (i) time for a quantity to halve

time for (radio)activity / count rate / number of atoms / number of nuclei to halve

(ii) any relevant halving seen, e.g. 16 000 / 2
1000

29.

(a) 127 n(eutrons) **and** 82 p(rotons) B1
 82 e(lectrons) B1
 electrons in orbit around nucleus/in shells around nucleus **or** around
 neutrons and protons **or** neutrons and protons in nucleus B1

(b) (i) more protons **and** fewer neutrons **or** one more proton C1
 one more proton **and** one fewer neutron **or** neutron becomes proton
 (and electron/beta-particle) A1

(ii)

		
gamma / γ	beta / β	alpha / α

reversed order 1 / 2

correct order 2 / 2

B2

(iii) (circular by eye) curve from beginning of field **not** with straight line
 initially B1
 upward curve (in field) B1

(c) (i) (radiation) that is always present **or** occurs everywhere **or** cannot be
 eliminated **or** from environment / surroundings / natural sources / air B1

(ii) two separate sources: rocks (e.g. radon / ground), outer space
 (e.g. cosmic rays), man-made sources (e.g. nuclear waste / fall-out) B2

(iii) 2 half-lives (implied) **or** $\frac{1}{4}$ seen **or** 76 (counts / minute) C1
 19 **or** 23 (counts / minute) C1
 35 counts / minute A1

30.		
(a)	(i) (atoms) 3 and 4	B1
	(ii) (atoms) 3 and 5	B1
	(iii) (atoms) 3 and 4	B1
(b)	(i) 17	B1
	(ii) 35	B1
(c)	(i) two separate sources: rocks (e.g. radon), outer space (e.g. cosmic rays), man-made sources (e.g. nuclear waste/fallout)	B2
	(ii) 22 counts/minute	B1
	(iii) 27 counts/minute	B1
	(iv) use of 27/2 or 27/4 or 27/8 from 85 to 90 days (inclusive)	B1 B1
(d)	(i) (background count-rate is) reduced not to zero or not stopped or (some) gamma-rays in background count	B1 B1
	(ii) not sensible all the beta-radiation would be absorbed or no beta-radiation reaches the detector	M1 A1

31.

- | | | |
|---------|--|----------------|
| (a) (i) | 51 | B1 |
| (ii) | more protons than electrons
or different number of protons and electrons
positive and negative do not cancel | B1
B1 |
| (iii) | 25 protons
a different number of neutrons | B1
B1 |
| (b) (i) | 147 | B1 |
| (ii) | α has mass number 4
α has proton number 2
correct proton number for U ecf their value for α | B1
B1
B1 |
| (c) (i) | alpha particles only travel a short distance in air
or alpha particles stopped / scattered / deflected by air
or alpha particles ionise air | B1 |
| (ii) | particles come off in different directions
or not emitted in one line / as a ray
or not all the particles pass through the slit | B1 |
| (iii) | B correct shape and deflected more than A | B1 |
| (iv) | particles close to / fired at the nucleus are deflected (back) / repelled
some particles pass (straight) through
a few particles come back / large deflection or most pass (straight) through
(with little deviation)
and how this explains the nucleus is small | B1
B1
B1 |

32.

- (a) same element **or** same number of protons / atomic number B1
different / particular number of neutrons **or** nucleons B1

- (b) (i) 38 **cao** B1

- (ii) 52 **cao** B1

- (c) $^{90}_{39}\text{(Y)}$ **or** $^{90}\text{(Y)}$ and $^0_{-1}\text{(}\beta\text{)}$ B1

- $^0_{-1}\text{(}\beta\text{)}$ $^{90}_{39}\text{(Y)}$ and $^0_{-1}\text{(}\beta\text{)}$ B1

- (d) (i) 87/29 **or** 3 (half-lives) **or** $6.0 \times 10^8/8$ C1
 7.5×10^7 A1

- (ii) any detector B1
corresponding detection method B1

detector	film	(solid-state) detector	GM-tube	ionisation chamber	scintillation counter	cloud chamber
detection	fogged	count/reading	count/reading	count/reading	count/reading	track seen

- no reduction with paper **or** (use of) electric/magnetic field **or** describe pattern of track M1

- complete reduction with aluminium/lead **or** correct deflection of track in electric/magnetic field **or** no other track A1

- (iii) 1. unpredictable **or** not be known in advance **or** no set time between emissions **or** fluctuates **or** not fixed **or** counts obtained varies B1

2. any **two** from:
direction / in space
time
which nucleus decays

B2

- 33.**
- (a) neutrons and protons together and alone in the middle B1
 5 protons B1
 7 neutrons (if protons and neutrons unlabelled 1/2) B1
 5 electrons **and** electrons surrounding nucleus B1
- (b) (i) 6 B1
 (ii) 12 B1
- 34.**
- (a) (i) electron(s) and proton(s) [B1]
 (ii) neutron(s) and proton(s) [B1]
- (b) (i) top box 14 [B1]
 bottom box 7 [B1]
- (c) (i) sensible halving seen, e.g. 2.4 → 1.2 **or** two halves clear **or** $\frac{1}{2} \times \frac{1}{2}$ seen [C1]
 11 400 **or** 11 000 years [A1]
- 35.**
- (a) (i) 53 protons **not** if also 53 electrons in nucleus [B1]
 78 neutrons **or** 131 protons and neutrons [B1]
- (ii) emission of at least one of alpha particle, beta particle or gamma ray [B1]
 emission from the nucleus **or** breakdown of nucleus [B1]
- (b) (i) random emission indicated [B1]
- (ii) average 2772 **or** 2773 **or** 2770 **or** 2800 **or** 1/8 used **or** 3 clear halvings seen [C1]
or $(\frac{1}{2})^3$ seen **not** halving of 131 or 53 [A1]
 value between 330 and 360

36.

- (a) (i) any two of:
 minimise time of exposure
 lead clothing (e.g. lead gloves **not** radioactive suit)
 tongs, manipulator, forceps, tweezers
 behind protective/lead glass/shield
 wear film badge B2
- (ii) (radioactive emission) random/unpredictable (process)
 (e.g. background radiation is random; **ignore** spontaneous) B1
- (b) penetration strong(er) **or** penetrates casing (accept α **or** β **or** both;
ignore larger range) B1
 (more) weakly/slowly ionising B1
 either explained: harms health **or** hazardous **or** dangerous
or air is not ionised **or** sounds all the time (accept doesn't work) B1

37.

- | (a) | Penetration | Magnetic/electric field | Cloud chamber | Spark counter | |
|-----|---|---|--|--|----|
| | diagram:
sample,
detector, small
gap | diagram:
sample, detector,
magnet | diagram:
sample, cloud
chamber | diagram:
sample, spark
counter, small
gap
labelled or clear | B1 |
| | (insert/remove)
(a sheet of)
paper/card/Al
foil (in gap) | (insert/remove)
magnet | sample in
cloud
chamber | sample near
to counter | B1 |
| | no change in
count | increased count in
correct direction | no short,
straight,
dense tracks | no sparks | B1 |
- (b) any **two** of:
 minimise time of exposure
 lead clothing (e.g. lead gloves **not** radioactive suit)
 forceps, tweezers, tongs, manipulator
 behind protective glass/shield
 wear film badge B2

- 38.
- (a) (i) 6 protons B1
 8 neutrons B1
 6 electrons outside nucleus **or** 6 electrons and protons & neutrons inside nucleus B1
- (ii) different number of neutrons B1
 same number of protons (ignore electrons) B1
- (b) (i) 2 half lives seen e.g. $8 \rightarrow 4 \rightarrow 2$ C1
 number of carbon atoms 2×10^{20} A1
 number of nitrogen atoms 6×10^{20} **or** $(8 \times 10^{20} - N_C)$ atoms B1
- (ii) many half lives **or** has decayed (too much) **or** very few atoms (of C) left B1
- (c) (i) background count/rate taken without source B1
 any count taken over any measured time e.g. 1 minute
or any rate determined (**allow** read ratemeter) B1
 take count/rate with aluminium **between source and detector** B1
 at 5 mm count/rate goes to background/constant/zero **when** corrected for background B1
- (ii) for protection (of the class/teacher e.g. to avoid cancer) **or** gamma-rays cannot pass through **or** to stop particles (hitting class/teacher) (**ign.** just "for safety") B1
- (iii) gamma-rays not stopped by/pass through (5–10 mm) aluminium B1

- 39.
- (a) solid-state detector/GM tube/ionisation chamber/scintillation counter/spark counter/spinthaloscope B1
 count **or** count-rate **or** reading referred to B1
 (some) detection with appropriate blocking in the way **or** same reading/track in electric/magnetic field B1
OR
 film B1
 develop B1
 (some) detection with appropriate blocking in the way **or** same reading/track in electric/magnetic field B1
OR
 (diffusion) cloud chamber B1
 track **seen/looked for/formed** B1
 pattern of track described B1
- (b) any **two** lines:
- one **distance** method: tongs/robotic arm/carry in large box
- one **protection** method: lead suit/lead gloves/lead boxes/shield
- one **time** method: reduced time/wear badge B2
- 40.
- (a) (i) two protons
 OR has charge +2(e)
 OR helium nucleus OR He nucleus B1
- (and) two neutrons
 OR has mass 4 (u)
 OR symbol ${}^4_2\text{He}$ B1
- (ii) electromagnetic (particle/wave) M1
 high frequency/high energy/low wavelength A1
- (b) (i) 1. alpha identified (as the reason) B1
- (alpha) particles **stopped/blocked/absorbed** (few cm air)
 OR distance covered by/range of (alpha) particles (in air) is small/a few cm B1
2. experiment takes time in some way
 OR otherwise count falls (during half life) B1

- (ii) keep distance (e.g. use forceps/tongs, do not point at person/eyes)
OR use absorber (e.g. lead covering)
OR place in store when not in use; use for short time
OR wear badge B1
- (c) (i) YES (alpha particles present)
and count falls **with paper** in some way B1
- (ii) NO (beta particles) M1
when (5 mm) Al used **and**
no **further/more/extra** reduction OR no difference A1
- (iii) YES (gammas present)
and gammas pass through (5 mm) Al **or** 820 after Al B1
- (d) ANY 2 lines
cosmic rays; the Sun; outer space
rocks (e.g. granite); stones; soil; buildings; food
radon/thoron/carbon-14 (gas)
weapons tests; nuclear bombs
leaks from (nuclear) power stations
nuclear waste B2
- (e) cancer (accept any specific cancer); tumours
radiation sickness; burns; mutations;
genetic problems; damage to DNA/chromosomes
cell damage (e.g. kills cells, cures cancer); birth defects
sterility; hair loss B1

41.

- (a) (gamma) produces little or no ionisation; passes out of detector; requires shielding; reaches people B1
- (b) (i) at least 3 lines between plates and in middle at least one **straight, vertical** line B1
 correct curvature at edges B1
 at least one arrow down and no arrows wrong B1
- (ii) alpha charged (positively); alpha repelled by positive/attracted by negative B1
 deflected down/towards positive (plate) B1
- (c) (gamma) undeviated; straight line B1
 (gamma) uncharged B1
- (d) (i) any attempt at halving or 3 **half-lives** seen C1
 $\frac{1}{8}$; 0.125; 12.5% A1
- (ii) decays too fast; have to replace source often; current falls too quickly; detector only works for a short time B1
- (iii) 1. any TWO of:
 number of protons
 number of electrons
 charge on nucleus B2
2. number of neutrons B1
 nucleon or mass number or mass B1
 (americium-242 has **one** extra neutron gets 2 marks)

42.

(a) (nuclear) fission B1

(b) (i) 1...143 B1

2...36 B1

3...141 B1

(ii) $(E =) mc^2$ C1

$3.1 \times 10^{-28} \times (3.0 \times 10^8)^2$ or $3.1 \times 10^{-28} \times 3.0 \times 10^8$ and $(E =) mc^2$ C1

$2.8(2.79) \times 10^{-11} \text{ J}$ A1

(c) any five of:



(one mark for three correct boxes)

(splitting produces) kinetic energy of neutrons

further splitting/chain reaction

energy/heat produced/from reactor/reaction or from neutrons

coolant gets hot

energy to boiler/water or water heated or heat in water implied

water boiled or steam produced B5

(d) (i) time for something to halve C1

time for (radio)activity/count rate/number of atoms/nuclei to halve A1

(ii) one appropriate precaution:

short exposure time

safety/protective suit/gloves/clothes or lead boxes

large distance/(long handled) tool/forceps/tongs

robotic/mechanical handling

film badge B1

- 43.
- | | | | | |
|-----|------------------------|----|---|----|
| (a) | $^{131}_{54}\text{Xe}$ | OR | ^{131}Xe and $^0_0\beta$ | B1 |
| | $^0_{-1}\beta$ | | $^{131}_{54}\text{Xe}$ and $^0_{-1}\beta$ | B1 |
- (b) (i) downward curve B1
- (ii) horizontal line B1
- (c) any **two** of:
 direction/space (of emission)
 time/frequency (of emission) **or** period/interval between emissions **or** different counts (in same time)
 nucleus that decays is unpredictable B2
- 44.
- (a) fission cao B1
- (b) neutron hits/goes inside (U) **nucleus** B1
 atom/nucleus/particle/uranium/nuclide splits/forms daughter nuclei **and** emits neutrons/energy B1
- (c) (i) emits particles // emits **ionising/nuclear** radiation // spontaneous or random emission (of radiation) // **atom/nucleus** decays B1
- (ii) long time to decay // radioactive for a long time // decays slowly B1
 long time for any quantity to halve
 halving of:
 count, count rate, emissions, (number of) nuclei, (number of) atoms, activity B1

45.		
(a)	(i) 24	B1
	(ii) charges (of electrons and protons) cancel // protons and electrons have the same (size of) charge but opposite in sign	B1
(b)	(i) nuclear at start // nucleus loses energy to kinetic/e.m. energy (condone light/photon/ γ)	B1 B1
	(ii) beta 0 at top	B1
	beta -1 at bottom	B1
	Mg 24 at top	B1
	Mg 12 at bottom	B1
(c)	(i) 600 at 15 hours	B1
	300 at 30 hours	B1
	line of decreasing gradient (not if it cuts time axis before $t = 30$)	B1
	(ii) emission is random // not predictable // not regular // exptl. error	B1
	(iii) 4800 (counts per minute) // 80 counts /sec	B1
	(iv) GM tube/solid state detector/cloud chamber	M1
	ratemeter/data logger // counter/scalar and stopwatch/timer	A1

46.		
(a) (i)	15	B1
(ii)	32	B1
(b) (i)	$^{32}_{15}\text{P} (\rightarrow)$	B1
	superscripts: 32 on S and 0 on beta (allow e)	B1
	subscripts: 16 on S and -1 on beta (allow e) (just $^{32}_{16}\text{S } 1/2$)	B1
(ii)	electron	M1
	high speed or from nucleus or causes ionisation	A1
(iii)	record/measure background reading/count/radiation	B1
	sample near named detector	B1
	interpose paper/card/less than 5 cm air and no change in reading	B1
	interpose 2 mm – 20 mm of aluminium and reading = background	
	n.b. points may be made on a diagram, other methods marked analogously	B1
(c) (i)	time for some measurable quantity to halve	M1
	number of atoms/no. of nuclei/activity/count rate	A1
(ii)	$350 \div 1400$ or $\frac{1}{4}$ or 2 (half-lives)	C1
	28.6 days	A1
47.		
(a) (i)	Geiger Muller/GM tube or any other gamma detector	B1
(ii)	keep distance e.g. forceps/tongs	
	barrier accept gloves, lead suit, metal container	
	time of use accept use badge ANY ONE	B1
(iii)	(with source) take a count for a time	
	or take count rate (from a ratemeter/meter/count meter/counter)	
	or read ratemeter (connected to GM tube) or record number of tracks (chamber)	
	or count clicks in a time or note time when click occurs	B1
	or several readings taken or readings fluctuate ignore readings random	
	or time when click occurs varies	B1
(b)	electromagnetic	B1
	(wave/ray/particle) high frequency or small wavelength	B1

- 48.
- (a) electron B1
 negative B1
 electromagnetic (high frequency wave/particle/photon) B1
 neutral/none B1
- (b) (i) time taken to halve M1
 activity **or** number of atoms/nuclei **or** count (rate)
 (**ignore** radioactivity/mass/volume/amount/number of particles/molecules/
 a nucleus to halve) A1
- (ii) alpha stopped by body/flesh/skin **or** cannot penetrate body/skin **or** causes
 damage to body (1 max for damage) B1
 gamma penetrates body/not absorbed **or** can be detected outside body **or**
 causes less/no damage to body (1 max for damage) B1
- (iii) takes time for isotope to spread/investigation/experiment (so 6 min too short) B1
- (iv) radioactive for longer/more dangerous/more damage/causes damage B1
- (c) (i) (radioactive emission is) random B1
- (ii) 3200 seen (as average) C1
 attempt to halve e.g. 3202 → 1601 **or** 4 half-lives C1
 52 hours A1
- (iii) rocks/cosmic rays/radon gas/nuclear fall out B1
- 49.
- (a) (i) 3 **cao** B1
- (ii) 208 **cao** B1
- (iii) 11 **cao** B1
- (b) (i) 17 **cao** B1
- (ii) 20 **cao** B1

50.		
(a)	number of protons and neutrons protons and neutrons in the nucleus	B1 B1
(b) (i)	2	B1
(ii)	4	B1
(iii)	90 or 92– (i) and (iv) 234 or 238– (ii)	B1
51.		
(a)	fusion	B1
(b) (i)	mass decreases or product/nuclei/atoms less massive mass becomes/converted to energy	B1 B1
(ii)	$E = mc^2$ $6.6 \times 10^{-29} \times (3.0 \times 10^8)^2$ 5.9×10^{-12} or 5.94×10^{-12} J	B1 C1 A1
52.		
(a)	different number of neutrons/ mass number	B1
(b)	time taken to halve (number of) nuclei/atoms/activity/count (rate) (not radioactivity/amount/mass/substance/clearly one nucleus/particles)	M1 A1
(c)	mark on graph at 2000 or at two suitable points 12 s	B1 B1
53.		
(a)	emission of at least one of alpha, beta, gamma (particles) from the nucleus or at random	M1 A1
(b) (i)	background stated or explained	B1
(ii)	not radioactive average the same or 93 total on both sides or two increase and one decreases or variation explained	B1 B1